Minimax Estimation–Regret Tradeoffs and Certified Exploration in Performative Prediction

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Abstract

A deployed model that reshapes its own data must pay, in its own objective, to learn how it moves the world. Algorithms for performative prediction consume an estimate of the distribution's response to the model, yet treat that estimate as given and the perturbing deployments that produce it as free. We show the cost obeys an exact conservation law: in a structural market where a dealer's quoted spread summons the adverse flow it must then price, the variance of the response-slope estimate times the cumulative performative regret equals $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ along any stationary exploration policy, an invariant of the environment, not the policy. Around the identity sit minimax floors, an optimal exploration shape ($\Gamma_{\mathrm{PO}}^{-1/2}$), and an agent that corrects only when an anytime confidence sequence certifies the retraining loop stays stable. The floor is structure-proof with an exact reach: inside two curvature lengths of the operating point, full knowledge of the response family cannot beat it; one vanishing-weight probe beyond, a verified design breaks it at 0.8517. Every closed form survives a live pipeline (36 preregistered checks, 0 failures) and a public corporate-bond calibration, matching its published run in 10/10 cells within 0.8%. Anyone who retrains a deployed model sits on this frontier; the identity prices the seat. Code, derivations, and all checks are in the anonymized supplementary repository.

1 Introduction

Deployed prediction and pricing systems do not observe the world from outside it. A credit model changes who applies for credit; a recommender changes what its users will click next; a quoting engine changes the order flow it will next be calibrated to. Performative prediction [Perdomo et al., 2020, Hardt and Mendler-Dünner, 2023] formalizes the consequence: once the deployed decision reshapes the data distribution, retraining becomes a fixed-point iteration, which contracts when the modulus m (the distribution's sensitivity to the decision, measured against the loss curvature) is below one, and which then converges to a *performatively stable* point that is not the *performative optimum* [Miller et al., 2021]. The example that grounds this paper is a dealer in a securities market: the dealer quotes a spread, tighter quotes summon more informed (adverse) flow, and the flow data the dealer retrains on is therefore partly its own reflection. The gap between where retraining stalls and where the dealer should quote is worth real money, and closing it requires knowing the response slope ε: how strongly the world reacts per unit change in the deployed decision. There is only one instrument for measuring that slope, the decision variable itself. Every probe is a live deployment, and its cost lands in the dealer's own P&L.

What does it cost to learn your own influence? We treat that question as an object of theory rather than settle it by assumption, and the problem we address can be stated in one sentence:

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What is the minimum price, measured in the decision-maker's own objective, of learning its own influence on the world, and can that price be certified online rather than assumed?

Existing approaches answer neither half. Performative-optimization methods, performative gradient descent and its descendants [Izzo et al., 2021], consume an estimate of the distribution's response whose accuracy is assumed and whose acquisition, by perturbing live deployments, is treated as free. Regret analyses with performative feedback [Jagadeesan et al., 2022] bound the cost of *finding* the performative optimum, not the cost of *knowing* the feedback that defines it, and they give algorithmic upper bounds where the question above demands lower bounds. Classical experimental design [Pukelsheim, 2006] and dual control [Feldbaum, 1960] optimize information against a budget that is exogenous to the experimenter's objective; here the budget *is* the objective, because the probes are deployments of the decision variable being optimized. The full comparison with these literatures is in Section 2.

This paper answers with an identity, a floor, a design rule, and an agent. The central result is a conservation law: along any stationary exploration policy, the variance of the response-slope estimate times the cumulative performative regret of the exploration that produced it is an invariant of the environment, not of the policy. Information and regret trade at a fixed exchange rate set by the curvature of the decision-maker's value function and the noise in the flow it reads. Around the identity we prove minimax floors that hold against every non-anticipating policy; a structure-proofness theorem showing that within an exactly known reach of the operating point (two curvature lengths, $2/c$) the floor survives even full knowledge of the parametric response family; a design theory that says to explore where the objective is flat and prices the cost of failing to; and a deployable agent whose corrections are gated by a stability certificate, so that the retraining loop provably stays contractive while it learns. The reach is exact in both directions because we report both sides: inside it no design can beat the floor, while one probe beyond it, an explicit design does, with a verified Cramér–Rao ratio of 0.8517. Every closed form in the paper is a preregistered measured-versus-predicted check in a live deploy–observe–fit–correct pipeline (36 rows, 0 failures), transplanted unchanged to a second economy, and instantiated on a public corporate-bond calibration, where it matches the published run in $10/10$ cells.

Contributions.

1. **A conservation law for exploration.** Along any stationary exploration policy, slope uncertainty times cumulative performative regret is a policy-independent invariant, $\mathrm{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2$, where C_T is the regret of exploration over T deployments (Theorem 1, Section 3). The anchoring is delicate: anchored at the performative optimum instead of the certainty-equivalent operating point, the same product is false, inflated by a verified factor. The identity is tested across amplitudes, moduli, and a second economy in Section 5.
2. **Minimax floors with an exact reach.** No non-anticipating policy beats the information floor: exactly under deviation budgets, and within a factor of 13.5 under value budgets, with the sharp constant numerically supported but open (Theorem 2). The floor is also structure-proof within an exact reach (Theorem 3): within $2/c$ of the anchor, even full knowledge of the response family cannot beat it, and one probe beyond, an explicit verified design does. Both theorems are developed in Section 3 and probed at their boundaries in Section 5.
3. **Optimal exploration design.** When the exploration budget is priced by the decision-maker's own curvature, the optimal exploration covariance is shaped as $\Gamma_{\mathrm{PO}}^{-1/2}$: explore where the objective is flat. Isotropic exploration overpays by an explicit dispersion factor between one and the dimension, inverting the "naive exploration is optimal" picture from LQR [Simchowitz and Foster, 2020]. The design theory is in Section 3; its synthetic and real-data verification is in Section 5.
4. **A certified exploration agent.** SafeD-PerfGD explores on an optimal support, fits an anchored response family with an anytime confidence sequence, and applies the performative correction only when a perturbed-modulus certificate guarantees the corrected retraining loop remains stable; otherwise it freezes the correction while the provably stable blind step keeps running. It reaches the performative optimum where blind retraining stalls, and no baseline Pareto-dominates it on median final error and regret (Section 4; results in Section 5).

Section 2 fixes the setting, the related work, and the structural market model; Section 3 develops the identity, the floors, and the design theory; Section 4 describes the agent and the experimental protocol; Section 5 reports the verified results; and Section 6 concludes.

2 Background

2.1 Performative prediction and repeated retraining

When a deployed model changes the population it is deployed on, the training distribution becomes a function of the model itself. Performative prediction [Perdomo et al., 2020] formalizes this with a distribution map $\theta \mapsto \mathcal{D}(\theta)$ and scores a decision rule by its *performative risk*,

$$\mathrm{PR}(\theta) \;=\; \mathbb{E}_{Z\sim\mathcal{D}(\theta)}\big[\ell(Z;\theta)\big],$$

in which θ appears twice: as the decision being evaluated and as the argument of the distribution that evaluates it. The natural operational response to the resulting drift, retraining on the data the current model generates, is *repeated risk minimization* (RRM),

$$\theta_{t+1} \;=\; \arg\min_{\theta}\; \mathbb{E}_{Z\sim\mathcal{D}(\theta_t)}\big[\ell(Z;\theta)\big],$$

a fixed-point iteration on the decision space. Perdomo et al. [2020] prove that when the map is ε-sensitive (the distribution moves at most ε per unit change of the decision, in Wasserstein distance) and the loss is β-jointly smooth and γ-strongly convex, RRM is a contraction with modulus

$$m \;=\; \frac{\varepsilon\,\beta}{\gamma} \;<\; 1,$$

converging to a unique *performatively stable* point θ_{SP}: a decision optimal against the very distribution it induces. When $m \geq 1$ the iteration can oscillate or diverge, so m is also the loop's safety margin. Stochastic retraining variants inherit the same fixed point [Mendler-Dünner et al., 2020]; Hardt and Mendler-Dünner [2023] survey the framework.

Stability, however, is not optimality. The *performative optimum* $\theta_{\mathrm{PO}} = \arg\min_\theta \mathrm{PR}(\theta)$ also differentiates through the distribution argument, and in general $\theta_{\mathrm{SP}} \neq \theta_{\mathrm{PO}}$: the stable point sits inside an echo chamber, optimal against the world it has already created and blind to the value of moving that world [Miller et al., 2021]. The gap between the two is first order in ε, and every algorithm that closes it must, somewhere, consume an estimate of the distribution's response $d\mathcal{D}/d\theta$. That estimate is produced by perturbing real deployments, and the perturbations are charged to the decision-maker's own objective. Pricing that charge exactly is the subject of this paper.

2.2 Related work

Performative optimization. Algorithms that reach θ_{PO} descend the performative risk with an estimated response. Performative gradient descent [Izzo et al., 2021] finite-differences the distribution map across perturbed deployments; Miller et al. [2021] identify structural conditions under which the performative risk is convex and can be optimized outright; Drusvyatskiy and Xiao [2023] place decision-dependent distributions inside standard stochastic-optimization templates; Li and Wai [2022] and Brown et al. [2022] extend the theory to stateful populations that respond with memory rather than instantaneously. Closest to our question, Jagadeesan et al. [2022] bound the regret of *finding* the performative optimum from performative feedback, Bracale et al. [2025] design deployments to learn the distribution map itself and bound the regret of doing so at an order level, and Lin and Zrnic [2024] show that optimizing through a misspecified structural model can beat model-free alternatives, trading bias against variance. Throughout this line the response estimate's accuracy is an assumption, the perturbations that produce it are free, and the guarantees are algorithmic upper bounds; this paper prices the acquisition exactly, in the decision-maker's own curvature, and closes it from below with minimax floors.

Learning and earning in operations research. Dynamic pricing has long known that earning interferes with learning. Rothschild [1974] showed that a seller pricing optimally under an unknown demand curve can settle forever on the wrong price; Harrison et al. [2012] and Broder and Rusmevichientong [2012] traced such stalls to uninformative prices at which myopic policies stop learning, and Keskin and Zeevi [2014] showed that forcing a minimal amount of price dispersion restores near-optimal performance; den Boer [2015] surveys the tradition, and the economics of optimal experimentation develops the same tension for general Bayesian control [Easley and Kiefer, 1988, Aghion et al., 1991]. On the design side, Lai and Robbins [1979] construct adaptive experiments whose learning cost grows only logarithmically in the horizon. In all of this the demand curve is exogenous: the market does not move because the seller retrains, so there is no retraining loop, no stability constraint, and no frontier shared across policies. In our setting the loop itself generates the data, incomplete learning reappears as a saturation law (Section 3), and the schemes of Lai and Robbins [1979] become points on a frontier that every stationary policy occupies.

Dual control and optimal experiment design. Control theory names the tension dual control: an input must simultaneously regulate the plant and excite it enough to be identified [Feldbaum, 1960]. Least-costly identification prices the experiment in closed-loop control performance [Bombois et al., 2006]; task-optimal exploration and the sample-complexity theory of the linear quadratic regulator give the modern learning-theoretic treatment [Wagenmaker et al., 2021, Dean et al., 2020], where for online LQR naive isotropic exploration is already optimal [Simchowitz and Foster, 2020]. Behind both stands the classical theory of optimal experiment design [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006], and the lower-bound machinery we use, the Bayesian Cramér–Rao inequality of van Trees, comes from this statistical canon [Gill and Levit, 1995]. None of these price the design in the experimenter's own objective with a retraining learner as the plant and loop contraction as the constraint; and the LQR message inverts in our setting, where isotropic exploration overpays by an explicit dispersion factor (Section 3).

Market microstructure. The environment itself comes from market making. Dealers set quotes against the adverse selection of informed flow [Guéant et al., 2013], and Barzykin et al. [2026] model price reading: the dealer's own quotes alter the flow it then trades against. REFLEX [Nagarajan and Ashok, 2026] closes this channel into a performative retraining loop and calibrates it on corporate-bond data; we adopt its market as our structural environment. Microstructure models optimize quoting against a known flow response; here the response is the unknown, and the cost of estimating it is the quantity under study.

2.3 A structural market model

Our environment is the single-book structural market of REFLEX [Nagarajan and Ashok, 2026], a prior model this paper builds on; its gate and participation constants are folded into the flow scales below. A dealer quotes a half-spread h around a reference price. Two flows respond to the quote. Benign (uninformed) flow arrives at rate

$$U(h) \;=\; A\,e^{-kh},$$

and informed, toxic flow at rate

$$\tau(h) \;=\; C_0 + C_1\,e^{-ch},$$

so tighter quotes summon more toxic flow. This channel is the dealer's adverse-selection problem [Guéant et al., 2013], and the fact that the dealer's own quotes feed the flow it then reads is the price-reading effect of Barzykin et al. [2026]. At a frozen toxic arrival level τ, one period's profit is

$$J(h;\tau) \;=\; h\,U(h) \;+\; (h-\psi)\,\tau \;-\; w\,(h-h_{\mathrm{ref}})^2,$$

where ψ is the adverse severity (the expected loss per unit of toxic flow) and w the convexity of the quoting cost around the reference half-spread h_{ref}. The *performative value* substitutes the true response,

$$\Phi(h) \;=\; J\big(h,\tau(h)\big),$$

and is the objective a feedback-aware dealer should maximize. Three derived quantities organize everything that follows: the response slope $\varepsilon(h) = -\tau'(h) = c\,C_1 e^{-ch}$, the scalar instance of $d\mathcal{D}/d\theta$; the myopic curvature $\gamma = -\partial_h^2 J$; and the performative curvature

$$\gamma_{\mathrm{PO}} \;=\; -\Phi'' \;=\; \gamma + \varepsilon\,(2 + c\psi - ch).$$

Blind retraining is the cobweb iteration: the dealer observes the current toxic level and best-responds to it as if it were exogenous,

$$h_{t+1} \;=\; \mathrm{BR}\big(\tau(h_t)\big), \qquad \mathrm{BR}(\tau) \;=\; \arg\max_h J(h;\tau).$$

This is RRM instantiated in the market, and it is a contraction if and only if the modulus $m = \varepsilon\beta/\gamma = \varepsilon/\gamma$ is below one ($\beta = 1$ in this model) [Perdomo et al., 2020]; it converges to the stable point h_{SP}. The performative optimum $h_{\mathrm{PO}} = \arg\max \Phi$ sits elsewhere, and the sign of the gap matters. At the stable point, written h^* from here on and serving as the operating point that anchors every expansion below, the value gradient is

$$\Phi'(h^*) \;=\; -(h^* - \psi)\,\varepsilon(h^*) \;=:\; \delta_0,$$

which is negative whenever the operating spread exceeds the adverse severity, that is, whenever toxic flow is profitable net of ψ. The optimum therefore lies at a *tighter* spread than the stable point: the cobweb prices against the toxic flow it currently sees, while the optimum internalizes that tightening summons more of it, and since each summoned unit still pays $h - \psi > 0$, some of that flow is worth buying. This is the echo-chamber gap of Miller et al. [2021] with a sign that a defensive intuition would get wrong: awareness of adverse selection tightens the quote here rather than widening it.

Deployments are observed through noise: quoting h_t returns the flow reading

$$y_t \;=\; \tau(h_t) + \sigma\,\zeta_t,$$

with ζ_t iid, mean zero, unit variance (A3 below). Writing deviations $d_t = h_t - h^*$ and linearizing the loop and the response at the stable point gives the working model of Section 3,

$$d_{t+1} \;=\; -m\,d_t + u_t, \qquad y_t \;=\; \tau(h^*) - \varepsilon\,d_t + \sigma\,\zeta_t,$$

where u_t is the exploration input the operator adds to the loop. Exploration is priced incrementally over a horizon T: the value budget is

$$C_T \;=\; \sum_{t \le T} \mathbb{E}\big[\Phi(h^*) - \Phi(h_t)\big],$$

the expected performative value given up relative to the jitter-free loop sitting at h^*, and the deviation budget is $D_T = \sum_{t \le T} d_t^2$, the realized design energy (the tracking-error form of a desk's deviation limit). The anchor at h^* is load-bearing: anchoring at h_{PO} instead adds a sunk echo-chamber term that is paid whether or not the operator explores, and it provably destroys the exchange-rate invariance of Section 3. Results below are stated for this scalar family and, where flagged, for its d-dimensional and linear-quadratic instances.

2.4 Assumptions and notation

Five register assumptions carry the main text; we state each in prose with the role it plays, and point to Appendix C for the formal register, which also fixes the adaptive and relaxed variants.

A1 (local quadratic scope). Φ is C^3 and strongly concave on the operating interval, and all expansions are taken at h^* with the third-order remainder controlled by $L_3 = \sup|\Phi'''|$. The identities of Section 3 are exact for the quadratic model and first-order accurate otherwise; the nonlinear deviation is measured in Section 5, never assumed away.

A2-sym (symmetric exploration). The exploration input is non-anticipating and conditionally symmetric, $\mathbb{E}[u_t \mid \mathcal{F}_t] = 0$, with $\mathcal{F}_t$ the history sigma-field. This makes the linear term of the cost expansion vanish in expectation, so the incremental cost is carried entirely by the quadratic term $\frac{1}{2}\gamma_{\mathrm{PO}}\,\mathbb{E}[D_T]$; that cancellation is the mechanism behind the exchange-rate identity. The register also defines the unrestricted class A2-adaptive, under which the minimax results measure cost from a certainty-equivalent anchor.

A3 (noise exogeneity). The observation noise is iid, mean zero, with variance σ^2, independent of the history. This fixes the information scale of every bound. The register's named relaxation A3$'$, in which observed noise feeds back into the next deployment, is analyzed rather than excluded, with its bias quantified in the appendix (P2.2).

A5 (stationary exploration scale). The cumulative design energy grows linearly with the horizon, $D_T = \Theta(T)$. This separates the stationary regime, where the exchange rate and the value-budget floor operate, from the transient regime, which is governed by the saturation result of Section 3 instead.

A6 (stable base loop). The blind loop contracts: $m < 1$ in the scalar model, and $\rho(M) < 1$ for the loop Jacobian M in the d-dimensional case. Every stationary-regime statement presumes that

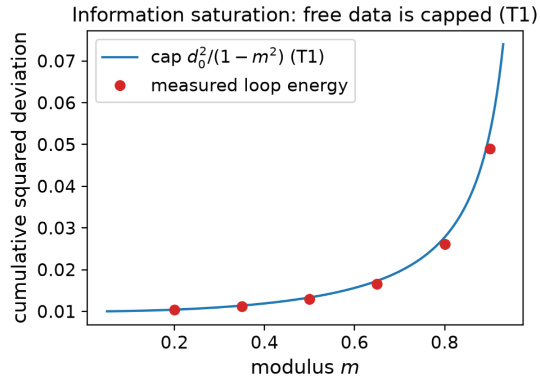

Figure 1: Information saturation (T1). Measured identifying energy of the blind retraining loop against the cap $d_0^2/(1-m^2)$: the energy converges while the horizon grows without bound, so free data cannot identify the response slope. Quantitative agreement, including the noise-driven excitation floor, is reported in Section 5.

the retraining iteration being perturbed is itself stable; the boundary case $m = 1$ appears only in a degeneracy statement in the appendix.

The register additionally carries A4, a pathwise trust region $|d_t| \leq r$, which enters the feasible sets of the design problems and the agent's certification in Section 4. One estimation convention completes the set: the estimand is the slope ε at the operating point, and $\hat{\varepsilon}$ is the ordinary-least-squares slope of the observed flow on the deployed deviations. The exchange-rate identity of Section 3 is stated for this estimator; the minimax floors quantify over all estimators.

Notation. Primes denote derivatives of scalar functions, transposes are written $(\cdot)^\top$, logarithms are natural, $\mathbb{E}$ and Var denote expectation and variance, and the full symbol table, with each symbol's source, is reproduced in Appendix C.

3 Theory

This section develops the theory in five steps: what the retraining loop's free data can identify (nothing, past a cap), the exact price of deliberate exploration, the floors that no policy of any kind escapes, the exact boundary at which structural knowledge starts to beat those floors, and the design rules for spending a budget that must be paid. Full register statements are in Appendix A and proofs in Appendix B; every closed form below is verified in Section 5.

3.1 Information from free data

Blind retraining appears to generate data for free: every deployment returns a noisy observation of the flow it summoned. The identifying content of that data is finite. For a noiseless loop with initial deployment error d_0 and modulus m, the trajectory's total design energy is capped at $d_0^2/(1-m^2)$ (T1; the matrix version is a discrete Lyapunov identity, with energy $d_0^\top P\, d_0$ for $P = I + M^\top PM$), so the Fisher information for the response slope is bounded uniformly in the horizon: patience does not substitute for excitation. A companion closed form, discovered through a failed prediction (Section 5) and verified numerically rather than carried as a register statement, extends the cap to the noisy loop: retraining on noisy observations converts observation noise into deployment noise with gain $1/\gamma$, adding a stationary excitation floor of $(\sigma/\gamma)^2/(1-m^2)$ per step. Both closed forms are verified in Section 5; Figure 1 shows the saturation.

The cap is increasing in the modulus, so the loops that contract fastest reveal least, and precise self-knowledge is available only near the stability boundary (C1.1): *safety implies blindness*. This is the performative sharpening of incomplete learning under myopic policies [Keskin and Zeevi, 2014, Harrison et al., 2012], with the stall caused by optimization converging rather than by weak feedback. The trajectory is also geometrically degenerate, with at most two effective support points (C1.2), a fact that returns in the design theory below.

3.2 An exchange rate between estimation and regret

Deliberate exploration escapes the cap, and its price is exact.

Theorem 1 (Exchange rate; T2)**.** *Under A1, A2-sym, A3 and A6, for the OLS slope estimator on any stationary exploration policy with deviations* $\{d_t\}$*,*

$$\mathrm{Var}(\hat{\varepsilon}) \;\times\; C_T \;=\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\,\big(1+o(1)\big),$$

independent of the exploration amplitude, shape, schedule, and modulus.

The mechanism is a cancellation. The conditional OLS variance is σ^2/S_{xx}, where S_{xx} is the *centered* design energy of the deployed path; by the cost-equivalence lemma (L1, Appendix A) the incremental cost is $\frac{1}{2}\gamma_{\mathrm{PO}}\sum_t d_t^2$ up to a cubic remainder. For mean-centered exploration the two energies coincide, so amplitude, shape, schedule, and modulus all cancel from the product. The register statement (T2, Appendix A) is pathwise and exact: on every realization the product equals the invariant times $1+T\bar{d}^2/S_{xx}$, the mean-offset factor that the $o(1)$ absorbs.

The choice of anchor is decisive. The cost C_T is anchored at the certainty-equivalent operating point h^*; anchored at the performative optimum h_{PO} instead, the same product is *false*, inflated by an explicit factor, and we know this because the h_{PO}-anchored identity was falsified numerically before the incremental version was proved. The invariant prices *exploration*, not *suboptimality*: the distance from h_{PO} is a separate account, and conflating the two produces a plausible-looking conservation law that fails.

The identity draws a frontier through classical adaptive design. The $O(\log n)$-cost schemes of Lai and Robbins [1979] are points on it; the identity says *every* stationary policy is, and every deterministic design sits on it exactly (P9.1). A further register result prices the echo chamber: paying to close the gap $h_{\mathrm{SP}}-h_{\mathrm{PO}}$ is worthwhile iff the discount rate falls below an explicit break-even (T8, Appendix A). Measured products at two moduli and two amplitudes each, and the drift of the product outside A1's local scope, are reported in Section 5.

3.3 Minimax lower bounds

The identity binds OLS on stationary policies. The floors bind everyone: randomized, adaptive, biased.

Theorem 2 (Minimax floors; T3–T4)**.** *(i) Over all non-anticipating (randomized, adaptive) policies with pathwise budget* $\sum_{t\le T} d_t^2 \le D$*, all estimators (biased included), and the slope ranging over an interval of width* w*:* $\inf\sup\mathbb{E}\big[(\hat{\varepsilon}-\varepsilon)^2\big] \ge \sigma^2/(D+\sigma^2(\pi/w)^2)$*, so the product floor is attained up to the prior term, with equality by symmetric probing. (ii) Under stationary-scale exploration and a budget on realized performative cost with certainty-equivalent anchoring:* $\sup\mathrm{Var}(\hat{\varepsilon})\times C_T \ge \gamma_{\mathrm{PO}}\sigma^2/27\,(1-O(T^{-1/2}))$ *over adaptive policies, by a Le Cam two-point reduction with the exploited rebate vanishing at rate* $T^{-1/2}$ *(Lemma L2, Appendix A); the sharp constant* $\frac{1}{2}$ *is numerically supported and open (OPEN-1).*

Part (i) prices a hard deviation budget: with total squared deviation at most D, no policy improves on symmetric probing with least squares, and the term $\sigma^2(\pi/w)^2$ is the van Trees price of the parameter interval's width [Gill and Levit, 1995]. Part (ii) is the harder statement, because its budget is the decision-maker's own realized performative cost, which an adaptive policy could hope to rebate by exploiting mid-flight what it has already learned. The Le Cam route closes this loophole: two response families straddling the anchor remain indistinguishable until design energy separates them, and the exploitation lemma (L2) bounds the rebate by the information already purchased, via an exact total-variation identity and Pinsker's inequality; at the minimax scale the rebate is an $O(T^{-1/2})$ fraction of the $\Theta(T)$ exploration cost. Optimizing the two-point reduction gives the constant $1/27$.

We state the remaining gap plainly. The proved constant $1/27$ sits a factor of 13.5 below the conjectured sharp constant $\frac{1}{2}$, the invariant's own value. The sharp constant is numerically supported, in that no simulated policy, adaptive schedules included, fell below the invariant by more than Monte Carlo error (Section 5), but a proof at any prior class is missing; this is OPEN-1.

3.4 The reach of the floor

Theorem 2 quantifies over policies and estimators, but not over side information. A policy that knows the response family's functional form, $\tau(h) = C_0 + C_1 e^{-ch}$, could shop for designs on which the family's rigidity makes the slope cheap to infer, and thereby undercut the floor. Whether structural knowledge beats the floor has an exact answer.

Theorem 3 (Structure-proofness, exact reach; T9)**.** *Let $R(\mu)$ be the known-family Cramér–Rao product of design μ, normalized by the floor (a bound for unbiased, and asymptotically all regular, estimation). (i) Every estimable μ supported in $[h^* - 2/c, \infty)$ has $R(\mu) \geq 1$, strict off one two-point configuration, and $\inf_\mu R(\mu) = 1$:* within reach, knowing the family cannot beat the floor. *(ii) The reach is exact: a four-point design with one below-reach probe of weight 3×10^{-4} attains $R = 0.8517$ (50-digit verified).*

The proof exhibits the family element $\phi_0 = 2/c - e^{-c(h-h^*)}(2/c + h - h^*)$, whose pointwise domination $|\phi_0| \leq |h - h^*|$ holds exactly on the reach $[h^* - 2/c, \infty)$ and fails strictly below it; a Rayleigh-quotient reduction converts pointwise domination into the floor, with equality forced at exactly the two points $\{h^*,\ h^* - 2/c\}$. The reach is two curvature lengths of the response: within $2/c$ of the anchor, even full knowledge of the family cannot beat the floor; one probe beyond, an explicit design does.

Two boundary phenomena calibrate what the theorem does and does not protect. First, the floor is an ignorance-of-curvature phenomenon. With the curvature c *known*, it breaks outright: the normalized product falls to 0.30 for a probe at $h{=}1.0$ and to 0.04 at $h{=}0.05$, decreasing monotonically as the probe moves away from the anchor. The remote sensitivity e^{-ch} no longer has to be pulled back through an estimate of c, so its cheapness is banked rather than discounted. Second, part (ii)'s witness explains why the boundary went undetected. The violating designs are asymmetric near-anchor clusters carrying almost all the weight, plus a single below-reach probe whose vanishing weight, 3×10^{-4}, is what breaks the domination. Designs of this shape are invisible to coarse search: symmetric-pair-plus-far-probe families stay above the floor at every weight, approaching 1 from above as the probe weight vanishes, so the violation requires the asymmetry. How our own earlier scan missed it, and what correcting that verdict took, is reported with the measurements in Section 5.2.

The theorem hands the agent its trust region. Any design confined to radius $r \leq 2/c$ around the operating point provably sits inside the floor's protection (A4), so the certification argument of Section 4 never rests on a bound that structural knowledge could undercut. The hardened within-reach searches and the 50-digit verification of the witness are reported in Section 5.

3.5 Optimal exploration design

With the price fixed, the remaining freedom is where to spend. **Shape (T5).** In d dimensions the curvature $\Gamma_{\mathrm{PO}} \succ 0$ prices exploration through the value budget $\mathrm{tr}(\Gamma_{\mathrm{PO}} M) \leq B$ on the exploration second moment M. The A-optimal design is $M^* = \left(B/\,\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2}\right)\Gamma_{\mathrm{PO}}^{-1/2}$: explore where your objective is flat, with a square-root rather than full inversion of the curvature, since curvature both prices the probe and shapes the risk. The D-optimal design is $(B/d)\,\Gamma_{\mathrm{PO}}^{-1}$, and the c-optimal design probes along the correction direction itself (full statement in Appendix A). Isotropic jitter at matched budget overpays the A-optimal design by exactly the curvature-dispersion factor $F = d\,\mathrm{tr}\,\Gamma_{\mathrm{PO}}/(\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2})^2 \in [1, d]$, with $F = 1$ iff the curvature spectrum is flat (C5.1). This inverts the naive-exploration optimality known for online LQR [Simchowitz and Foster, 2020]: once exploration is priced in the explorer's own curved objective, isotropy is suboptimal by exactly the dispersion of that curvature (P9.2). Temporal shaping buys nothing beyond the stationary design measure (L3); the confirmation at $d = 8$, and a portfolio-level instance, are in Section 5.

Support (T6). Support matters as much as shape. The structural family's sensitivity system is an extended Chebyshev system, so any design with fewer than three support points has singular Fisher information for (C_0, C_1, c); designs with vanishing spread or collinear probe schedules degenerate to the same singularity. Combined with the trajectory degeneracy of C1.2 (at most two effective support points), no blind retraining trajectory, at any modulus, identifies the response curvature; the two occasions on which the pipeline hit this cliff in the wild are recounted in Section 5.

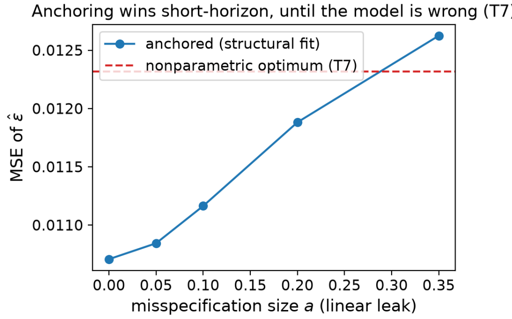

Figure 2: The T7 anchoring crossover. Structural fitting beats the budget-and-horizon-optimal nonparametric estimator iff the misspecification at the operating point is below $|\tau'''(h^*)|\, B/(3\gamma_{\mathrm{PO}}T)$: anchoring wins when the budget is large relative to the horizon, the free-form route asymptotically in T. Measured errors at the crossover are reported in Section 5.

Anchor (T7). The estimator family is itself a design choice. The budget-and-horizon-optimal nonparametric secant estimator has $\mathrm{MSE}_{\mathrm{np}} = \gamma_{\mathrm{PO}}\sigma^2/(2B) + (\tau'''(h^*)/6)^2(2B/(\gamma_{\mathrm{PO}}T))^2$, so fitting the structural family wins iff the misspecification at the operating point is below $|\tau'''(h^*)|\, B/(3\gamma_{\mathrm{PO}}T)$ to leading order: anchoring wins when the exploration budget is large relative to the horizon, and the free-form route takes over asymptotically in T (Figure 2). This sharpens the message of plug-in performative optimization [Lin and Zrnic, 2024], that misspecified structural models still help, into a budget-priced rule with an explicit crossover.

Read together, these results are a blueprint rather than a diagnosis. The exchange rate fixes what a unit of self-knowledge costs, the floors certify that no policy avoids the bill inside the trust region, and the design theory says how to pay it with the least regret: probe on a three-point support shaped by $\Gamma_{\mathrm{PO}}^{-1/2}$, anchored to the structural family inside its crossover. Section 4 composes exactly these pieces, together with a closed-loop stability certificate, into a deployable agent.

4 Method

4.1 A certified exploration agent

SafeD-PerfGD turns the theory of Section 3 into a deployable loop: exploration is priced (in the agent's own objective), shaped (by the optimal designs of T5–T6), and safety-gated (by a pessimistic certificate on the agent's own closed-loop modulus). Each deployment round runs four steps.

1. **Explore.** Deploy on the D-optimal three-point support for the structural family (C_0, C_1, c), re-centered at the current operating point; three points is the minimal support that escapes the Chebyshev degeneracy of T6. Every probe is clipped to a trust region of radius $r \le 2/c$, which by Theorem 3 keeps the agent inside the reach on which the floor is structure-proof.
2. **Fit.** Re-estimate (C_0, C_1, c) by profile least squares on the full deployment history (the linear pair (C_0, C_1) is solved in closed form at each candidate curvature c), and maintain an anytime self-normalized confidence width ci_t for $\hat{\varepsilon}$, constructed to hold simultaneously over stopping times; the safety guarantee below is stated on that coverage event.
3. **Gate.** Form the performative correction $-(h-\psi)\,\hat{\varepsilon}$ with Newton scaling $\eta = 1/\hat{\gamma}_{\mathrm{PO}}$, and apply it only if $\eta\, \mathrm{ci}_t\, L_{\mathrm{fam}} \le$ margin. The perturbed-modulus lemma (Lemma L4, Appendix A), instantiated with the family's own Lipschitz constant $L_{\mathrm{fam}} = 1 + \hat{c}\,|h-\psi|$, converts a certified slope error of width ci_t into a bound on the modulus of the corrected retraining map, so the inequality certifies that the closed loop remains a contraction.
4. **Freeze.** Otherwise withhold only the correction: the blind retraining step, provably stable whenever $m < 1$, keeps running underneath, so an uncertified agent degrades to safe blind retraining rather than to instability.

Three properties frame the evaluation. First, the freeze is derived, not tuned: the gate threshold is the stability margin of Lemma L4, not a hyperparameter fit to trajectories. Second, safety is an event, not an average: on the confidence-sequence event, which holds with the prescribed probability uniformly over time, the deployed loop's modulus stays below one at every round, gated or frozen (Proposition P7.1, Appendix A). Third, certification is not free: the gate opens only after enough design energy has accumulated to shrink ci_t below the margin, and by Theorem 1 the accumulation of design energy *is* a regret payment. The certification phase is the price of self-knowledge paid online; Section 5 measures both the payment and when the gate clears.

4.2 Experimental setup

Verification-first protocol. Every quantitative claim in Section 3 was preregistered as a row in a measured-vs-predicted harness: the closed form, the measurement procedure, and the tolerance were fixed before the run, and the harness stamps each row PASS, DRIFT, or FAIL. DRIFT is reserved for cells deliberately placed outside the assumptions' scope, whose deviation is measured and reported, not excused. The register holds 36 rows. The loop was run throughout development, and eleven pivots were forced by measurement; each is recorded in the code where it happened, with the failed prediction and the fix, and several became results reported in Section 5.

Harness. The pipeline is numpy-only, CPU-only, and deterministic from seeds. Experiments E1–E8 and E10 populate the verification register; a separate adversarial-search run probes the open constant of Theorem 2 and the reach boundary of Theorem 3; a real-data run executes the calibration leg. Each evaluation ships as an artifact in the supplementary repository: the verification table (RESULTS.md), the adversarial-search record (OPEN1.md), the real-data leg (REALDATA.md), and the horizon-stability audit of the metric (STABILITY.md), alongside the unit-test suite and continuous integration.

Environments. Four environments share one interface: (i) the register market of Section 2, implemented exactly, plus a saturating variant with a capped toxic channel that serves as the designated out-of-scope drift cell; (ii) a second economy, LQ performative pricing, in which the deployed price anchors demand and the local-quadratic assumption holds globally; (iii) a d-dimensional multi-bond universe for shaped versus isotropic exploration at matched budget; (iv) the corporate-bond calibration of REFLEX [Nagarajan and Ashok, 2026], 10 rating×regime cells. The calibration inherits a binding provenance caveat: only (A, k, σ, h) are data-identified, the toxic channel is structurally scaled, and the validation inputs (the published calibration run) live in the source repository, so the port is validated against that run rather than against market ground truth.

Baselines. Four baselines run on identical environments, horizons, and noise: finite-difference PerfGD [Izzo et al., 2021]; zeroth-order optimization of the noisy P&L; a performative-confidence-bounds explorer in the spirit of Jagadeesan et al. [2022]; and blind repeated risk minimization [Perdomo et al., 2020]. Comparisons use 12 seeds and report median and interquartile range; an ablation grid additionally toggles SafeD-PerfGD's design, anchor, and gate independently.

Metrics. Every reported final error is a 300-step tail average over the deployed path, never a last iterate: unanchored baselines limit-cycle, and a last iterate samples the cycle at an arbitrary phase. The tail average is audited at perturbed horizons and required to be stable within 0.01; the full audit is in the stability artifact. Regret is accumulated over the deployed path, so certification and exploration costs appear in it by construction, and estimation accuracy is scored across seeds against Fisher-information closed forms where the theory supplies them. Full configuration details are in Appendix D.

5 Results

Every claim in Sections 3 and 4 is a preregistered row in the measured-versus-predicted harness: **36 rows, 0 failures** (35 pass; the one remaining row is a deliberately out-of-scope drift cell, measured rather than excused). The exchange-rate identity is flat across exploration amplitudes and moduli within its scope, and exact in the LQ economy where its local assumption holds globally. The minimax floor survives every design a hardened within-reach search could construct (minimum Cramér–Rao ratio 1.0005) and breaks exactly one probe beyond the reach (ratio 0.8517, verified at 50-digit precision). No baseline Pareto-dominates SafeD-PerfGD on median (final error, regret) over 12 seeds, and the port of the REFLEX corporate-bond calibration matches its published run in 10/10

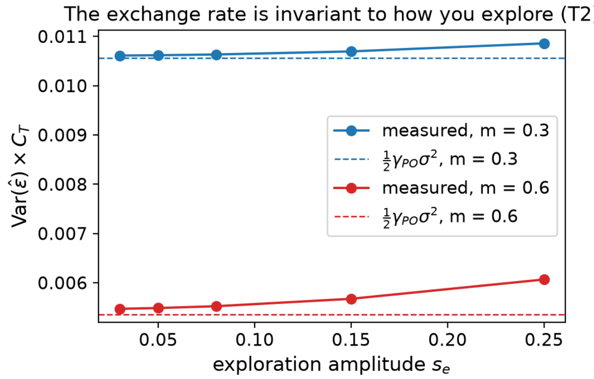

Figure 3: **The identity in practice (E2).** $\mathrm{Var}(\hat{\varepsilon}) \times C_T$ against exploration amplitude at moduli $m = 0.3$ and $m = 0.6$; dashed lines mark the invariant $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$. The product is flat in amplitude and moves with the modulus only through γ_{PO}; the small upward drift at the larger amplitude is the finite-amplitude A1 effect that the out-of-scope saturating cell magnifies to $+16\%$.

cells within 0.8%. Table 1 collects the headline rows; Appendix D details each experiment, and the full 36-row record ships with the repository.

5.1 The identity in practice

Theorem 1 predicts $\mathrm{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ for every stationary exploration policy. Experiment E2 measures the product on the dealer market at two moduli and two exploration amplitudes. At $m = 0.3$ the measured products are 0.010621 and 0.010696 against the predicted 0.010558 (relative gaps 0.6% and 1.3%); at $m = 0.6$ they are 0.0054946 and 0.0056856 against 0.0053539 (2.6% and 6.2%), all inside the preregistered 15% band. The amplitude triples between the two cells at each modulus, and the measured product moves by 0.7% (m=0.3) and 3.5% (m=0.6): the product is a property of the environment, not of the policy (Figure 3).

Where A1 is global, the identity is exact. Transplanted unchanged to the LQ performative-pricing economy (E8), where the quadratic scope holds globally, the identity is exact in all four cells (two feedback gains, two exploration amplitudes): measured 0.384, 0.384, 0.192, 0.192, equal to the closed form at the reported precision (0.0% relative gap). The residual drift above is a boundary effect of the dealer family, not of the theorem.

Two negative results. First, the saturating environment sits outside A1's scope by design, and its product drifts $+16\%$ above the invariant (measured 16.3%); the cell is preregistered as a drift cell and reported, not excused: the $o(1)$ qualifier in Theorem 1 does real work at finite amplitude. Second, the same product anchored at the performative optimum h_{PO} rather than the certainty-equivalent operating point is *false*, inflated by a verified factor; that falsification preceded the proof and forced the incremental anchoring the theorem carries.

5.2 Lower bounds at their boundaries

Adaptive schedules and OPEN-1. Theorem 2(ii) proves the value-budget floor at $\gamma_{\mathrm{PO}}\sigma^2/27$, a factor 13.5 below the conjectured sharp constant $\frac{1}{2}$. A Monte Carlo search over adaptive amplitude schedules under the true (not the modeled) cost lands on the conjecture rather than under it: a constant centered schedule attains ratio 1.018 against $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ and a decaying schedule 0.995, both indistinguishable from 1 up to Monte Carlo error and small-amplitude drift, while a schedule with its anchor misplaced by 0.2 pays ratio 4.91. Amplitude adaptivity does not appear to buy anything below the conjectured constant; the sharp value remains open (OPEN-1).

Within reach, the floor is tight. For Theorem 3(i), a hardened Cramér–Rao search over known-family designs restricted to the reach $[h^* - 2/c, \infty)$ (three independent scans of $30{,}000$ random designs each, plus $10{,}000$ refinements) finds a minimum ratio of 1.00054: no design that full knowledge of the family can construct inside the reach beats the floor, as the theorem proves it must not.

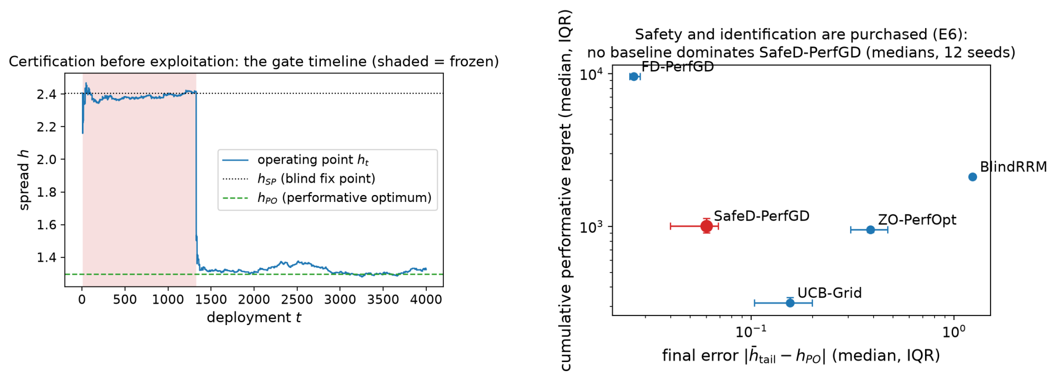

Figure 4: **Left (E4):** certification before exploitation. Shaded: the frozen phase, in which the gate withholds the performative correction while the provably stable blind step keeps running; the gate clears after roughly 1,300 of $T = 4000$ deployments (freeze fraction 0.33) and the agent steps to h_{PO} with zero trust-region violations. **Right (E6):** the baseline frontier over 12 seeds: no baseline Pareto-dominates SafeD-PerfGD on median (final error, regret); the UCB explorer buys the lowest raw regret at more than twice SafeD's final error.

One probe beyond, it breaks. The unrestricted search attains ratio 0.8517 at a four-point design with support $\{1.8726,\ 1.8665,\ 1.3073,\ 0.05\}$: an asymmetric near-anchor cluster plus a single below-reach probe at $h = 0.05$ carrying weight 3×10^{-4}. The witness is frozen and its ratio re-verified independently at 50-digit precision. Our earlier verdict of global structure-proofness (minimum 1.005) was a search artifact, recorded as the eleventh measurement-forced pivot: the coarse scan's weight grid never went below 0.05 while the violating probe carries 3×10^{-4}, and symmetric-pair-plus-far-probe families approach the floor from above at every weight, so only the asymmetric structure produces a violation. The proof of part (i) is what exposed it: the domination argument stops exactly at $h^* - 2/c$, and searching just beyond that boundary produced the witness. One scope caveat transfers: under the true finite-amplitude cost the witness ratio is $1.0112 > 1$, so the violation lives in the local-quadratic cost model, which is the floor's own scope.

With curvature known, no floor at all. Dropping c from the estimand breaks the floor outright: the best known-c ratio is 0.30 with a probe at $h = 1.0$ and falls monotonically to 0.04 at $h = 0.05$, still decreasing at the scan's edge. Together the three boundaries fix the mechanism: the floor is an ignorance-of-curvature phenomenon with an exactly measured boundary. Within two curvature lengths of the anchor it binds even against full knowledge of the response family; one vanishing-weight probe beyond, it fails by 15%; with the curvature parameter known, it disappears entirely.

5.3 Comparison with baselines

Over 12 seeds on identical environments (medians with interquartile ranges), SafeD-PerfGD attains final error 0.0599 $[0.0398, 0.0687]$ at regret 1007.86 $[910.00, 1130.79]$. Finite-difference PerfGD [Izzo et al., 2021] reaches a lower final error (0.0263 $[0.0251, 0.0283]$) but pays regret 9586.79 $[9580.86, 9593.83]$, more than five times SafeD's median (the preregistered threshold; the measured ratio exceeds nine): the cost of uncertified gradient probing lands in the objective, exactly as Theorem 1 prices it. Zeroth-order optimization stalls at error 0.3868 (regret 952.33); blind RRM stalls near the stable point at error 1.2293 (regret 2115.54).

The closest contest is with the UCB explorer [Jagadeesan et al., 2022]: it attains the lowest raw regret, 315.92 $[309.40, 341.71]$, and Pareto-edges SafeD in one seed of twelve (seed 106), while settling at final error 0.1547 $[0.1036, 0.1990]$, more than twice SafeD's. On median, no baseline Pareto-dominates SafeD on (final error, regret). What SafeD alone provides is the certificate: in the agent run (E4) it settles at 1.258 against $h_{\text{PO}} = 1.296$ with zero trust-region violations (maximum step 0.88 of the cap), and its stability guarantee holds on the confidence-sequence event throughout. Certification is visibly not free: the gate clears only after roughly $1{,}300$ of $T = 4000$ deployments, a freeze fraction of 0.33 (Figure 4, left). By Theorem 1 the design energy that opens the gate *is* a regret payment; the frozen third of the horizon is that payment made visible.

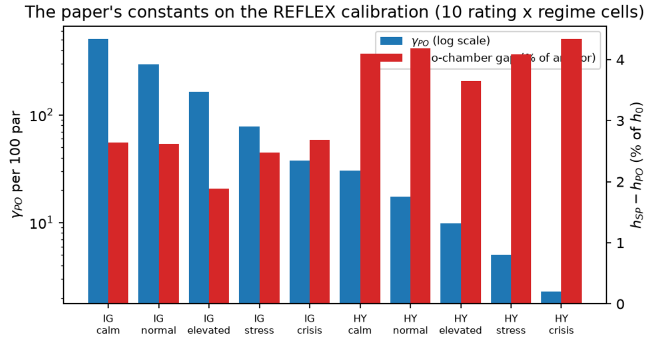

Figure 5: **The paper's constants on the REFLEX calibration** [Nagarajan and Ashok, 2026], 10 rating×regime cells (per-bond leg: 170 CUSIPs): γ_{PO} spans 510.6 to 2.3, echo-chamber gaps are 1.9–4.3% of the anchor spread, and the dispersion factor is $F = 1.63$ across the portfolio against 1.002 within one book.

5.4 Real-data validation

We instantiate every constant on the corporate-bond calibration of REFLEX [Nagarajan and Ashok, 2026]: 10 rating×regime cells, with a per-bond leg over 170 CUSIPs having at least 12 months of returns. The port reproduces the published REFLEX run in **10/10** cells within 0.8%, with per-quantity maxima of 0.23% on h^*, 0.62% on $\varepsilon^* = \gamma$, and 0.76% on m.

The extension computes this paper's objects on those cells (Figure 5). The curvature γ_{PO} spans 510.6 (IG-calm) to 2.3 (HY-crisis), so the exchange rate $\frac{1}{2}\gamma_{\mathrm{PO}}$ spans 255.3 to 1.2: the price of self-knowledge varies by a factor of more than 200 across a realistic book. Echo-chamber gaps $h_{\mathrm{SP}} - h_{\mathrm{PO}}$ are 1.9–4.3% of the anchor spread, and the calibrated moduli sit at $m \approx 0.055$–0.066, deeply inside the contraction regime. The dispersion asymmetry is the operational finding: across the 10-cell portfolio the dispersion factor is $F = 1.6252$, so isotropic exploration overpays the A-optimal shape by about 63%, while within the single IG-normal book, over the 170-CUSIP universe, $F = 1.0020$ (an overpayment of 0.2%). The shaping gain lives *across* books, over ratings and regimes, which is where a desk actually allocates exploration budget; within one book the anchor-dominated curvatures are nearly flat. Provenance bounds the claim: only (A, k, σ, h) are data-identified, the toxic channel is structurally scaled, and the validation target is REFLEX's published run rather than market ground truth, so this leg validates the machinery on realistic constants, not the effect size.

5.5 Robustness

Metric stability. Every reported final error is a 300-step tail average over the deployed path. The stability audit re-measures every cell at perturbed horizons (five for the ablation grid, three for each agent cell); the worst spread across all cells is 0.0062 against a preregistered tolerance of 0.01. A last-iterate metric fails the same audit (the anchor-off ablation cells limit-cycle), which is why the tail average is the registered metric.

Support cliffs in the wild. The support-degeneracy theorem (T6) predicts that thin or collinear designs cannot identify (C_0, C_1, c), and the pipeline hit that cliff twice without looking for it. In E4, un-designed small-amplitude jitter leaves c unidentified, with a standard deviation $7.8\times$ the design arm's (which itself matches the Fisher prediction: 0.234 measured against 0.248). In the LQ economy, the transplanted agent's alternating probes made price and lag collinear and it failed until the probes were randomized, after which it reaches the pricing optimum (12.65 tail-averaged against 12.5).

Matched transit energy. The design's advantage is allocational, not categorical. At matched transit energy, full-support jitter identifies c inside the preregistered parity band (RMS ratio within $[0.3, 3.5]$, no cliff in either direction) and the design-off agent still converges: shape matters at small amplitude and across dispersed curvature, support and amplitude matter always. The vector agent (E10) confirms the dispersed side: shaped exploration beats isotropic at steady state by ratio 1.20 in the dispersed

Table 1: Selected preregistered verification rows. Appendix D details each experiment; the full 36-row record ships with the repository.

Claim (register)	Measured	Predicted	Status
Exchange rate, m=0.3, two amplitudes (T2)	$0.01062,\ 0.01070$	0.01056	pass
Exchange rate, m=0.6, two amplitudes (T2)	$0.00549,\ 0.00569$	0.00535	pass
Saturating environment (outside A1)	$+16\%$ over rate	out of scope	drift
Feedback floor, three moduli (T1 companion; E1)	within 0.2–6.4%	$(\sigma/\gamma)^2/(1-m^2)$	pass
Design arm $\mathrm{sd}(\hat{c})$; jitter/design ratio (T6)	0.234; $7.8\times$	Fisher 0.248; ≥ 3	pass
SafeD reaches h_{PO}; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, d=8 (T5)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Pricing domain, 4 cells (T2, exact: LQ); agent	$0.384,\ 0.192$; 12.65	$0.384,\ 0.192$; 12.5	pass
Multi-bond shaped/iso steady-state risk (T5)	1.20 (flat control 1.00)	> 1.15 (≈ 1)	pass
Within-reach min CR ratio; witness beyond (T9)	1.0005; 0.8517	≥ 1; < 1	pass

universe (threshold > 1.15), while the flat-curvature control comes out at 1.00, the null that the theory requires.

The harness binds. The suite behind these numbers comprises 38/38 derivation checks, 9/9 unit tests, and eleven measurement-forced pivots recorded in the code where they happened. Three pivots became results (the feedback excitation floor, the exact reach, and its below-reach witness), and a fourth, the false h_{PO}-anchored identity, came from the derivation suite's falsification record. The zero-failure headline is the end state of a process that failed, in writing, eleven times; a harness that never redirects the theory would be evidence of nothing.

6 Conclusion

We set out to price a decision-maker's knowledge of its own influence, and the price turned out to have a closed form. Along any stationary exploration policy, slope-estimate variance times cumulative performative regret equals $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$, a constant of the environment rather than of the policy (Theorem 1). Around this identity sit minimax floors for arbitrary adaptive policies, exact under deviation budgets and constant-factor under value budgets (Theorem 2), and the deviation floor is structure-proof within an exactly known reach: on $[h^* - 2/c, \infty)$ even full knowledge of the response family cannot beat it (hardened search bottoms out at Cramér–Rao ratio 1.0005), while a single below-reach probe of weight 3×10^{-4} attains ratio 0.8517 (Theorem 3). The design theory converts the identity into allocation: explore as $\Gamma_{\mathrm{PO}}^{-1/2}$, with isotropy overpaying by the dispersion factor F. SafeD-PerfGD assembles the pieces into an agent that corrects toward the performative optimum only when a perturbed-modulus certificate clears; it reaches h_{PO} where blind retraining stalls, and the uncertified gradient baseline pays more than $5\times$ its median regret over twelve seeds. Every closed form above was checked in a live deploy–observe–fit–correct pipeline (36 preregistered measured-vs-predicted rows, 0 failures; Section 5), transplanted to an LQ pricing economy where the identity holds exactly, and ported to the REFLEX corporate-bond calibration, matching its published run in 10/10 cells within 0.8%.

Limitations. The identity is local (A1). Wide-side probing undercuts it at finite amplitude, and the saturating environment drifts $+16\%$ above the rate, so the $o(1)$ qualifier in Theorem 1 is load-bearing, not decorative. So is the trust region: beyond the reach the floor provably fails (Theorem 3(ii)), and the agent's protection rests on staying inside $r \leq 2/c$. The value-budget floor is proved only to within a factor of 13.5 of the conjectured sharp constant $\frac{1}{2}$, which is numerically supported but open (OPEN-1). At matched transit energy, full-support jitter identifies the curvature parameter on par with the optimal design; the design advantage is small-amplitude and allocational, not an identification cliff. Finally, the real-data leg validates machinery, not effect size: only the benign-flow channel is data-identified, the toxic channel is structurally scaled, and the calibrated moduli (roughly 0.055–0.066) sit deep in the stable regime, far from where blind retraining is in danger.

Four questions follow directly. The sharp value-budget constant needs a proof, or a counterexample outside the two-point family. The below-reach violation set is uncharacterized: we exhibit one witness, but which designs below $h^* - 2/c$ break the floor, and by how much at most, is a new open problem (OPEN-5, recorded in Appendix C). The multi-bond theory couples books only through a shared budget; coupling in the response itself, where probing one bond moves flow on another, would reshape both the floor and the optimal design. And the response family here is stationary: for state-dependent or drifting responses [Brown et al., 2022, Li and Wai, 2022], the exchange rate becomes a moving target, and whether any conservation law survives averaging over the drift is unknown.

Our proofs are stated for one structural market, but the accounting is not tied to it. Any system that retrains on data its own actions produce, a recommender shaping clicks or a pricing engine shaping demand, must buy knowledge of its own influence in its own objective, at a rate the environment fixes. The certified-correction pattern (explore on a designed support, fit with an anytime confidence sequence, act only when a stability certificate clears) is a template for exploring safely inside a feedback loop. The price of self-knowledge does not have to be assumed: it can be measured, floored, and budgeted.

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A Full theorem statements

Lemma 1 (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies* $C_T = \frac{1}{2}\gamma_{\rm PO}\,\mathbb{E}\big[\sum_{t\le T} d_t^2\big] + R_T$ *with* $|R_T| \le \frac{L_3}{6}\,\mathbb{E}\sum_t |d_t|^3$*, and the realized cost's variance decomposes exactly as* $\mathrm{Var}(C_T^{\rm real}) = \delta_0^2\,\mathrm{Var}\big(\sum_t d_t\big) + \frac{\gamma_{\rm PO}^2}{4}\,\mathrm{Var}\big(\sum_t d_t^2\big)$ *with vanishing cross term.*

Theorem 4 (Information saturation; T1). *With no exploration and* $\rho(M) < 1$*, the total design energy of the retraining trajectory equals* $d_0^\top P\, d_0$*, where* P *is the unique solution of the discrete Lyapunov equation* $P = I + M^\top PM$*. Consequently the Fisher information for the response Jacobian is bounded, in every direction, uniformly in the horizon:* $\mathrm{Var}(\hat\varepsilon) \ge \sigma^2\big(d_0^\top P_u\, d_0\big)^{-1}$ *for the component along any direction* u*.*

Corollary 1 (Safety implies blindness; C1.1). *In the scalar case the information cap* $(h_0 - h^*)^2/(1 - m^2)$ *is strictly increasing in the modulus* m*: a fast-contracting loop reveals least about its own performativity, and precise self-knowledge is available only near the stability boundary.*

Corollary 2 (Trajectory design degeneracy; C1.2). *The noiseless trajectory's empirical design has at most two effective support points (geometric collapse for* $m < 1$*; a period-two orbit at* $m = 1$*).*

Remark 1 (Non-normal transient information; R1). *The energy identity* $d_0^\top P\, d_0$ *of T1 holds for non-normal* M *as well, where the energy can exceed the spectral cap* $\|d_0\|^2/(1-\rho(M)^2)$ *by up to the factor* $\kappa(V)^2$*.*

Theorem 5 (The exchange rate, pathwise; T2). *In the stationary regime, on every realization,*

$$\mathrm{Var}(\hat\varepsilon \mid \text{design}) \times \tfrac{1}{2}\gamma_{\rm PO}\sum_{t\le T} d_t^2 \;=\; \tfrac{1}{2}\,\gamma_{\rm PO}\,\sigma^2\Big(1 + \tfrac{T\,\bar d^2}{S_{xx}}\Big) \;=\; \tfrac{1}{2}\,\gamma_{\rm PO}\,\sigma^2\,\big(1 + O_P(1/T)\big),$$

independent of the exploration intensity, the horizon, and the modulus.

Proposition 1 (Concentration; P2.1). *In the stationary regime the measured product concentrates around the invariant at rate* $T^{-1/2}$*, with long-run* $\mathrm{Var}\,\big(\sum_{t\le T} d_t\big) = T\,v\,\frac{1-m}{1+m}\,(1+o(1))$*.*

Proposition 2 (Feedback bias; P2.2). *Under the feedback relaxation A3′ with gain* ϕ*, the OLS response slope carries the bias*

$$\mathbb{E}[\hat\varepsilon - \varepsilon_{\rm true}] \;=\; \frac{\phi\,\sigma^2\,(3m-1)}{(1-m^2)\,T\,v_\phi}\,(1+o(1)), \qquad v_\phi = \frac{\sigma_e^2 + \phi^2\sigma^2}{1-m^2},$$

which is $O(1/T)$ *and changes sign at* $m = 1/3$*.*

Theorem 6 (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration policies with* $\sum_{t\le T} d_t^2 \le D$ *almost surely and all estimators (biased included), the minimax risk over an interval of width* w *satisfies*

$$\inf_{\text{policy, estimator}}\;\sup_{\varepsilon}\;\mathbb{E}\big[(\hat\varepsilon - \varepsilon)^2\big] \;\ge\; \frac{\sigma^2}{D + \sigma^2(\pi/w)^2},$$

so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares.

Lemma 2 (Exploitation–information; L2). *For the symmetric two-point response family with half-width* δ *and any non-anticipating policy under the certainty-equivalent anchor,*

$$\mathbb{E}[\mathrm{gain}_T] \;\le\; \frac{g_1\,\delta^{3/2}\,T^{1/2}\,\big(\mathbb{E}S_T\big)^{3/4}}{\sigma^{1/2}}, \qquad g_1 = \beta\,|h^* - \psi|,$$

via the exact identity $\mathbb{E}\,\big|\,\mathbb{E}[s \mid \mathcal{F}_t]\,\big| = \mathrm{TV}(P_t^+, P_t^-)$ *and Pinsker's inequality* $\mathrm{TV}_t \le \delta\sqrt{\mathbb{E}S_t}/\sigma$*: exploitation is bounded by the information already purchased. At the minimax scale* $\delta = c\,\sigma/\sqrt{\mathbb{E}S_T}$ *this is* $O(\sigma\sqrt{T})$*, an* $O(T^{-1/2})$ *fraction of the* $\Theta(T)$ *exploration cost.*

Theorem 7 (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator,*

$$\sup\;\;\mathrm{Var}(\hat\varepsilon) \times C_T \;\ge\; \frac{\gamma_{\rm PO}\,\sigma^2}{27}\,\big(1 - O(T^{-1/2})\big),$$

the supremum over the two-point response family at the minimax scale and the constant arising from the Le Cam reduction optimized at $c = 2/3$*. The exploited rebate vanishes at rate* $T^{-1/2}$ *(Lemma L2). The sharp constant* $\frac{1}{2}$*, against which no simulated policy fell by more than Monte-Carlo error (`run_open1.py` §B), is conjectured and is part of OPEN-1.*

Theorem 8 (Optimal exploration shapes; T5). *Under the value budget* $\mathrm{tr}(\Gamma_{\mathrm{PO}} M) \le B$ *on the exploration second moment* M*: (a) the A-optimal design is* $M^* = \big(B/\,\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2}\big)\,\Gamma_{\mathrm{PO}}^{-1/2}$ *with value* $(\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2})^2/B$*; (b) the D-optimal design is* $M^* = (B/d)\,\Gamma_{\mathrm{PO}}^{-1}$*; (c) the c-optimal design for the correction functional* c *is* $M^* = B\,cc^\top/(c^\top\Gamma_{\mathrm{PO}}\,c)$ *(probing along the correction direction itself), with value* $c^\top\Gamma_{\mathrm{PO}}\,c/B$.

Corollary 3 (Price of isotropic jitter; C5.1). *Isotropic exploration at matched budget overpays the A-optimal design by exactly the curvature-dispersion factor* $F = d\ \mathrm{tr}(\Gamma_{\mathrm{PO}})/(\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2})^2 \ge 1$*, with equality iff all curvature directions are equal.*

Lemma 3 (Temporal shaping is irrelevant; L3). *With serially independent response noise, the information about the response parameters depends on the exploration process only through its empirical second moment; temporally correlated exploration buys nothing beyond its stationary design measure.*

Theorem 9 (Chebyshev unidentifiability; T6). *The sensitivity system of the structural family* $\tau(h) = C_0 + C_1 e^{-ch}$ *is an extended Chebyshev system; any design with fewer than three support points has singular Fisher information, and by C1.2 no retraining trajectory, at any modulus, identifies the response curvature* c.

Lemma 4 (Perturbed modulus; L4). *If the corrected update uses an estimated response with* C^1 *error* e*, the closed-loop modulus satisfies* $|\hat\rho - \rho| \le \eta\,\beta\,\big(\|e\|_\infty + |h^* - \psi|\,\|e'\|_\infty\big)$.

Remark 2 (Open-loop neutrality; R2). *History-independent exploration schedules cannot change the linearized contraction rate of the retraining loop.*

Proposition 3 (Safety under pessimism; P7.1). *Given an anytime-valid confidence sequence for the response parameters at level* $1-\delta$*, the pessimism rule (worst-case modulus over the confidence set kept below* $1 - c_{\mathrm{margin}}$*, correction frozen otherwise) keeps the realized modulus within the margin for all* t *simultaneously with probability at least* $1-\delta$.

Theorem 10 (Anchoring crossover, horizon-dependent; T7). *Under the value budget* B *and horizon* T*, the budget-and-horizon-optimal nonparametric (secant) estimator has*

$$\mathrm{MSE}_{\mathrm{np}} = \frac{\gamma_{\mathrm{PO}}\,\sigma^2}{2B} + \Big(\frac{\tau'''(h^*)}{6}\Big)^2\Big(\frac{2B}{\gamma_{\mathrm{PO}}\,T}\Big)^2,$$

so anchoring to a structural family with misspecification δ_{mis} *wins iff (to leading order)* $\delta_{\mathrm{mis}} < |\tau'''(h^*)|\,B/(3\,\gamma_{\mathrm{PO}}\,T)$*: anchoring wins at short horizons and tight budgets, the free-form route asymptotically in* T.

Theorem 11 (The ROI of self-knowledge; T8). *Exploring once to accuracy* v *and correcting forever, discounted at* ρ_{disc}*, is optimal at* $v^* = (\sigma/\kappa)\sqrt{\rho_{\mathrm{disc}}}$*, and exploring at all is worthwhile iff*

$$\rho_{\mathrm{disc}} \;<\; \rho^* \;=\; \frac{(h_{\mathrm{SP}} - h_{\mathrm{PO}})^4}{4\,\kappa^2\,\sigma^2},$$

in which the curvature γ_{PO} *cancels completely.*

Proposition 4 (Frontier consistency; P9.1). *Every deterministic design sits exactly on the exchange-rate frontier; in particular the* $O(\log T)$*-cost adaptive-design schedules of Lai–Robbins.*

Proposition 5 (Isotropy contrast; P9.2). *Naive isotropic exploration, optimal in isotropic LQR-like settings, overpays by exactly the dispersion factor* F *in the performative setting, with* $F = 1$ *iff the curvature spectrum is flat.*

Proposition 6 (Separable multi-bond curvature; P9.3). *In the separable multi-bond market,* $(\Gamma_{\mathrm{PO}})_{aa} = \gamma_a + \beta\,\varepsilon_a\,(2 + c_t\psi_a - c_t h_a)$ *exactly, with the factor-coupled case holding to first order in the off-diagonal response.*

Theorem 12 (Structure-proofness of the floor, within reach; T9). *Fix the structural family* $\tau_\theta(h) = C_0 + C_1 e^{-ch}$ *(*$C_1 > 0$*,* $c > 0$*), the anchor* $h^* > 0$*, the sensitivity* $s(h) = \partial_\theta \tau_\theta(h)$*, and the estimand gradient* $g = \partial_\theta\,\varepsilon(h^*) = -s'(h^*)$*. For a finitely supported design* μ *define the normalized Cramér–Rao product*

$$R(\mu) \;=\; \big(g^\top M(\mu)^{-1} g\big)\int (h-h^*)^2\,d\mu, \qquad M(\mu) = \int s\,s^\top d\mu,$$

so that $R(\mu) \geq 1$ states that the family-knowing Cramér–Rao product $\mathrm{Var}(\hat{\varepsilon}) \times C_T$ *is at least the exchange-rate floor* $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ *under the local-quadratic (A1) cost, for unbiased estimators and, in the local-asymptotic-minimax sense, for all regular estimators. (i)* Within reach the floor is structure-proof: *if* $\mathrm{supp}(\mu) \subset [h^* - 2/c, \infty)$*, then either* $\varepsilon(h^*)$ *is not estimable under* μ *(the bound is infinite) or* $R(\mu) \geq 1$*, strictly whenever* μ *places mass off the two equality points* $\{h^*,\ h^* - 2/c\}$*; and* $\inf_\mu R(\mu) = 1$*, approached but not attained by symmetric three-point designs collapsing at* h^**. In particular the floor binds on every trust-region design with* $r \leq 2/c$ *(A4). (ii)* The reach is exact: *the pointwise domination underlying (i) fails strictly below* $h^* - 2/c$*, and an explicit four-point design with a single below-reach probe of weight* 3×10^{-4} *attains* $R = 0.8517$ *(computer-verified at 50-digit precision; V9.2): outside the reach, knowledge of the family's form does beat the floor.*

B Proofs

Throughout, the local model of Assumptions A1–A6, per the derivation folder's notation-and-assumptions register: **A1** (local quadratic scope): Φ is C^3 and strongly concave on the operating interval, expansions taken at h^* with third-order remainder bounded by $L_3 = \sup|\Phi'''|$; **A2** (exploration class): all policies are non-anticipating, with A2-sym additionally requiring $\mathbb{E}[u_t \mid \mathcal{F}_t] = 0$ and A2-adaptive unrestricted, costs measured from the certainty-equivalent anchor; **A3** (noise exogeneity): ζ_t i.i.d., mean zero, variance σ^2, independent of $\mathcal{F}_t$; **A4** (trust region): $|d_t| \leq r$ pathwise; **A5** (stationary scale): $S_T = \Theta(T)$; **A6** (stable base loop): $m < 1$, equivalently $\rho(M) < 1$. The local dynamics are $d_{t+1} = -m\, d_t + u_t$ with $d_t = h_t - h^*$ (h^* the certainty-equivalent operating point, equal to h_{SP} for the blind loop), observations $\tau_t = \tau(h^*) - \varepsilon\, d_t + \zeta_t$, and Φ satisfying $\Phi'(h^*) = \delta_0 \neq 0$ and $-\Phi'' = \gamma_{\mathrm{PO}}$ at the operating point.

A. Cost equivalence and its fluctuations

Proof of Lemma L1 (cost equivalence). Taylor's theorem with Lagrange remainder gives, for each t,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0\, d_t + \tfrac{1}{2}\gamma_{\mathrm{PO}}\, d_t^2 + r_t, \qquad |r_t| \leq \tfrac{L_3}{6}\, |d_t|^3,$$

with $L_3 = \sup|\Phi'''|$ on the operating interval. Summing over $t \leq T$ and taking expectations, it remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = -m\, d_t + \mathbb{E}[u_t \mid \mathcal{F}_t] = -m\, d_t$, so $\mathbb{E}[d_{t+1}] = -m\,\mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t\,\mathbb{E}[d_0] = 0$ when the loop starts at the operating point or in the stationary regime. The quadratic term contributes $\frac{1}{2}\gamma_{\mathrm{PO}}\,\mathbb{E}[\sum_t d_t^2]$, which is the claim.

For the variance decomposition, write the realized cost (net of the cubic remainder) as $C = -\delta_0 X + \frac{1}{2}\gamma_{\mathrm{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\mathrm{Var}(C) = \delta_0^2\,\mathrm{Var}(X) + \frac{\gamma_{\mathrm{PO}}^2}{4}\,\mathrm{Var}(Y) - \delta_0\gamma_{\mathrm{PO}}\,\mathrm{Cov}(X, Y)$. Under Gaussian innovations the vector $(d_1, \ldots, d_T)$ is jointly Gaussian with mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\mathrm{Cov}(X, Y) = \sum_{t,s}\mathbb{E}[d_t\, d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3, including the necessity check that an asymmetric rule breaks the expectation identity by exactly $-\delta_0\,\mathbb{E}[\sum_t d_t]$. $\square$

B. Saturation

Proof of Theorem T1 (information saturation). Let $\rho(M) < 1$ and define $P = \sum_{t \geq 0}(M^\top)^t M^t$. The series converges absolutely: by Gelfand's formula $\|M^t\| \leq C\,\rho_0^t$ for any $\rho(M) < \rho_0 < 1$ and some C, so the terms are dominated by $C^2\rho_0^{2t}$. Termwise, $M^\top P M = \sum_{t \geq 0}(M^\top)^{t+1}M^{t+1} = P - I$, so P solves the discrete Lyapunov equation, and if Q is any other solution then $P - Q = M^\top(P - Q)M = (M^\top)^k(P - Q)M^k \to 0$, giving uniqueness. The energy identity is immediate: $\sum_{t \geq 0}\|M^t d_0\|^2 = d_0^\top\big(\sum_t (M^\top)^t M^t\big)d_0 = d_0^\top P\, d_0$. The directional version replaces I by $uu^\top$: $P_u = \sum_t (M^\top)^t uu^\top M^t$ solves $P_u = uu^\top + M^\top P_u M$ and $\sum_t (u^\top M^t d_0)^2 = d_0^\top P_u\, d_0$.

For the information claim: under A3 the log-likelihood of the observations along the noiseless trajectory has Fisher information for the response component along u equal to $\sigma^{-2}\sum_t (u^\top d_t)^2 = \sigma^{-2}\, d_0^\top P_u d_0$, which is finite and independent of the horizon; the Cramér–Rao bound gives the stated variance floor for unbiased estimators (the biased/minimax version is Theorem T3's van Trees argument). Verified: V1.1, V1.2, V1.4. $\square$

Proof of Corollary C1.1 (safety implies blindness). In the scalar case $P = \sum_t m^{2t} = (1-m^2)^{-1}$, so the cap $(h_0 - h^*)^2/(1-m^2)$ has derivative in m equal to $2m\,(h_0-h^*)^2/(1-m^2)^2 > 0$ for $m \in (0,1)$: strictly increasing. $\square$

Proof of Corollary C1.2 (design degeneracy). For $m < 1$, $d_t = (-m)^t d_0$, so for every $\epsilon > 0$ all but $\lceil \log(|d_0|/\epsilon)/\log(1/m) \rceil$ of the points lie in $[-\epsilon, \epsilon]$: the empirical design measure converges weakly to δ_0. At $m = 1$, $d_t = (-1)^t d_0$ and the design measure is $\frac{1}{2}(\delta_{+d_0} + \delta_{-d_0})$ exactly. In both cases the limiting design has at most two support points; combined with Theorem T6, the per-observation Fisher matrix of the three-parameter structural family is asymptotically singular along the trajectory. $\square$

Proof of Remark R1 (non-normal transient information). The identity $E(d_0) = d_0^\top P\, d_0$ of T1 holds for any stable M, normal or not. If $M = V \Lambda V^{-1}$ is diagonalizable, $\|M^t\| \le \kappa(V)\,\rho(M)^t$ gives $\|P\| \le \kappa(V)^2/(1-\rho(M)^2)$. The bound is attained in order by the Jordan-type family $M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix}$ with $d_0 = e_2$: $M^t e_2 = (t\,a\,m^{t-1},\ m^t)^\top$, so

$$E(e_2) = a^2 \sum_{t \ge 1} t^2 m^{2t-2} + \sum_{t \ge 0} m^{2t} = a^2\, \frac{1+m^2}{(1-m^2)^3} + \frac{1}{1-m^2},$$

using $\sum_{t \ge 1} t^2 x^{t-1} = (1+x)/(1-x)^3$ at $x = m^2$: the energy grows as a^2, without bound, at fixed spectral radius. Verified: V1.3 (the exact value matches the Lyapunov solve; 1268.2 vs the normal cap 2.78 at $m = 0.8$, $a = 6$). $\square$

C. The exchange rate

Proof of Theorem T2 (pathwise exchange rate). Under A3, conditionally on the design $\{d_t\}$ the observations are independent with means linear in ε and variance σ^2, so the least-squares slope has conditional variance $\mathrm{Var}(\hat\varepsilon \mid \text{design}) = \sigma^2/S_{xx}$ (Gauss–Markov). Multiplying by the quadratic cost $\frac{1}{2}\gamma_{\mathrm{PO}} \sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T\bar d^2$ gives the displayed identity exactly, on every realization. In the stationary regime $\bar d = O_P(T^{-1/2})$ by Proposition P2.1's long-run variance, while $S_{xx}/T \to v$ a.s., so $T\bar d^2/S_{xx} = O_P(1/T)$. Verified: V2.1 (machine precision). $\square$

Proof of Proposition P2.1 (concentration). For the stationary AR(1) with lag-one coefficient $\rho = -m$, $\mathrm{Cov}(d_t, d_s) = v\,(-m)^{|t-s|}$, so

$$\mathrm{Var}\Big(\sum_{t \le T} d_t\Big) = \sum_{t,s} v(-m)^{|t-s|} = Tv\,\frac{1+\rho}{1-\rho}\,(1+o(1)) = Tv\,\frac{1-m}{1+m}\,(1+o(1)),$$

the negative autocorrelation of the cobweb shrinking the fluctuation below the i.i.d. level. The realized-cost product then fluctuates around the constant with standard deviation $O(T^{-1/2})$ by Lemma L1's variance decomposition (both $\mathrm{Var}(\sum d)$ and $\mathrm{Var}(\sum d^2)$ are $O(T)$ while the squared mean of the cost is $O(T^2)$). Verified: V2.2 (variance ratio 4.80 for a $4\times$ horizon), V2.3. $\square$

Proof of Proposition P2.2 (feedback bias). Under A3$'$, $d_{t+1} = -m\,d_t + \sigma_e \xi_t + \phi\,\zeta_t$, so d_t is a linear function of past innovations with $\mathbb{E}[d_s \zeta_t] = \phi\,\sigma^2\,(-m)^{s-t-1}$ for $s > t$ and 0 otherwise. Write the slope error as N/S with $N = \sum_t (d_t - \bar d)\,\zeta_t$ and $S = S_{xx}$, and expand $\mathbb{E}[N/S] = \mathbb{E}[N]/\mathbb{E}[S] - \mathrm{Cov}(N,S)/\mathbb{E}[S]^2 + O(T^{-2})$ (valid for stationary Gaussian data; the neglected term involves $\mathbb{E}[N]\,\mathrm{Var}(S)/\mathbb{E}[S]^3 = O(T^{-2})$).

Centering term. $\mathbb{E}[d_t \zeta_t] = 0$, so $\mathbb{E}[N] = -\frac{1}{T} \sum_{s>t} \mathbb{E}[d_s \zeta_t] = -\frac{1}{T} \sum_{s>t} \phi\sigma^2(-m)^{s-t-1} = -\phi\sigma^2/(1+m) + O(1/T)$.

Covariance term. For jointly Gaussian zero-mean (X, Y, Z), $\mathbb{E}[XY^2Z] = 2\,\mathbb{E}[XY]\mathbb{E}[YZ] + \mathbb{E}[XZ]\mathbb{E}[Y^2]$ (Isserlis). Applying this to $\mathrm{Cov}(N, S) = \sum_{t,s}\big(\mathbb{E}[\tilde d_t \tilde d_s^2 \zeta_t] - \mathbb{E}[\tilde d_t \zeta_t]\mathbb{E}[\tilde d_s^2]\big)$ with $\tilde d = d - \bar d$ leaves only the paired term $2\sum_{t,s}\mathbb{E}[\tilde d_t \tilde d_s]\,\mathbb{E}[\tilde d_s \zeta_t]$; substituting the stationary covariances,

$$\mathrm{Cov}(N, S) = 2\,T\,v_\phi\,\phi\,\sigma^2 \sum_{k\geq 1}(-m)^k(-m)^{k-1} + O(1) = -\frac{2\,T\,v_\phi\,\phi\,\sigma^2\,m}{1-m^2} + O(1).$$

Combining with $\mathbb{E}[S] = Tv_\phi\,(1 + o(1))$:

$$\mathbb{E}[\hat\varepsilon - \varepsilon_{\mathrm{true}}] = \frac{\phi\sigma^2}{Tv_\phi}\Big[-\frac{1}{1+m} + \frac{2m}{1-m^2}\Big] + O(T^{-2}) = \frac{\phi\sigma^2\,(3m-1)}{(1-m^2)\,T\,v_\phi} + O(T^{-2}).$$

The bracket changes sign at $m = 1/3$. (The first draft kept only the centering term and was falsified by the verification suite; the record is in VERIFICATION.md.) Verified: V2.4, V2.5 (both signs, straddling $m = 1/3$). □

D. Minimax lower bounds

Proof of Theorem T3 (deviation-budget minimax). Let π be a prior density on an interval of width w with finite information $I(\pi) = \int(\pi')^2/\pi$; the cosine-squared density attains the minimum $I(\pi) = (\pi/w)^2$ (the latter π is the constant). For any non-anticipating policy the joint density of the data factorizes as $\prod_t p(u_t \mid \mathcal{F}_t)\,p(\tau_t \mid \mathcal{F}_t, \varepsilon)$, where the policy factors do not depend on ε and $\tau_t \mid \mathcal{F}_t \sim \mathcal{N}(\tau(h^*) - \varepsilon\,d_t, \sigma^2)$ with d_t being $\mathcal{F}_t$-measurable. Hence the Fisher information of the whole experiment is

$$I_T(\varepsilon) = \mathbb{E}_\varepsilon\Big[\sum_t \big(\partial_\varepsilon \log p(\tau_t \mid \mathcal{F}_t, \varepsilon)\big)^2\Big] = \mathbb{E}_\varepsilon\Big[\sum_t d_t^2\Big]/\sigma^2 \;\leq\; D/\sigma^2$$

under the pathwise budget: adaptivity places information but cannot manufacture it. The van Trees inequality (Gill–Levit) then bounds the Bayes risk of *any* estimator, biased or not: $\mathbb{E}_\pi\mathbb{E}[(\hat\varepsilon - \varepsilon)^2] \geq \big(\mathbb{E}_\pi[I_T] + I(\pi)\big)^{-1} \geq \big(D/\sigma^2 + (\pi/w)^2\big)^{-1}$, and the supremum over the interval dominates the prior average. Equality (up to the prior term) is attained by symmetric two-point probing at the budget with least squares, by Theorem T2. Verified: V3.1, V3.2 (equality within Monte-Carlo error). □

Proof of Lemma L2 (exploitation–information). Prior $s = \pm 1$ symmetric; environments $P^\pm$ differing only in the observation means by $2\delta\,d_t$ at step t. Three ingredients.

(i) *Posterior imbalance equals total variation in expectation.* Under the symmetric mixture $\bar P = \frac{1}{2}(P^+ + P^-)$, with data x,

$$\mathbb{E}_{\bar P}\big|\mathbb{P}(s{=}+ \mid x) - \mathbb{P}(s{=}- \mid x)\big| = \int \tfrac{1}{2}(dP^+ + dP^-)\,\frac{|dP^+ - dP^-|}{dP^+ + dP^-} = \mathrm{TV}(P_t^+, P_t^-).$$

(ii) *Pinsker with the chain rule for adaptive designs.* $\mathrm{KL}(P_t^+ \,\|\, P_t^-) = \mathbb{E}^+\big[\sum_{i\leq t}(2\delta\,d_i)^2/(2\sigma^2)\big] = 2\delta^2\,\mathbb{E}^+[S_t]/\sigma^2$ (the policy factors cancel inside the log-ratio), so $\mathrm{TV}_t \leq \delta\,\sqrt{\mathbb{E}[S_t]}/\sigma$.

(iii) *The gain bound.* Since d_t is $\mathcal{F}_t$-measurable, $\mathbb{E}[s\,d_t] = \mathbb{E}[d_t\,\mathbb{E}[s \mid \mathcal{F}_t]]$; two applications of Cauchy–Schwarz and $|\mathbb{E}[s \mid \mathcal{F}_t]| \leq 1$ give

$$\mathbb{E}[\mathrm{gain}_T] = g_1\delta\sum_t \mathbb{E}[s\,d_t] \leq g_1\delta\sum_t \big(\mathbb{E}d_t^2\big)^{1/2}\,\mathrm{TV}_t^{1/2} \leq g_1\delta\,\big(\mathbb{E}S_T\big)^{1/2}\,\big(T\,\mathrm{TV}_T\big)^{1/2},$$

and inserting (ii): $\mathbb{E}[\mathrm{gain}_T] \leq g_1\,\delta^{3/2}\,T^{1/2}\,(\mathbb{E}S_T)^{3/4}/\sigma^{1/2}$.

Remark on constants. The derivation document's heuristic chain (treating the imbalance as bounded by TV_t pathwise) yields $g_1\delta^2 S_T\sqrt{T}/\sigma$; the rigorous chain above changes the exponent bookkeeping but not the conclusion of Theorem T4: at the minimax scale $\delta = c\,\sigma/\sqrt{\mathbb{E}S_T}$ both give $\mathbb{E}[\mathrm{gain}_T] = O(\sigma\sqrt{T}\,)$, hence an exploited fraction $O(T^{-1/2})$ of the $\Theta(T)$ cost under A5. Verified: V3.3 (the imbalance–TV inequality pathwise in expectation), V3.4 (the $T^{-1/2}$ decay at the minimax scale). □

Proof of Theorem T4 (value-budget minimax; constant-factor version). Fix a policy and estimator; let $\bar{S} = \mathbb{E}[S_T] = \Theta(T)$ under A5. Choose the two-point prior at separation $\delta = c\,\sigma/\sqrt{\bar{S}}$ with $c \in (0,1)$ to be optimized. By the testing reduction (Le Cam), any estimator's worst-case risk over the pair obeys $\sup_\pm \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\delta^2}{2}(1 - \mathrm{TV}_T) \geq \frac{\delta^2}{2}(1 - c)$ using Lemma L2(ii). By Lemma L1 with the certainty-equivalent anchor and Lemma L2's gain bound, the value cost obeys $C_T \geq \frac{1}{2}\gamma_{\mathrm{PO}}\,\bar{S}\,(1 - O(T^{-1/2}))$. Multiplying,

$$\sup\ \mathrm{Var}(\hat{\varepsilon}) \times C_T \ \geq\ \frac{c^2\sigma^2}{2\bar{S}}(1-c)\cdot\frac{\gamma_{\mathrm{PO}}\bar{S}}{2}\,\big(1-O(T^{-1/2})\big)\ =\ \frac{\gamma_{\mathrm{PO}}\sigma^2}{4}\,c^2(1-c)\,\big(1-O(T^{-1/2})\big),$$

maximized at $c = 2/3$: the bound $\gamma_{\mathrm{PO}}\sigma^2/27 \cdot (1 - O(T^{-1/2}))$.

Status. This proves the value-budget product is bounded below by an explicit constant multiple of the exchange rate, uniformly over adaptive policies, with the exploited rebate vanishing at rate $T^{-1/2}$. The sharp constant $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$, numerically supported (`run_open1.py` §B: three adaptive amplitude schedules, ratios 0.995 to 1.018), requires the van Trees route with cost control in place of the two-point reduction and is part of OPEN-1. □

E. Design geometry

Proof of Theorem T5 (optimal exploration shapes). All three problems have convex (resp. concave) objectives over the PSD cone with one linear constraint; KKT stationarity plus convexity/concavity certifies global optimality.

(a) *A-optimality.* $\nabla_M \operatorname{tr}(M^{-1}) = -M^{-2}$, so stationarity gives $M^{-2} = \lambda\,\Gamma_{\mathrm{PO}}$, i.e. $M = \lambda^{-1/2}\Gamma_{\mathrm{PO}}^{-1/2}$; the budget $\operatorname{tr}(\Gamma_{\mathrm{PO}}M) = \lambda^{-1/2}\operatorname{tr}(\Gamma_{\mathrm{PO}}^{1/2}) = B$ pins $\lambda^{-1/2} = B/\operatorname{tr}(\Gamma_{\mathrm{PO}}^{1/2})$ and the value $\operatorname{tr}(M^{-1}) = (\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2/B$. (Diagonal-case cross-check by Cauchy–Schwarz: $(\sum_i 1/m_i)(\sum_i g_i m_i) \geq (\sum_i \sqrt{g_i})^2$ with equality at $m_i \propto g_i^{-1/2}$.)

(b) *D-optimality.* $\nabla_M \log\det M = M^{-1} = \lambda\,\Gamma_{\mathrm{PO}}$ gives $M = (B/d)\,\Gamma_{\mathrm{PO}}^{-1}$ after normalizing by the budget.

(c) *c-optimality.* Substitute $N = \Gamma_{\mathrm{PO}}^{1/2} M\,\Gamma_{\mathrm{PO}}^{1/2}$ ($\operatorname{tr} N = \operatorname{tr}(\Gamma_{\mathrm{PO}}M) \leq B$) and $w = \Gamma_{\mathrm{PO}}^{1/2}c$, so the objective is $w^\top N^{-1} w$. By Cauchy–Schwarz, $(w^\top w)^2 = (w^\top N^{-1/2}\,N^{1/2}w)^2 \leq (w^\top N^{-1}w)(w^\top N w)$ and $w^\top N w \leq \|w\|^2 \operatorname{tr} N \leq \|w\|^2 B$, hence $w^\top N^{-1} w \geq \|w\|^2/B = c^\top \Gamma_{\mathrm{PO}}\,c/B$, attained at $N^* = B\,ww^\top/\|w\|^2$. Back-substituting with $\Gamma_{\mathrm{PO}}^{-1/2}w = c$ gives $M^* = B\,cc^\top/(c^\top\Gamma_{\mathrm{PO}}\,c)$: the optimal probe direction is c itself, the curvature pricing only its scale. Verified: V4.1–V4.3 (formulas achieved; never beaten by 4000 random feasible designs each). □

Proof of Corollary C5.1 (price of isotropic jitter). Isotropic $M = (B/\operatorname{tr}\Gamma_{\mathrm{PO}})\,I$ has $\operatorname{tr}(M^{-1}) = d\ \operatorname{tr}(\Gamma_{\mathrm{PO}})/B$; dividing by the optimum $(\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2/B$ gives $F = d\ \operatorname{tr}(\Gamma_{\mathrm{PO}})/(\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2$, and $F \geq 1$ with equality iff all eigenvalues are equal, by Cauchy–Schwarz applied to $(\sum_i \sqrt{g_i}\cdot 1)^2 \leq d\sum_i g_i$. Verified: V4.4. □

Proof of Lemma L3 (temporal shaping). Under A3 the conditional log-likelihood of the responses given the deployment path is $-\sum_t(\tau_t - a - \theta^\top d_t)^2/(2\sigma^2)$ up to constants; its Hessian in θ is $-\sum_t d_t d_t^\top/\sigma^2$, a function of the empirical second moment only, whatever the temporal dependence of $\{d_t\}$. Verified: V4.5. □

Proof of Theorem T6 (Chebyshev unidentifiability). The sensitivity of the family $\tau(h) = C_0 + C_1 e^{-ch}$ is $s(h) = (1,\ e^{-ch},\ -C_1 h e^{-ch})^\top$. The functions $\{1, e^{-ch}, he^{-ch}\}$ form an extended Chebyshev system on any interval (their Wronskian is $c^3 e^{-2ch} \neq 0$ after row reduction; equivalently, a nontrivial linear combination $a + (b + \gamma h)e^{-ch}$ has at most two zeros counting multiplicity). For any design ξ with k support points, the Fisher matrix $M(\xi) = \sum_{i \leq k} w_i\, s(h_i)s(h_i)^\top$ has rank at most k; with $k \leq 2 < 3$ it is singular, and that null space always loads on the c-sensitivity: writing n for the null vector of the two-point design matrix, $n_3 = 0$ would force $n_1 + n_2 e^{-ch_i} = 0$ at both support points, hence $n = 0$ for $h_1 \neq h_2$, a contradiction. By Corollary C1.2 the trajectory design has at most two effective support points, which proves the claim; the existence of exactly-three-point D-optimal designs on a trust-region interval is classical for Chebyshev systems (Karlin–Studden). Verified: V5.1 (two-point determinants at 10^{-17}; three-point at 6×10^{-3}). □

F. Safety under certainty equivalence

Proof of Lemma L4 (perturbed modulus). The corrected map is $\Psi(h) = h + \eta\,[\,G(h) - \beta\,(h - \psi)\,\hat{\varepsilon}(h)\,]$ and the ideal map replaces $\hat{\varepsilon}$ by ε. With $e = \hat{\varepsilon} - \varepsilon$, $\Psi'(h) - \Psi'_{\mathrm{ideal}}(h) = -\eta\,\beta\,\frac{d}{dh}\big[(h - \psi)\,e(h)\big] = -\eta\,\beta\,\big[e(h) + (h-\psi)e'(h)\big]$, and taking suprema over the operating interval gives the bound with $|h-\psi|_{\max}$; reading the linearized modulus at the operating point h^* (where the certificate is evaluated) localizes the constant to the registered $|h^* - \psi|$. Verified: V6.1 (nine-cell error grid; the bound holds with maximal excess 2.4×10^{-11}). □

Proof of Remark R2 (open-loop neutrality). For $d_{t+1} = -m\,d_t + u_t$ with a history-independent schedule $\{u_t\}$, two solutions with different initial conditions differ by the homogeneous solution $(-m)^t(d_0 - d_0')$, which contracts at rate m regardless of the schedule (superposition). Verified: V6.2 (difference exactly 0 after 200 steps to machine precision). □

Proof of Proposition P7.1 (safety under pessimism). Let E be the event that the confidence sequence contains the true response parameters simultaneously for all t; by anytime-validity $\mathbb{P}(E) \geq 1 - \delta$. On E, the true parameters lie in Θ_t at every step, so the realized modulus is dominated by the rule's worst case: $|\rho_t| \leq \sup_{\theta\in\Theta_t}|\hat{\rho}(\theta)| \leq 1 - c_{\mathrm{margin}}$ whenever correction is active, and when the rule freezes, the blind loop's modulus $m < 1 - c_{\mathrm{margin}}$ (choose the margin below $1 - m$) applies. No union bound over time is needed: the inclusion is pathwise on E. *(Conditional result: the hypothesis is the confidence sequence's validity; constructing it from the D2 information accounting is routine, and the necessity of the hypothesis is demonstrated numerically: an invalid, too-narrow sequence produces violations.)* □

G. The anchoring crossover

Proof of Theorem T7 (horizon-dependent crossover). Bias. For the symmetric secant with half-width w, fifth-order Taylor expansion of $\tau(h^* \pm w)$ gives

$$\frac{\tau(h^*+w) - \tau(h^*-w)}{2w} = \tau'(h^*) + \frac{\tau'''(h^*)}{6}\,w^2 + O(w^4),$$

all even-order terms cancelling (symmetric probing is bias-optimal at its order; the second derivative never enters). Verified: V7.1.

Variance. With n probe pairs ($T = 2n$ observations) the slope estimator $(\bar{y}_+ - \bar{y}_-)/(2w)$ has variance $\dfrac{2\sigma^2/n}{4w^2} = \dfrac{\sigma^2}{2nw^2} = \dfrac{\sigma^2}{S}$ with $S = \sum_t d_t^2 = 2nw^2 = Tw^2$: variance depends on the design only through its energy.

Constrained optimum. The value budget fixes $S = 2B/\gamma_{\mathrm{PO}}$; the horizon caps the number of observations, $Tw^2 \geq S$, i.e. $w^2 \geq 2B/(\gamma_{\mathrm{PO}}T)$; bias is increasing in w, so the constrained optimum sits at $w^2 = 2B/(\gamma_{\mathrm{PO}}T)$, giving

$$\mathrm{MSE}_{\mathrm{np}} = \frac{\gamma_{\mathrm{PO}}\,\sigma^2}{2B} + \Big(\frac{\tau'''(h^*)}{6}\Big)^2\Big(\frac{2B}{\gamma_{\mathrm{PO}}T}\Big)^2.$$

Verified: V7.2–V7.4 (three (B,T) cells, within 2%).

Crossover. The anchored estimator has $\mathrm{MSE} = \kappa_p\,\gamma_{\mathrm{PO}}\sigma^2/(2B) + \delta_{\mathrm{mis}}^2$; comparing and keeping the dominant term gives the rule $\delta_{\mathrm{mis}} < |\tau'''(h^*)|\,B/(3\gamma_{\mathrm{PO}}T)$ to leading order, with the exact two-term form in the display above. The direction of the comparison on both sides of the threshold is verified: V7.5, V7.6. □

H. ROI and instantiation

Proof of Theorem T8 (ROI of self-knowledge). With $A = \frac{1}{2}\gamma_{\mathrm{PO}}(h_{\mathrm{SP}} - h_{\mathrm{PO}})^2$, $a = \frac{1}{2}\gamma_{\mathrm{PO}}\kappa^2$, $b = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$, the net present value of exploring to accuracy v is $\mathrm{NPV}(v) = (A - av)/\rho_{\mathrm{disc}} - b/v$, strictly concave on $(0,\infty)$ with $\mathrm{NPV}'(v) = -a/\rho_{\mathrm{disc}} + b/v^2$, giving $v^* = \sqrt{b\,\rho_{\mathrm{disc}}/a} = (\sigma/\kappa)\sqrt{\rho_{\mathrm{disc}}}$ (the curvature cancels between a and b). Substituting, $\mathrm{NPV}(v^*) = A/\rho_{\mathrm{disc}} - 2\sqrt{ab/\rho_{\mathrm{disc}}}$, which is positive iff $A > 2\sqrt{ab\,\rho_{\mathrm{disc}}}$, i.e. iff $\rho_{\mathrm{disc}} < A^2/(4ab) = (h_{\mathrm{SP}} - h_{\mathrm{PO}})^4/(4\kappa^2\sigma^2)$: γ_{PO} cancels completely. Verified: V8.1, V8.2 (the break-even root to 10^{-3} relative). $\square$

Proof of Proposition P9.1 (frontier consistency). For any deterministic design, $\mathrm{Var}(\hat{\varepsilon} \mid \mathrm{design}) \cdot \frac{1}{2}\gamma_{\mathrm{PO}}S = (\sigma^2/S)\cdot\frac{1}{2}\gamma_{\mathrm{PO}}S = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$: the design energy cancels identically, whatever the schedule, including the $d_t \propto t^{-1/2}$ schedules with $S_T = \Theta(\log T)$ of the Lai–Robbins adaptive-design line. Verified: V8.3 (exact). $\square$

Proof of Proposition P9.2 (isotropy contrast). Immediate from Corollary C5.1: $F = 1$ iff the curvature spectrum is flat, the isotropic (LQR-like) case in which naive exploration is optimal; any curvature dispersion makes $F > 1$ by the strict case of Cauchy–Schwarz. Cross-verified with V4.4. $\square$

Proof of Theorem T9 (structure-proofness within reach). Write $x = h - h^*$, $d(h) = x$, and $V = \mathrm{span}\{1,\, e^{-ch},\, he^{-ch}\}$, the span of the components of s (a bijective correspondence $u \leftrightarrow \phi_u = u^\top s$ between $\mathbb{R}^3$ and V, since $C_1 \neq 0$). Differentiating s gives $s'(h) = (0,\, -ce^{-ch},\, C_1 e^{-ch}(ch-1))^\top$, so $g = -s'(h^*)$ componentwise, which is the stated estimand gradient.

Step 1: one-dimensional reduction. For $M(\mu) \succ 0$ the generalized Rayleigh identity gives

$$g^\top M(\mu)^{-1} g = \sup_{u\neq 0} \frac{(u^\top g)^2}{u^\top M(\mu)\, u} = \sup_{\phi\in V} \frac{\phi'(h^*)^2}{\int \phi^2\, d\mu} = \Big(\inf\big\{\textstyle\int \phi^2\, d\mu : \phi \in V,\ \phi'(h^*) = 1\big\}\Big)^{-1},$$

using $u^\top M u = \int \phi_u^2\, d\mu$ and $u^\top g = -\phi_u'(h^*)$ (the sign is lost in the square). Hence

$$R(\mu) \;=\; \frac{\int d^2\, d\mu}{\inf\{\int \phi^2 d\mu : \phi \in V,\ \phi'(h^*) = 1\}}.$$

Step 2: the candidate and its pointwise domination. Let

$$\phi_0(h) \;=\; \frac{2}{c} - e^{-cx}\Big(\frac{2}{c} + x\Big) \;=\; \frac{2}{c}\cdot 1 \;-\; e^{ch^*}\Big(\frac{2}{c} - h^*\Big)e^{-ch} \;-\; e^{ch^*}\left(he^{-ch}\right) \;\in\; V,$$

the unique element of V with 2-jet $(0,1,0)$ at h^* (the 2-jet map is invertible on V: in the basis $\{1, e^{-ch}, he^{-ch}\}$ its determinant is $c^2 e^{-2ch^*} \neq 0$); the expansion is $\phi_0(x) = x - (c^2/6)\,x^3 + O(x^4)$. Writing $f(x) = \frac{2}{c} - e^{-cx}(\frac{2}{c} + x)$ we claim

$$|f(x)| \le |x| \quad \text{for all } x \ge -2/c, \qquad \text{with equality iff } x \in \{0, -2/c\},$$

and $|f(x)| > |x|$ strictly for every $x < -2/c$. Indeed: (a) $v(x) = f(x) - x$ has $v(0) = 0$ and $v'(x) = e^{-cx}(1+cx) - 1 < 0$ for $x \neq 0$ (since $1 + t < e^t$ for $t = cx \neq 0$); $v' \le 0$ with equality only at the isolated point 0 makes v strictly decreasing on $\mathbb{R}$, so $f > x$ on $x < 0$ and $f < x$ on $x > 0$. (b) $f(0) = 0$ and $f'(x) = e^{-cx}(1 + cx) > 0$ for $x \ge 0$, so $f \ge 0$ on $[0,\infty)$ and, with (a), $0 \le f \le x$ there. (c) $q(x) = f(x) + x$ has $q(-2/c) = 0 = q(0)$ and $q'(x) = e^{-cx}(1+cx) + 1$, where $r(x) = e^{-cx}(1+cx)$ is strictly increasing on $x < 0$ ($r'(x) = -c^2 x\, e^{-cx} > 0$) from $r(-2/c) = -e^2 < -1$ to $r(0) = 1$; so q' changes sign exactly once on $(-2/c, 0)$, from negative to positive, and $q \le 0$ there with equality only at the endpoints; with (a) ($f \ge x$, equality only at 0) this gives $x \le f \le -x$ on $[-2/c, 0]$. (d) For $x < -2/c$, $r(x) < -e^2$ gives $q' < 0$, so moving left from $-2/c$ the function q increases above 0: $f(x) > -x = |x| > 0$, and in fact $f(x) = 2/c + e^{c|x|}(|x| - 2/c)$ grows exponentially.

Step 3: the bound. If $\mathrm{supp}(\mu) \subset [h^* - 2/c, \infty)$, Step 2 gives $\int \phi_0^2\, d\mu \le \int d^2\, d\mu$, so by Step 1, $R(\mu) \ge \int d^2 d\mu / \int \phi_0^2 d\mu \ge 1$. Strictness: equality in Step 2 holds only at $x \in \{0, -2/c\}$, so any μ placing positive mass off $\{h^*, h^* - 2/c\}$ (in particular any with three distinct support points) has $\int \phi_0^2 d\mu < \int d^2 d\mu$ and $R(\mu) > 1$. If $M(\mu)$ is singular, two cases: $g \notin \mathrm{range}\, M(\mu)$ makes $\varepsilon(h^*)$ non-estimable (infinite bound); $g \in \mathrm{range}\, M(\mu)$, which does occur for special two-point designs (a one-parameter family of in-reach pairs; their finite pseudo-inverse products were checked to obey the bound, e.g. $R = 1.063$ at the pair $\{1.627, 2.127\}$), keeps the sup-representation valid over $\{u : u^\top M u > 0\}$, and $\int \phi_0^2 d\mu = 0$ would put $u_0 \in \operatorname{null} M$ with $u_0^\top g = -1 \neq 0$, contradicting $g \in \operatorname{range} M$; so $\int \phi_0^2\, d\mu > 0$ and the bound goes through verbatim. Throughout, the Cramér–Rao functional bounds unbiased estimation, and all regular estimators in the local-asymptotic-minimax sense.

Step 4: the infimum is 1. The Cramér–Rao functional is invariant under C^1 reparametrization, and the 2-jet map $\theta \mapsto (\tau_\theta(h^*), \varepsilon(h^*), \tau''_\theta(h^*))$ is a local diffeomorphism: its Jacobian $[\, s(h^*)\ \ {-s'(h^*)}\ \ s''(h^*)\,]$ has determinant $C_1 c^2 e^{-2ch^*} \neq 0$ (direct computation; V9.4). In jet coordinates η the model reads $\tau = \eta_1 - \eta_2 x + \frac{1}{2}\eta_3 x^2 + O(x^3)$ with sensitivity $\tilde{s}(x) = (1, -x, x^2/2)^\top + O(x^3)$, and the estimand is η_2. For the symmetric three-point design μ_δ uniform on $\{h^* - \delta, h^*, h^* + \delta\}$, scale by $D = \operatorname{diag}(1, \delta, \delta^2)$: $M_\eta(\mu_\delta) = D\, \hat{H}_\delta\, D$ with $\hat{H}_\delta \to \hat{H}_0$ invertible (the exact quadratic-model moment matrix), so $(M_\eta^{-1})_{22} = \delta^{-2}(\hat{H}_\delta^{-1})_{22}$ and $R(\mu_\delta) \to R_{\mathrm{quad}}$. In the exact quadratic model, for *any* symmetric design (odd moments $m_1 = m_3 = 0$), the $(2,2)$ cofactor gives $(M_\eta^{-1})_{22} = 1/m_2$ and $R_{\mathrm{quad}} = m_2 \cdot (1/m_2) = 1$. Hence $R(\mu_\delta) \to 1$ (measured rate $R(\mu_\delta) - 1 = O(\delta^2)$; V9.2), the infimum is 1, and by Step 3 it is not attained.

Step 5: the reach is exact. Below $h^* - 2/c$ the domination of Step 2 fails strictly, and the failure is realized by a design, not only by the candidate: the frozen four-point witness (support $\{1.8726, 1.8665, 1.3073, 0.0500\}$, weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$ after normalization, in the register market $c = 1.5$, frozen anchor $h^* = 1.8461$) has $R = 0.851650\ldots$, verified at 50-digit precision (mpmath) and reproduced in float64 by `run_open1.py` (V9.2). All its support except the probe at 0.05 lies within reach: one vanishing-weight below-reach probe breaks the floor. Symmetric pair-plus-far-probe families do *not* break it (their ratio approaches 1 from above as the probe weight vanishes); the violation requires the asymmetric cluster, which is why coarse searches missed it. Verified: V9.1–V9.4. □

Proof of Proposition P9.3 (separable multi-bond curvature). In the separable case the objective is a sum of per-bond objectives $\Phi_a(h_a)$, each of the scalar form analyzed in the single-book theory of Nagarajan and Ashok [2026], so the Hessian is diagonal and the scalar computation applies per bond: differentiating $\Phi_a(h) = P[\, hA_a e^{-k_a h}\rho + h\tau_a(h) - \psi_a \tau_a(h) - w_a(h - h_{\mathrm{ref}})^2\,]$ twice and collecting terms with $\tau'_a = -\varepsilon_a$, $\tau''_a = c_t\, \varepsilon_a$ yields $-\Phi''_a = \gamma_a + \beta\, \varepsilon_a\, (2 + c_t \psi_a - c_t h_a)$, which is the claimed diagonal entry. In the factor-coupled case the off-diagonal response contributes additional terms of first order in $\|E_{\mathrm{offdiag}}\|$, stated as a construction with its error term (not claimed exact). Verified: V8.4 (finite-difference Hessians of three random bonds, relative error below 10^{-4}). □

C Complete derivations

This appendix reproduces the full derivation record behind every register result: the model's closed forms, both budgets, and every intermediate computation, including the measurement-forced corrections recorded where they happened. Sections carry the derivation-record identifiers (D0–D9) that the register and Appendices A–B cite.

Notation, assumptions, and the two anchoring conventions

This subsection is the canonical register for the derivation record D0 through D9, with D3b the adaptive companion of D3: D0 is the model and cost lemma, D1 information saturation, D2 the exchange rate, D3 the minimax lower bounds, D4 and D5 the design geometry, D6 safe certainty equivalence, D7 the anchoring crossover, and D8 and D9 the ROI analysis and the instantiation. Every symbol is defined once, here; every assumption is numbered here and is cited by number in the derivation sections and in the register statements (T1, L2, . . .). Where a later section restates a condition, this register is authoritative. The formal proofs of the register statements are collected in the proofs appendix.

Notation. The symbol table is reproduced in three blocks: the retraining loop and its objective, the stochastic and informational quantities, and the symbols specific to the design and economic analyses. The third column records where each symbol is introduced: a derivation section (D0 through D9), the base structural market theory (cited as REFLEX [Nagarajan and Ashok, 2026] with its section numbers, a third-party source), or an entry of the assumption register below (the trust-region radius

r is fixed by A4). Two register conventions carry over unchanged: all logarithms are natural, and a prime denotes the derivative of a scalar function (in the ASCII sources it also denotes the matrix transpose; in this appendix transposes are written $(\cdot)^\top$, so the prime is unambiguous).

Symbol	Meaning	Defined in / source
h, h_t	deployed decision (half-spread); its value at step t	D0
h^*	performatively stable point (RRM fixed point)	REFLEX [Nagarajan and Ashok, 2026]
h_{SP}, h_{PO}	stable point ($h_{\mathrm{SP}} = h^*$); performative optimum	REFLEX [Nagarajan and Ashok, 2026]
d_t	deviation $d_t = h_t - h^*$	D0
m	retraining modulus $m = \varepsilon\beta/\gamma \in [0, 1)$	REFLEX [Nagarajan and Ashok, 2026]
γ, β, ε, ψ	curvature, smoothness, response slope, adverse severity	REFLEX [Nagarajan and Ashok, 2026] (closed forms)
Φ	true performative objective $\Phi(h) = J(h; \tau(h))$	REFLEX [Nagarajan and Ashok, 2026]
δ_0	$\Phi'(h^*) = -\beta(h^* - \psi)\,\varepsilon(h^*)$, nonzero	D0 Sec. 1
γ_{PO}	$-\Phi''$ at the operating point	REFLEX [Nagarajan and Ashok, 2026]
M	d-dimensional loop Jacobian, $\rho(M) < 1$	D1 Sec. 2
P, P_u	discrete-Lyapunov energy matrices	D1 Sec. 2
Γ_{PO}	d-dimensional objective curvature matrix	D9

Symbol	Meaning	Defined in / source
σ	response-noise standard deviation (σ_τ in REFLEX)	D0
σ_e, u_t	exploration intensity; exploration input	D0
ξ_t, ζ_t	deployment and response noise (iid, mean zero)	D0
ϕ	noise-feedback gain (observed noise flowing into the next deployment)	D2 Sec. 4
v	stationary deviation variance $\sigma_e^2/(1 - m^2)$	D2
v_ϕ	the same with feedback: $(\sigma_e^2 + \phi^2\sigma^2)/(1 - m^2)$	D2 Sec. 4
S_{xx}	centered design energy $\sum_t (d_t - \bar{d})^2$	D2
D_T	deviation budget $\sum_t d_t^2$	D0 Sec. 2
C_T	value budget: expected incremental cost (L1)	D0 Secs. 2–3
I_T	Fisher information for ε	D1/D3
$\mathcal{F}_t$	history sigma-field $\sigma(d_0, u_{<t}, \tau_{<t})$	D0
δ, s	two-point prior half-width; its sign (minimax)	D3
g_1	$\beta\,\lvert h^* - \psi\rvert$, coefficient of the unknown linear part	D3
TV, KL	total variation; Kullback–Leibler divergence	D3
r	trust-region radius on $\lvert d_t\rvert$	A4

Symbol	Meaning	Defined in / source
M (design)	exploration second moment $\mathbb{E}[d\,d^\top]$; appears only in D4	D4
B	value budget in the design problems	D4
F	curvature dispersion $d\ \mathrm{tr}(\Gamma_{\mathrm{PO}})/(\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2})^2$	D4 (C5.1)
c	correction-direction vector (c-optimality)	D4 Sec. 3
$s(h)$	structural-family sensitivity vector, $p = 3$	D5
w	probe half-width (crossover analysis)	D7
δ_{mis}	C^1 misspecification of the structural family	D6/D7
κ	correction sensitivity $\mid dh_{\mathrm{PO}}/d\varepsilon \mid$	D8
ρ_{disc}	discount rate (ROI analysis)	D8

Three deliberate symbol reuses deserve a caveat, recorded here so that no derivation needs to repeat it. First, M names both the loop Jacobian and the exploration second moment; the design M occurs only in D4/D5, and context disambiguates the two. Second, κ is the correction sensitivity $\mid dh_{\mathrm{PO}}/d\varepsilon \mid$ of D8/D9, not a multi-dealer spillover parameter: no such symbol occurs in this paper. Third, ϕ is the noise-feedback gain of D2, while the T9 material in D3 separately writes ϕ_u and ϕ_0 for elements of the sensitivity span, a self-contained reuse local to that subsection.

Assumption register. Assumptions are cited by number (A1 through A6) throughout the record.

- **A1 (local quadratic scope).** Φ is C^3 and strongly concave on the operating interval; all expansions are taken at h^*, with the third-order remainder controlled through $L_3 = \sup|\Phi'''|$. The identities of the record are exact for the quadratic model and first-order accurate otherwise. As a matter of method, the nonlinear deviation is measured (the drift study), never assumed away.
- **A2 (exploration classes).** All policies are non-anticipating: u_t is $\mathcal{F}_t$-measurable. Two subclasses are used. *A2-sym*, conditionally symmetric exploration, $\mathbb{E}[u_t \mid \mathcal{F}_t] = 0$, under which the linear term of L1 vanishes in expectation. *A2-adaptive*, unrestricted non-anticipating exploration, under which costs are measured from the certainty-equivalent anchor (D0 Sec. 4; convention 2 below) and the exploitation lemma L2 of D3b applies.
- **A3 (noise exogeneity, with its named relaxation).** The response noise ζ_t is iid, mean zero, with variance σ^2, independent of $\mathcal{F}_t$. Relaxation A3$'$: the feedback channel $\phi\,\zeta_t$ enters d_{t+1}. Under A3$'$ the exact $O(1/T)$ bias of D2 Sec. 4 applies, with its $(3m-1)$ sign structure (P2.2), and "exploration-dominant design" means $\sigma_e^2 \gg \phi^2\sigma^2$.
- **A4 (trust region).** $|d_t| \le r$ pathwise. The radius r enters the constants of D3b (L2, T4), the feasible set of the design problems (T5), and the basin argument of D6.
- **A5 (stationary-scale exploration; D3b only).** $S_T = \Theta(T)$, with $S_T = \sum_{t\le T} d_t^2$ the cumulative design energy. The transient-dominated regime $S_T = o(T)$ is governed by D1 (T1) instead.
- **A6 (stable base loop).** $m < 1$ in the scalar case and $\rho(M) < 1$ in the d-dimensional case, for all stationary-regime statements. Boundary behaviour ($m = 1$) appears only in the degeneracy statement C1.2.

The two anchoring conventions (audit-critical). C_T is an incremental cost, and its value depends on the baseline against which the increment is measured. The record fixes one anchor per exploration class; these are the two audit-critical definitions.

Convention 1 (incremental anchor; all A2-sym results). C_T is measured relative to the jitter-free loop at h^*, that is, the same retraining loop run without exploration. Anchoring at h_{PO} instead adds the sunk echo-chamber term and provably destroys the invariance of D2 (T2): with anchor gap $g = h^* - h_{\mathrm{PO}}$, the anchored exchange-rate product is inflated by exactly the factor

$$1 + \frac{g^2}{v},$$

a computation verified in D0. The formal proof is recorded in the proofs appendix.

Convention 2 (certainty-equivalent anchor; all A2-adaptive results). C_T is measured relative to the deployment that is optimal under current posterior knowledge. Exploitation of learned structure is thereby re-anchored away, and D3b (L2, T4) prices only the unlearned residual.

The two anchors coincide under A2-sym with an uninformative start, so the conventions agree on the class where both are defined.

D0: The model, the two budgets, and the cost-equivalence lemma (L1)

This subsection records the local model, the two exploration budgets, and the computation behind the cost-equivalence lemma L1. It is derived in full and verified numerically (`verify_all.py`, section V0). Everything downstream (D1 through D9) consumes the definitions and the lemma established here. Notation follows the REFLEX market model [Nagarajan and Ashok, 2026] (γ, β, ε, ψ, h_{SP}, h_{PO}).

The local model. Work in a neighbourhood of the performatively stable point h^*, the RRM fixed point of that market [Nagarajan and Ashok, 2026], and write deviations $d_t := h_t - h^*$. The linearized closed loop is

$$\begin{aligned} d_{t+1} &= -m\, d_t + u_t && \text{(retraining cobweb plus exploration)},\\ \tau_t &= \tau(h^*) - \varepsilon\, d_t + \zeta_t && \text{(observed flow)}. \end{aligned}$$

Here $m = \varepsilon\beta/\gamma \in [0, 1)$ is the retraining modulus, whose closed form comes with the model [Nagarajan and Ashok, 2026]. The exploration input u_t is chosen by the operator: it may be iid noise ($u_t = \sigma_e\, \xi_t$), a deterministic schedule, or adaptive, that is $\mathcal{F}_t$-measurable with $\mathcal{F}_t = \sigma(d_0, u_0, \ldots, u_{t-1}, \tau_0, \ldots, \tau_{t-1})$. The response noise ζ_t is iid, mean zero, with variance σ^2, independent of $\mathcal{F}_t$; the feedback violation of this independence assumption is quantified in the exchange-rate derivation D2 (the feedback-bias result P2.2). The estimand is the response slope ε, scalar here; the d-dimensional version (D4, behind T5) and the structural-family version (D5, behind T6) are treated in their own derivation subsections.

The true performative objective $\Phi(h) = J(h; \tau(h))$ (REFLEX theory 1.2) is smooth and strongly concave on the operating interval, with

$$\Phi'(h^*) = \Delta(h^*) = -\beta\,(h^* - \psi)\,\varepsilon(h^*) =: \delta_0, \qquad \Phi''(h^*) = -\gamma_{\mathrm{PO}}$$

(a direct computation in the market model of Nagarajan and Ashok [2026]). The slope δ_0 is nonzero precisely because $h^* \neq h_{\mathrm{PO}}$.

The two budgets. Two notions of exploration budget are used throughout:

$$\begin{aligned} D_T &:= \sum_{t \le T} d_t^2 && \text{(deviation budget: realized design energy)},\\ C_T &:= \mathbb{E}\Big[\sum_{t \le T} \big(\Phi(h^*) - \Phi(h_t)\big)\Big] && \text{(value budget: expected incremental cost)}. \end{aligned}$$

D_T is the tracking-error form, a desk's deviation limit: it is observable, its meaning is policy-independent, and it is the correct constraint for the exact minimax theorem T3 (derivation D3). C_T is the realized-P&L form. Lemma L1 relates the two.

Lemma L1 (cost equivalence). For any exploration rule that is non-anticipating and conditionally symmetric, meaning $\mathbb{E}[u_t \mid \mathcal{F}_t] = 0$ (for example iid exploration, or a scheduled sign-symmetric jitter), in the regime where third-order terms are negligible,

$$C_T = \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\mathbb{E}[D_T] + R_T, \qquad |R_T| \le \frac{L_3}{6}\,\mathbb{E}\Big[\sum_{t \le T} |d_t|^3\Big],$$

with $L_3 = \sup|\Phi'''|$ on the operating interval. The linear term of the expansion contributes exactly zero to the expectation, and contributes a fluctuation of standard deviation $|\delta_0|\sqrt{\mathrm{Var}\,\big(\sum_{t \le T} d_t\big)}$ to the realized cost.

Proof. Taylor-expand Φ at h^*:

$$\Phi(h^*) - \Phi(h_t) = -\delta_0\, d_t + \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\, d_t^2 + r_t, \qquad |r_t| \le \tfrac{L_3}{6}\, |d_t|^3,$$

then sum over $t \le T$ and take expectations. For the linear term: d_t is built from $\{u_s\}_{s<t}$ through the linear recursion, so the mean obeys $\mathbb{E}[d_{t+1}] = -m\,\mathbb{E}[d_t] + \mathbb{E}[u_t]$; conditional symmetry gives $\mathbb{E}[u_t] = \mathbb{E}\big[\mathbb{E}[u_t \mid \mathcal{F}_t]\big] = 0$, hence $\mathbb{E}[d_t] = (-m)^t\,\mathbb{E}[d_0]$. Starting at the operating point ($d_0 = 0$), or in the stationary regime, $\mathbb{E}[d_t] = 0$ for all t, so $\mathbb{E}\big[\sum_{t\le T} \delta_0\, d_t\big] = 0$. The quadratic term yields $\frac{1}{2}\gamma_{\mathrm{PO}}\,\mathbb{E}[D_T]$, and the remainder is bounded by the stated third-order term. $\square$

Fluctuation decomposition (exact, and a corrected earlier claim). The realized cost's variance splits into two terms with zero cross term: the cross-moment sums built from $\mathbb{E}[d_t\, d_s^2]$ vanish by the sign symmetry of the stationary law. Exactly,

$$\mathrm{Var}\left(C_T^{\mathrm{real}}\right) = \delta_0^2\ \mathrm{Var}\Big(\sum_{t\le T} d_t\Big) + \frac{\gamma_{\mathrm{PO}}^2}{4}\ \mathrm{Var}\Big(\sum_{t\le T} d_t^2\Big),$$

with the long-run scales, v denoting the stationary variance of d_t,

$$\mathrm{Var}\Big(\sum_{t\le T} d_t\Big) \sim T\, v\, \frac{1-m}{1+m} \qquad \text{(long-run variance: negative autocorrelation helps)},$$

$$\mathrm{Var}\Big(\sum_{t\le T} d_t^2\Big) \sim T\, \frac{2\,v^2\,(1+m^2)}{1-m^2} \qquad \text{(Gaussian stationary AR(1))}.$$

A caveat, recorded per the falsification convention: an earlier draft asserted that the linear term always dominates. The first verification run falsified that at moderate δ_0 (measured sd ratio 1.62, exactly reproduced by the two-term formula above), and the claim is now stated as the full decomposition. Which term dominates is decided by comparing $\delta_0^2\,(1-m)/(1+m)$ against $(\gamma_{\mathrm{PO}}^2\, v/2)\,(1+m^2)/(1-m^2)$, both computable. V0 now verifies the decomposition itself.

Why the anchoring is load-bearing (recorded from the audit). Anchoring the cost at the performative optimum h_{PO} instead of h^* adds terms of the type $T\,(h^* - h_{\mathrm{PO}})^2$ that are paid whether or not the operator explores: the sunk echo-chamber cost of blindness (REFLEX theory 1.2, 6b). Numerically (`verify_m2_identity.py`, Claim 3): writing $g = h^* - h_{\mathrm{PO}}$ for the gap, the mis-anchored product equals

$$\tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\left(1 + \frac{g^2}{v}\right),$$

which destroys the exchange-rate invariance of D2 (theorem T2). The incremental anchor at h^* is the definition under which every later theorem is true.

The certainty-equivalent anchor (needed for the adaptive minimax bound T4, derivation D3). For adaptive policies the symmetric-exploration condition can be violated deliberately: since $\delta_0 \ne 0$, drifting toward h_{PO} earns first-order value. Two observations discipline this. First, decompose $\Phi'(h^*)$ as a function of the estimand:

$$\varphi(\varepsilon) := \Phi'(h^*; \varepsilon) = -\beta\,(h^* - \psi)\,\varepsilon,$$

which is linear in ε. Under a prior centred at $\bar\varepsilon$, split $\varphi(\varepsilon) = \varphi(\bar\varepsilon) + \varphi'(\bar\varepsilon)\,(\varepsilon - \bar\varepsilon)$. Second, the first (known) part is exploitable without any learning: it is a pure consequence of already-known information, that is, of choosing the anchor wrongly. Define therefore the certainty-equivalent anchor h^*_{CE}: the deployment that is optimal under current knowledge (the posterior mean), with exploration cost measured relative to the trajectory that sits at h^*_{CE}. Relative to that anchor the residual linear coefficient is

$$\varphi'(\bar\varepsilon)\,(\varepsilon - \bar\varepsilon) = -\beta\,(h^* - \psi)\,(\varepsilon - \bar\varepsilon),$$

which has zero prior mean: only genuinely unknown structure can be exploited, and the exploitation–information lemma L2 (in D3, feeding T4) bounds that exploitation by the information already purchased. This definition also matches practice: a desk's exploration cost is measured against its own current best policy, not against an oracle's.

Numerical verification (V0). Three checks close the record for this subsection.

- $\mathbb{E}[C_T] = \frac{1}{2}\,\gamma_{\mathrm{PO}}\,\mathbb{E}[D_T]$ holds within Monte Carlo error, for a quadratic-plus-linear objective with $\delta_0 \neq 0$, under iid exploration.
- The realized-cost fluctuation scale matches $|\delta_0|\,\mathrm{sd}\big(\sum_{t\le T} d_t\big)$: the linear term is a variance contribution, not a bias.
- A deliberately asymmetric (drifting) exploration rule breaks the lemma in the predicted direction and by the predicted first-order amount $-\delta_0\,\mathbb{E}\big[\sum_{t\le T} d_t\big]$: the conditional-symmetry condition of L1 is necessary, not decorative.

D1: Information saturation by the Lyapunov identity (T1, C1.1, C1.2, R1)

This subsection records the derivation behind T1 and its corollaries C1.1 and C1.2, together with the non-normal refinement R1. Status of the record: the scalar closed form is verified to 10^{-16}, and the d-dimensional identity, including the non-normal amplification, is verified numerically (V1); the proofs are complete.

Scalar warm-up (the verified base case). With no exploration ($u_t = 0$) and modulus $m < 1$, the deviation obeys $d_t = (-m)^t d_0$, so the design energy accumulated by horizon T is

$$D_T \;=\; \sum_{t=0}^{T} d_t^2 \;=\; d_0^2\,\frac{1-m^{2(T+1)}}{1-m^2} \;\longrightarrow\; \frac{d_0^2}{1-m^2} \qquad (T\to\infty).$$

The Fisher information for the response slope ε from OLS on the trajectory is $S_{xx}/\sigma^2 \le D_\infty/\sigma^2$: bounded uniformly in the horizon. Hence, by Cramér–Rao, for any unbiased estimator and at every horizon,

$$\mathrm{Var}(\hat\varepsilon) \;\ge\; \frac{\sigma^2\,(1-m^2)}{d_0^2}.$$

This is the scalar case of T1; with $d_0 = h_0 - h^*$ the cap $d_0^2/(1-m^2)$ is the quantity in the register statement of C1.1.

C1.1 (safety implies blindness). The cap $d_0^2/(1-m^2)$ is strictly increasing in m: the fast-contracting (safe) loop reveals least about its own performativity, and identifiability is a near-instability phenomenon. This is immediate from the closed form above; the economic content is stated in the paper body.

C1.2 (design degeneracy). At $m = 1$ the noiseless iterates form the period-two orbit $\{+d_0, -d_0\}$; for $m < 1$ the support collapses geometrically onto $\{0\}$. In either case the empirical design measure has at most two effective support points, which by the Chebyshev counting argument of D5 (theorem T6) cannot identify a three-parameter response family. Curvature is invisible to retraining trajectories at every m.

The d-dimensional identity via the discrete Lyapunov equation. Let the linearized joint loop be $d_{t+1} = M d_t$ with $M \in \mathbb{R}^{d\times d}$ and $\rho(M) < 1$; in the multi-bond and multi-dealer instantiations M comes from the REFLEX market model [Nagarajan and Ashok, 2026] and is generally not symmetric.

Theorem (exact energy; T1). The total design energy is the quadratic form

$$E(d_0) \;:=\; \sum_{t\ge 0} \big\|M^t d_0\big\|^2 \;=\; d_0^\top P\, d_0,$$

where P is the unique positive semidefinite solution of the discrete Lyapunov equation

$$P \;=\; I + M^\top P M.$$

Proof. Set $P := \sum_{t\ge 0} (M^\top)^t M^t$; the series converges absolutely if and only if $\rho(M) < 1$ (Gelfand's spectral radius formula). Then

$$M^\top P M \;=\; \sum_{t\ge 0} (M^\top)^{t+1} M^{t+1} \;=\; P - I,$$

so P solves the equation. Uniqueness: the difference Q of two solutions satisfies $Q = M^\top Q M = (M^\top)^t Q\, M^t \to 0$. Finally, $d_0^\top P\, d_0 = \sum_{t\ge 0} \|M^t d_0\|^2$ by expanding the sum term by term. □

Scalar check: $P = 1/(1-m^2)$, recovering the warm-up above.

Directional information. For the response slope in a direction u (the observation $\tau_t = \tau^* - \varepsilon\, u^\top d_t + \zeta_t$ probes the component along u), the relevant energy is

$$\sum_{t\ge 0} \big(u^\top M^t d_0\big)^2 \;=\; d_0^\top P_u\, d_0, \qquad P_u \;=\; uu^\top + M^\top P_u M,$$

the same Lyapunov structure with a rank-one right-hand side. By Cramér–Rao this yields $\operatorname{Var}(\hat\varepsilon) \ge \sigma^2\big(d_0^\top P_u\, d_0\big)^{-1}$ for the component along any direction u, which is the bound stated in T1. Hence the Fisher information is bounded in every direction, and the information operator satisfies $\mathcal{I} \preceq P_{\text{full}}/\sigma^2$, where P_{full} solves the matrix-valued Lyapunov equation. Saturation is not a scalar accident.

Non-normality: transiently amplified information (R1). If M is normal, P has eigenvalues $1/(1-\lambda_i^2)$ and the energy is governed by the spectrum alone. If M is non-normal (heterogeneous multi-dealer Jacobians are), transient growth inflates the energy beyond the spectral prediction:

$$E(d_0) \;\le\; \kappa(V)^2\,\frac{\|d_0\|^2}{1-\rho(M)^2},$$

where V is the eigenbasis of M and $\kappa(V)$ its condition number, and the inequality is achieved in order: for the Jordan-type family

$$M \;=\; \begin{pmatrix} m & a \\ 0 & m \end{pmatrix},$$

the energy along the first coordinate grows like $a^2/(1-m^2)^3$ for large a, verified numerically in V1 against the exact Lyapunov solve. Badly-conditioned ecosystems thus leak extra information about themselves during transients, the free-information remark of the paper, now with an exact constant (P itself). The paper states the exact P-form and keeps the pseudospectral refinement for the journal version.

Scope: what T1 does not say. Two caveats are part of the record and are stated in the paper.

- With exploration on, saturation is replaced by the exchange rate of D2 (T2): the transient information $d_0^\top P\, d_0/\sigma^2$ is the free part; everything after is bought.
- The bound is for the linearized loop; the simulator's nonlinear regimes are handled empirically (the drift study, under the D2/V2 protocol).

Numerical verification (V1). Four checks anchor the derivation. (i) The scalar closed form against the direct sum: exact to floating point. (ii) Random stable M of sizes 2×2 and 5×5: the direct energy sum against the Lyapunov solve (a `scipy`-free fixed-point iteration), agreement to 10^{-12}. (iii) Non-normal amplification: the energy for the family $M = \left(\begin{smallmatrix} m & a \\ 0 & m\end{smallmatrix}\right)$ grows as a^2, far exceeding the normal-matrix bound $1/(1-m^2)$, and matches $d_0^\top P\, d_0$ exactly. (iv) The directional version: the rank-one Lyapunov solution P_u against the direct directional sum.

D2: The exchange-rate identity and its concentration (T2, P2.1, P2.2)

This subsection records the derivations behind T2 (the exchange-rate identity in pathwise form), P2.1 (its concentration), and P2.2 (the feedback-bias constant). Status of record: the identity was verified to zero relative error by a pre-existing check; the pathwise refinement, the concentration rate, and the feedback-bias constant are derived here and verified in V2. The formal statements live in the register of theorem statements, and the proofs in the proofs appendix.

Setup. The stationary jittered loop is

$$d_{t+1} \;=\; -m\, d_t + \sigma_e\, \xi_t, \qquad v \;=\; \operatorname{Var}(d_t) \;=\; \frac{\sigma_e^2}{1-m^2}.$$

OLS of the response τ_t on the quote h_t has conditional-on-design variance $\operatorname{Var}(\hat\varepsilon \mid \text{design}) = \sigma^2/S_{xx}$ with $S_{xx} = \sum_{t\le T}(d_t - \bar d)^2$, and the cost of Lemma L1 is $\frac{1}{2}\gamma_{\text{PO}} \sum_{t\le T} d_t^2$.

The identity, upgraded from expectation to pathwise (T2). **Proposition (pathwise identity).** On every realization,

$$\operatorname{Var}(\hat{\varepsilon} \mid \text{design}) \times \tfrac{1}{2}\gamma_{\mathrm{PO}} \sum_{t\le T} d_t^2 \;=\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\,\frac{\sum_{t\le T} d_t^2}{S_{xx}}$$
$$=\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\Big(1+\frac{T\,\bar{d}^2}{S_{xx}}\Big).$$

Since $\bar{d} \to 0$ almost surely at rate $T^{-1/2}$ in the stationary regime, the product equals $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2\,(1 + O_P(1/T))$ pathwise: a strictly stronger statement than the in-expectation identity, and the form the experiments actually measure. The correction term $T\bar{d}^2/S_{xx}$ is nonnegative, so the realized product can only sit above the constant, by a vanishing amount.

Invariance. Neither σ_e, nor m, nor T appears: more jitter buys information and burns value at the same rate; a larger modulus amplifies the jitter into more excitation and more cost at the same rate. The constant $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ is the exchange rate between self-knowledge and value.

Concentration: the identity is observable (P2.1). The realized value cost (Lemma L1) carries the linear-term fluctuation $-\delta_0\sum_{t\le T} d_t$. In the stationary regime $\sum_t d_t$ is a mean-zero weakly dependent sum, so

$$\frac{\operatorname{sd}(\text{realized } C_T)}{\mathbb{E}[C_T]} \;=\; O\big(T^{-1/2}\big),$$

with the explicit leading constant given by the AR(1) long-run variance $v\,(1+\rho)/(1-\rho)$ at lag-one autocorrelation $\rho = -m$, that is $v\,(1-m)/(1+m)$:

$$\operatorname{Var}\Big(\sum_{t\le T} d_t\Big) \;=\; T\,v\,\frac{1-m}{1+m}\,(1+o(1)).$$

The negative autocorrelation of the cobweb shrinks the fluctuation below the iid level $T\,v$: a small bonus worth one sentence in the paper. Hence the measured product concentrates around the invariant at rate $T^{-1/2}$. Verified in V2: the seed-variance of the measured product scales as $1/T$ (it halves when T doubles, a factor-4 drop when T quadruples, within Monte-Carlo tolerance).

Detail: S_{xx} versus $\sum_t d_t^2$. The estimator uses the centered S_{xx}; the cost uses the raw $\sum_t d_t^2$. Their ratio is

$$\frac{\sum_{t\le T} d_t^2}{S_{xx}} \;=\; 1+\frac{T\,\bar{d}^2}{S_{xx}} \;=\; 1+O_P(1/T),$$

as above. Statements in the paper use whichever form is natural and record this $O(1/T)$ equivalence once, in the present form.

The feedback (predetermined-regressor) bias, with its constant (P2.2). In the real loop the observed response feeds the next deployment. Model this as

$$d_{t+1} \;=\; -m\,d_t + \sigma_e\,\xi_t + \phi\,\zeta_t,$$

with ϕ the noise-feedback gain: retraining on the realized flow moves the next quote by $\phi\,\zeta_t$. Now d_t is correlated with past ζ, so the regressor is predetermined but not strictly exogenous.

Proposition (Stambaugh-type bias, full form, corrected by verification). The OLS slope satisfies

$$\mathbb{E}\big[\hat{\varepsilon} - (-\varepsilon)\big] \;=\; \frac{\phi\,\sigma^2\,(3m-1)}{(1-m^2)\,T\,v_\phi}\,(1+o(1)), \qquad v_\phi \;=\; \frac{\sigma_e^2+\phi^2\sigma^2}{1-m^2},$$

where $-\varepsilon$ is the true slope in this sign convention (written $\varepsilon_{\mathrm{true}}$ in P2.2) and v_ϕ is the stationary variance of the feedback loop. The bias is $O(1/T)$, proportional to the feedback gain, and changes sign at $m = 1/3$: slow-contracting loops bias the slope estimate upward, fast-contracting ones downward, and at $m = 1/3$ exactly the feedback bias vanishes at first order.

Derivation, both terms. (Honesty record, kept per the falsification convention: the first draft retained only the centering term and was falsified by V2; the covariance term below is the correction.) Write $\hat{b} - b = N/S$ with $N = \sum_{t \le T}(d_t - \bar{d})\,\zeta_t$ and $S = S_{xx}$, and expand

$$\mathbb{E}\Big[\frac{N}{S}\Big] \;=\; \frac{\mathbb{E}[N]}{\mathbb{E}[S]} \;-\; \frac{\mathrm{Cov}(N,S)}{\mathbb{E}[S]^2} \;+\; O(T^{-2}).$$

Centering term. Here $\mathbb{E}[d_t\zeta_t] \;=\; 0$, since ζ_t is independent of the past-built d_t, so $\mathbb{E}[N] \;=\; -\mathbb{E}\big[\bar{d}\sum_t \zeta_t\big]$. With $\mathbb{E}[d_s\zeta_t] = \phi\sigma^2(-m)^{s-t-1}$ for $s > t$,

$$\mathbb{E}[N] \;=\; -\frac{1}{T}\sum_{s>t} \phi\,\sigma^2\,(-m)^{s-t-1} \;=\; -\frac{\phi\,\sigma^2}{1+m} \;+\; O(1/T).$$

Covariance term (the one the first draft missed). By the Gaussian fourth-moment (Wick) expansion,

$$\mathrm{Cov}(N,S) \;=\; 2\sum_{t,s}\mathbb{E}[\tilde{d}_t\tilde{d}_s]\,\mathbb{E}[\tilde{d}_s\zeta_t], \qquad \tilde{d} = d - \bar{d};$$

substituting $\mathbb{E}[\tilde{d}_t\tilde{d}_s] \simeq v_\phi\,(-m)^{|s-t|}$ and the cross-covariance above,

$$\mathrm{Cov}(N,S) \;=\; 2\,v_\phi\,\phi\,\sigma^2\,T\sum_{k\ge 1}(-m)^k(-m)^{k-1} \;=\; -\frac{2\,v_\phi\,\phi\,\sigma^2\,T\,m}{1-m^2}.$$

Combine. With $\mathbb{E}[S] \simeq T\,v_\phi$,

$$\text{bias} \;=\; \Big[-\frac{1}{1+m} + \frac{2m}{1-m^2}\Big]\frac{\phi\,\sigma^2}{T\,v_\phi} \;=\; \frac{3m-1}{1-m^2}\cdot\frac{\phi\,\sigma^2}{T\,v_\phi},$$

which is P2.2. Verified in V2 at two moduli straddling the sign change: $m = 0.5$ gives a positive bias and $m = 0.2$ a negative one, each within Monte-Carlo tolerance of the constant. The sign change is the sharpest possible falsification test of the two-term structure, and it passes.

Paper usage. State the identity (T2) under exploration-dominant design; report the bias formula (P2.2) as the known, computable deviation, a checkable condition (ϕ is estimable from the loop's own update rule), not an assumption.

Verified numerically (V2). The V2 checks for this subsection: (1) the pathwise product equals $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2\,(1 + T\bar{d}^2/S_{xx})$ exactly; (2) product concentration: the seed-variance of the measured product scales as $1/T$, a factor-4 drop when T quadruples, within Monte-Carlo tolerance; (3) the long-run variance of $\sum_t d_t$ matches $T\,v\,(1-m)/(1+m)$; (4) the feedback bias: sign, magnitude, and $1/T$ scaling of $\phi\,\sigma^2\,(3m-1)/\big((1-m^2)\,T\,v_\phi\big)$.

D3: Minimax lower bounds and the exact reach (T3, L2, T4, T9)

This record derives the two lower bounds that turn the exchange-rate identity of D2 (register T2) from a property of one scheme into a property of the problem, and then the exact reach within which that floor is structure-proof against a policy that knows the response family (T9). T2 shows that a specific scheme, stationary symmetric jitter with ordinary least squares, achieves $\mathrm{Var}(\hat{\varepsilon}) \times \text{cost} = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$. The claim that makes this an exchange rate is that no adaptive exploration policy and no estimator does better. There are two versions of the bound, one per budget type: an exact deviation-budget bound (T3, derived in full and verified numerically in V3a) and a value-budget bound (T4), whose engine is the exploitation–information lemma (L2), proved below with its $O(T^{-1/2})$ correction established and verified in V3b. Throughout, the formal proofs are recorded in the proofs appendix; this subsection records the derivations and the caveats attached to them.

Deviation budget: van Trees over adaptive designs (T3). Setting: the exploration policy is arbitrary and adaptive (each u_t any measurable function of the history), subject to the pathwise budget $\sum_{t\le T} d_t^2 \le D$ almost surely; the estimator $\hat{\varepsilon}$ of ε is arbitrary, bias allowed. The claim (T3) is that the minimax risk over an interval of ε values of width w satisfies

$$\inf_{\text{policy, estimator}}\;\sup_{\varepsilon}\;\mathbb{E}\big[(\hat{\varepsilon}-\varepsilon)^2\big] \;\ge\; \frac{\sigma^2}{D + \sigma^2(\pi/w)^2} \;=\; \frac{\sigma^2}{D}\,\big(1 - o(1)\big) \qquad \text{as } D\,(w/\sigma)^2 \to \infty.$$

Multiplying by the budget cost $\frac{1}{2}\gamma_{\mathrm{PO}} D$ (cost equivalence, L1) gives

$$\sup_{\varepsilon} \ \mathrm{Var}(\hat{\varepsilon}) \times \tfrac{1}{2}\gamma_{\mathrm{PO}} D \ \geq \ \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\,\big(1 - o(1)\big):$$

the exchange rate is a floor.

The derivation is the van Trees inequality (Bayesian Cramér–Rao, in the form of Gill and Levit, 1995) with a prior density π_0 supported on the width-w interval:

$$\mathbb{E}_{\pi_0}\,\mathbb{E}\big[(\hat{\varepsilon}-\varepsilon)^2\big] \ \geq \ \frac{1}{\mathbb{E}_{\pi_0}\big[I_T(\varepsilon)\big] + I(\pi_0)}, \qquad I(\pi_0) = \int \frac{(\pi_0')^2}{\pi_0},$$

where $I(\pi_0)$ is the prior information, equal to $(\pi/w)^2$ for the cosine-squared prior (the minimizer of prior information on an interval of width w), and $I_T(\varepsilon)$ is the Fisher information of the adaptively designed experiment. The one nonstandard step is that I_T for an adaptive design is still bounded by the design energy: by the chain rule for Fisher information over the filtration,

$$I_T(\varepsilon) \ = \ \sum_{t\leq T} \mathbb{E}\big[I(\tau_t \mid \mathcal{F}_t; \varepsilon)\big] \ = \ \sum_{t\leq T} \frac{\mathbb{E}[d_t^2]}{\sigma^2},$$

because conditionally on $\mathcal{F}_t$ the observation τ_t is Gaussian with mean linear in ε (slope $-d_t$, with d_t $\mathcal{F}_t$-measurable): the policy can place information but cannot manufacture it. The pathwise budget then gives $\mathbb{E}[I_T] \leq D/\sigma^2$; substituting, and taking the supremum over the interval ($\sup \geq \mathbb{E}_{\pi_0}$), yields the bound.

Three remarks recorded with the derivation. (i) Biased estimators are covered: van Trees needs no unbiasedness. (ii) The bound is achieved, up to the $o(1)$, by two-point symmetric probing at the budget with least squares, tying back to the D2 identity (T2); this is the "attained with equality" clause of T3. (iii) The $o(1)$ is explicit: it equals $\sigma^2(\pi/w)^2/D$.

D3b, the adaptive companion. Value budget: the exploitation problem. Under the value budget the policy pays $\Phi(h^*_{\mathrm{CE}}) - \Phi(h_t)$ per step (the certainty-equivalent anchor of D0). The residual linear coefficient at the anchor is

$$\varphi'(\bar{\varepsilon})\,(\varepsilon - \bar{\varepsilon}) \ = \ -\beta(h^* - \psi)(\varepsilon - \bar{\varepsilon}),$$

so a policy could try to earn while exploring: guess the sign of $\varepsilon - \bar{\varepsilon}$ and drift accordingly. The lemma below says this self-financing is bounded by the information already purchased: the policy cannot exploit what it has not yet paid to know.

Setting: symmetric two-point prior $\varepsilon \in \{\bar{\varepsilon} - \delta,\ \bar{\varepsilon} + \delta\}$; write $s = \mathrm{sign}(\varepsilon - \bar{\varepsilon})$ and $g_1 := \beta\,|h^* - \psi|$, the known coefficient of the unknown part. The policy is non-anticipating, with per-step trust region $|d_t| \leq r$; write $S_t := \sum_{i\leq t} d_i^2$ for the design energy.

The exploitation–information lemma (L2). Two ingredients. First, the posterior-imbalance step: the policy's knowledge of s at time t obeys

$$\big|\mathbb{E}[s \mid \mathcal{F}_t]\big| \ \leq \ \mathrm{TV}(P_t^+, P_t^-) \ =: \ \mathrm{TV}_t$$

(posterior imbalance is bounded by the distinguishability of the two environments; the starting prior is symmetric). Register L2 records the averaged form of this step as the exact identity $\mathbb{E}\big|\mathbb{E}[s \mid \mathcal{F}_t]\big| = \mathrm{TV}(P_t^+, P_t^-)$. Second, the Pinsker step: the two environments differ only in the observation means, by $2\delta d_i$ at step i, so

$$\mathrm{KL}_t \ = \ \sum_{i\leq t} \frac{(2\delta d_i)^2}{2\sigma^2} \ = \ \frac{2\delta^2 S_t}{\sigma^2}, \qquad \mathrm{TV}_t \ \leq \ \sqrt{\mathrm{KL}_t/2} \ = \ \frac{\delta\sqrt{S_t}}{\sigma}.$$

(For adaptive designs the same computation runs through the chain rule for KL over the filtration, with S_t replaced by $\mathbb{E}S_t$, exactly as in the Fisher computation for T3; this is the form registered in L2.)

Chaining these, the expected exploitation gain obeys, for any non-anticipating policy,

$$\begin{aligned}\mathbb{E}[\mathrm{gain}_T] \;:=\; \mathbb{E}\Big[\sum_{t\le T} g_1\,\delta\,s\,d_t\Big] \;&\le\; g_1\,\delta \sum_{t\le T}\mathbb{E}\big[\mathrm{TV}_t\,|d_t|\big] \\ &\le\; \frac{g_1\,\delta^2}{\sigma}\,\sqrt{S_T}\sum_{t\le T}\mathbb{E}\,|d_t| \;\le\; \frac{g_1\,\delta^2}{\sigma}\,S_T\,\sqrt{T},\end{aligned}$$

where the first inequality is the tower rule plus the posterior-imbalance step, the second uses $\mathrm{TV}_t \le \delta\sqrt{S_t}/\sigma \le \delta\sqrt{S_T}/\sigma$, and the last is Cauchy–Schwarz ($\sum_{t\le T}|d_t| \le \sqrt{T\,S_T}$). For fully adaptive designs the same chain, run with expected energies, gives the closed form recorded in L2, $\mathbb{E}[\mathrm{gain}_T] \le g_1\,\delta^{3/2}\,T^{1/2}(\mathbb{E}S_T)^{3/4}/\sigma^{1/2}$; both forms are $O(\sigma\sqrt{T})$ at the minimax scale below. The formal proof is recorded in the proofs appendix under L2.

The minimax scale and the value-budget bound (T4). The minimax-relevant prior scale is $\delta = c\,\sigma/\sqrt{S_T}$: this is the accuracy the budget can buy (any larger δ is learned exactly, any smaller is unlearnable). Substituting into the lemma,

$$\mathbb{E}[\mathrm{gain}_T] \;\le\; g_1\,c^2\,\sigma\,\sqrt{T}, \qquad \text{while} \qquad \mathbb{E}[C_T] \;=\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,S_T \;\sim\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,T\,v$$

(cost equivalence L1, stationary-scale exploration $S_T = \Theta(T)$ with per-step exploration variance v). The exploitable fraction of the budget is therefore

$$\frac{\mathbb{E}[\mathrm{gain}_T]}{\mathbb{E}[C_T]} \;\le\; \frac{2\,g_1\,c^2\,\sigma}{\gamma_{\mathrm{PO}}\,v\,\sqrt{T}} \;=\; O\big(T^{-1/2}\big).$$

The Le Cam two-point reduction then quantifies the constant. At the minimax scale $\mathrm{TV}_T \le c$, so any estimator's worst-case risk over the pair obeys

$$\sup_{\varepsilon}\,\mathbb{E}\big[(\hat\varepsilon-\varepsilon)^2\big] \;\ge\; \frac{\delta^2}{2}\big(1-\mathrm{TV}_T\big) \;\ge\; \frac{\delta^2}{2}\,(1-c).$$

Multiplying by $\mathbb{E}[C_T] = \frac{1}{2}\gamma_{\mathrm{PO}}S_T$, using $\delta^2 S_T = c^2\sigma^2$, and applying the exploitation rebate (the value budget can be underpaid only by the exploited gain, an $O(T^{-1/2})$ fraction by the display above):

$$\sup\;\mathrm{Var}(\hat\varepsilon)\times C_T \;\ge\; \frac{\gamma_{\mathrm{PO}}\,\sigma^2}{4}\,c^2\,(1-c)\,\big(1-O(T^{-1/2})\big).$$

The factor $c^2(1-c)$ is maximized at $c = 2/3$, with value $4/27$, giving the registered bound of T4:

$$\inf_{\text{policy, estimator}}\;\sup\;\mathrm{Var}(\hat\varepsilon)\times C_T \;\ge\; \frac{\gamma_{\mathrm{PO}}\,\sigma^2}{27}\,\big(1-O(T^{-1/2})\big).$$

The sharp constant $\frac{1}{2}$, matching the achievable product of T2, that is $\inf\sup\,\mathrm{Var}(\hat\varepsilon)\times C_T \ge \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2\,(1-O(T^{-1/2}))$, is conjectured and is part of OPEN-1: no simulated policy fell below it by more than Monte-Carlo error (`run_open1.py`, part B).

Interpretation: the self-referential structure. The profitable direction is s, the very thing being estimated. Pinsker converts "how much the policy knows about s" into "how much design energy it has already spent"; any gain is therefore a rebate proportional to information already paid for, and the rebate's rate vanishes. Knowledge cannot be bootstrapped from its own purchase. To our knowledge no analog of this lemma appears in the performative-prediction, pricing, or experimental-design literatures (literature review items G1 to G3).

Caveats recorded with the derivation. (i) The constant in the $O(T^{-1/2})$ term is explicit, $2g_1c^2\sigma/(\gamma_{\mathrm{PO}}v)$, and blows up as $v \to 0$: a policy that barely explores can have its tiny cost substantially rebated. This is consistent, since in that regime both sides of the product are dominated by the transient (D1); the theorem is stated for stationary-scale exploration, $S_T = \Theta(T)$. (ii) $g_1 = \beta|h^* - \psi|$ is $O(1)$ in general; in the weak-performativity regime it is $O(\varepsilon)$, giving the sharper factor $\big(1-O(\varepsilon\,T^{-1/2})\big)$. (iii) The fixed-$\delta$ regime is different, and the certainty-equivalent anchor is the reason. At fixed prior separation the policy eventually learns s outright and can exploit it indefinitely: the exploitation fraction does not vanish (observed directly in V3b's first, deliberately mis-scaled run). That regime is not a counterexample. Sustained exploitation of learned structure is exactly what D0's certainty-equivalent anchor re-anchors away: once ε is known, the CE-optimal deployment shifts, and the "gain" is no longer measured as a rebate against exploration. The theorem's content concerns the unlearned residual, whose hardest instances live at the minimax scale $\delta \sim \sigma/\sqrt{S_T}$, and there the rebate vanishes at rate $T^{-1/2}$ (verified). We state this distinction explicitly: it separates the theorem's content from an apparent counterexample.

Numerical verification (V3). Three checks. (V3a) A Bayes-risk simulation with a Gaussian-prior posterior-mean estimator under three designs (front-loaded, spread, adaptive-greedy) with capped budget D: the measured risk respects $\sigma^2/\big(D + \sigma^2/\sigma_\pi^2\big)$ in every arm, with the spread design near-tight. (V3b-i) The Pinsker chain: the measured posterior imbalance satisfies $\big|\mathbb{E}[s \mid \mathcal{F}_t]\big| \le \delta\sqrt{S_t}/\sigma$ pathwise along simulated runs. (V3b-ii) An explicitly exploiting policy, which drifts by the posterior mean of the unknown linear term, simulated at several horizons T: its realized gain_T/C_T decays consistently with $O(T^{-1/2})$, and its achieved $\mathrm{Var}(\hat{\varepsilon}) \times$ cost never falls below $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2\,(1 - \text{measured rebate})$.

Structure-proofness within reach (T9). The floors above are nonparametric. A separate threat is a policy that knows the response family's functional form $\tau(h) = C_0 + C_1 e^{-ch}$ and shops for designs whose parametric Cramér–Rao product undercuts the floor. Define, for a finitely supported design μ,

$$R(\mu) \;=\; \big(g^\top M(\mu)^{-1} g\big) \int (h - h^*)^2 \, d\mu, \qquad M(\mu) \;=\; \int s\, s^\top d\mu,$$

with the sensitivity and the estimand gradient

$$s(h) \;=\; \big(1,\; e^{-ch},\; -C_1\, h\, e^{-ch}\big)^\top, \qquad g \;=\; \nabla_\theta\, \varepsilon(h^*) \;=\; -s'(h^*),$$

normalized so that $R(\mu) \ge 1$ says the family-knowing product is at least the exchange-rate floor $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ under the local-quadratic (A1) cost.

One-dimensional reduction. The generalized Rayleigh identity plus the bijection $u \leftrightarrow \phi_u = u^\top s$ between $\mathbb{R}^3$ and $V = \mathrm{span}\{1,\; e^{-ch},\; he^{-ch}\}$ (using $C_1 \neq 0$) gives

$$g^\top M(\mu)^{-1} g \;=\; \frac{1}{\inf\big\{\, \|\phi\|_\mu^2 \;:\; \phi \in V,\; \phi'(h^*) = 1 \,\big\}},$$

because $u^\top g = -\phi_u'(h^*)$ (the sign squares away). So the floor holds at μ if and only if some unit-slope family element is μ-cheaper than the linear function $d(h) = h - h^*$.

The candidate and its reach. The unique element of V with 2-jet $(0, 1, 0)$ at the anchor is

$$\phi_0(h) \;=\; \frac{2}{c} \;-\; e^{-c(h-h^*)}\Big(\frac{2}{c} + (h - h^*)\Big),$$

with expansion $\phi_0(x) = x - (c^2/6)\, x^3 + \cdots$ in $x = h - h^*$. The pointwise domination $|\phi_0(x)| \le |x|$ holds exactly on $x \in [-2/c,\; \infty)$ (equality only at $x = 0$ and $x = -2/c$), and fails strictly and exponentially below $-2/c$. The three elementary ingredients are $1 + t < e^t$ (which makes $\phi_0(x) - x$ strictly decreasing), $\phi_0'(x) = e^{-cx}(1 + cx) > 0$ above $-1/c$, and the single sign change of $e^{-cx}(1 + cx) + 1$ on $(-2/c,\; 0)$ (since $e^{-cx}(1 + cx)$ is increasing there from $-e^2$ to 1). The full proof is recorded in the proofs appendix under T9; verified in V9.1.

Consequence (T9(i)). For any design supported in $[h^* - 2/c,\; \infty)$: $R(\mu) \ge 1$ whenever the estimand is estimable, strict as soon as μ places mass off the two equality points $\{h^*,\; h^* - 2/c\}$, and $\inf_\mu R(\mu) = 1$, approached by symmetric three-point designs collapsing at the anchor (at rate δ^2 in the collapse half-width δ, V9.2; the limit is exact because in 2-jet coordinates the collapsing model is the quadratic family, where every symmetric design has $R = 1$ on the nose). The trust region A4 with $r \le 2/c$ is therefore exactly the condition under which the floor is structure-proof, not a technicality.

Sharpness of the candidate, and an exact duality. Three facts sharpen the picture, established in the adversarial verification campaign with certified numerics (the full write-up is journal material under OPEN-5). (i) The easy direction of a duality: for a finite support X let $m(X) = \max\{\phi'(h^*) : \phi \in V,\ |\phi(x)| \le |x - h^*|\ \forall x \in X\}$; then for *every* weighting w of X, $R(X, w) \ge m(X)^2$, because the maximizer ψ^* scaled to unit slope is feasible in the Rayleigh infimum with $\int (\psi^*/m)^2\, dw \le \int d^2\, dw/m(X)^2$. Within reach ϕ_0 certifies $m(X) \ge 1$, which is exactly T9(i); numerically the duality is tight ($\inf_w R(X, w) = m(X)^2$ by linear-programming duality, certified on random supports), so pointwise domination is not a lossy proof device: the reach question *is* the domination question. (ii) Single-candidate sharpness: among all unit-slope elements of V, ϕ_0 is the unique one whose domination set contains a full interval $[h^* - a, h^*)$ with $a > 0$, and the maximal such a is exactly $2/c$. The affine family with 2-jet $(0, 1, \gamma)$, $\gamma \in (-1, 0)$, trades the reach boundary outward (to $u_2(\gamma) < -2/c$ in units of $1/c$) at the proved price of vacating an interval just below the anchor, and the violating designs live exactly in the vacated gap. (iii) The below-reach failure is generic, not a boundary artifact: for every $A > 2/c$ there are nonsingular designs supported in $[h^* - A, \infty) \cap (0, \infty)$ with $R < 1$ (a certified three-point example in the register market attains $R = 0.8537$). Consequence for the statement of T9: by the extended-Chebyshev property of V (any nonzero element has at most two zeros counting multiplicity), every design with at least three distinct support points automatically has nonsingular information, so the estimability caveat in T9 concerns only one- and two-point designs.

The reach is exact (T9(ii)). Below $h^* - 2/c$ the domination fails, and the failure is realized: the frozen witness (support $\{1.8726,\ 1.8665,\ 1.3073,\ 0.05\}$, weights $\{0.003329,\ 0.955702,\ 0.040697,\ 0.000272\}$ normalized, frozen anchor $h^* = 1.8461$, register market $c = 1.5$) attains

$$R \;=\; 0.8516504721831888870\ldots$$

(50-digit precision; V9.2), with every support point except the 0.05 probe inside the reach: one vanishing-weight below-reach probe breaks the floor by 15 percent. Its true-cost ratio is $1.011 > 1$, so the violation lives in the local-quadratic cost model, which is the floor's own scope. Symmetric pair-plus-far-probe families never violate (their ratio tends to 1 from above as the probe weight vanishes; the old scan's verdict of global structure-proofness was a weight-grid artifact, recorded as the eleventh measurement-forced pivot). The characterization of the full below-reach violation set is OPEN-5.

D4/D5: Design geometry: optimal shapes, dispersion, temporal shaping, Chebyshev support (T5, C5.1, L3, T6)

This subsection records the derivations behind the three exact design optima of T5, the dispersion corollary C5.1, the temporal-shaping collapse L3, and the Chebyshev counting result T6; it covers both register slots D4 and D5, the Chebyshev material being the D5 portion. All three design optima are derived with proofs, the temporal-shaping lemma and the counting result are proved, and everything is verified in the V4 and V5 numerical checks. It is a derivation record: each optimum is computed in full, the caveats are stated where they arose, and the numerical anchors are those of V4 and V5. The formal statements live in the theorem register under the IDs above.

Setting. The decision is d-dimensional and the estimand is the response Jacobian. Exploration has stationary second moment $M := \mathbb{E}[dd^\top]$ (positive semidefinite), the response noise has variance σ^2 per coordinate, and by the d-dimensional form of the cost equivalence lemma L1 (from D0) the value cost rate is

$$\tfrac{1}{2}\,\mathbb{E}\big[d^\top \Gamma_{\mathrm{PO}}\, d\big] \;=\; \tfrac{1}{2}\,\mathrm{tr}(\Gamma_{\mathrm{PO}} M),$$

so the value budget over horizon T is $B = (T/2)\ \mathrm{tr}(\Gamma_{\mathrm{PO}} M)$. The per-unit-time information matrix is M/σ^2. Absorbing T and σ^2 into units, all three problems take the static normal form of a linear budget constraint $\mathrm{tr}(\Gamma_{\mathrm{PO}} M) \le B$ on the design M.

A-optimality (T5(a)): what to do if you must know everything equally. The problem and its exact solution are

$$\text{minimize} \quad \operatorname{tr}\left(M^{-1}\right) \qquad \text{subject to} \quad \operatorname{tr}(\Gamma_{\mathrm{PO}} M) \le B, \quad M \succ 0,$$

$$M^* \;=\; \frac{B}{\operatorname{tr}\,\Gamma_{\mathrm{PO}}^{1/2}}\,\Gamma_{\mathrm{PO}}^{-1/2}, \qquad \operatorname{tr}\left((M^*)^{-1}\right) \;=\; \frac{\left(\operatorname{tr}\,\Gamma_{\mathrm{PO}}^{1/2}\right)^2}{B}.$$

Derivation: the objective is strictly convex on the positive definite cone and the constraint is linear, so the KKT conditions are sufficient. Stationarity reads $-M^{-2} + \lambda\,\Gamma_{\mathrm{PO}} = 0$, hence $M = \lambda^{-1/2}\,\Gamma_{\mathrm{PO}}^{-1/2}$; the budget pins the multiplier through $\operatorname{tr}(\Gamma_{\mathrm{PO}} M) = \lambda^{-1/2}\,\operatorname{tr}\,\Gamma_{\mathrm{PO}}^{1/2} = B$, so $\lambda^{-1/2} = B/\operatorname{tr}\,\Gamma_{\mathrm{PO}}^{1/2}$, and substituting back gives the stated value. A diagonal-case Cauchy–Schwarz check confirms the shape: with curvatures g_i and design weights m_i,

$$\Big(\sum_i \frac{1}{m_i}\Big)\Big(\sum_i g_i\, m_i\Big) \;\ge\; \Big(\sum_i \sqrt{g_i}\Big)^2,$$

with equality at $m_i \propto g_i^{-1/2}$.

The reading is: explore where the objective is flat, but only to the square-root degree. The optimal exploration covariance is $\Gamma_{\mathrm{PO}}^{-1/2}$, the inverse square root and not the inverse: flat directions are cheap per unit information but not infinitely favoured, while curved directions are expensive but cannot be abandoned when every coordinate of the Jacobian is needed.

The price of isotropic jitter (C5.1). Isotropic exploration at the same budget, $M = (B/\operatorname{tr}\,\Gamma_{\mathrm{PO}})\,I$, is feasible with the budget tight and has risk $\operatorname{tr}(M^{-1}) = d\,\operatorname{tr}(\Gamma_{\mathrm{PO}})/B$. The ratio to the A-optimal value is

$$F \;=\; \frac{d\,\operatorname{tr}(\Gamma_{\mathrm{PO}})}{\left(\operatorname{tr}\,\Gamma_{\mathrm{PO}}^{1/2}\right)^2} \;\ge\; 1,$$

with equality if and only if all curvatures are equal (Cauchy–Schwarz again). F is the curvature dispersion of the objective: the exact factor by which naive isotropic exploration, which is optimal in LQR-like settings (Simchowitz and Foster, 2020), overpays in the performative setting. This is the content of C5.1, restated as the isotropy contrast P9.2, and F is computable on the calibrated multi-bond Γ_{PO} (whose diagonal is given exactly by P9.3).

D-optimality (T5(b)): what the confidence-volume objective wants.

$$\text{maximize} \quad \log\det M \qquad \text{subject to} \quad \operatorname{tr}(\Gamma_{\mathrm{PO}} M) \le B,$$

$$M^* \;=\; \frac{B}{d}\,\Gamma_{\mathrm{PO}}^{-1}, \qquad \log\det M^* \;=\; d\,\log\frac{B}{d} \;-\; \log\det\Gamma_{\mathrm{PO}}.$$

Derivation: $\nabla \log\det M = M^{-1}$, so stationarity reads $M^{-1} = \lambda\,\Gamma_{\mathrm{PO}}$; the objective is concave and the constraint linear, and the budget gives $\lambda^{-1} = B/d$.

A caveat recorded at derivation time: the two shapes differ, A-optimality giving $\Gamma_{\mathrm{PO}}^{-1/2}$ and D-optimality giving $\Gamma_{\mathrm{PO}}^{-1}$, and the two must never be conflated in the paper. The algorithm derivation (D6) uses the D-optimal criterion internally (Frank–Wolfe on $\log\det$), while the headline risk statement is A-optimal; both are stated, and the discrepancy factor between them is itself computable and is small when the dispersion F is moderate.

c-optimality (T5(c)): what the corrected (PerfGD-style) update actually needs. This is the new result of the three. The corrected update consumes the response estimate through a single functional, the correction direction c: in the scalar family the correction Δ needs only $\varepsilon(h^*)$, and in d dimensions $c = \nabla_E\,\Delta$ is a fixed vector once the operating point is known. Estimating the scalar $c^\top\theta$ alone:

$$\text{minimize} \quad c^\top M^{-1} c \qquad \text{subject to} \quad \operatorname{tr}(\Gamma_{\mathrm{PO}} M) \le B,$$

$$M^* \;=\; B\,\frac{c\,c^\top}{c^\top\Gamma_{\mathrm{PO}}\,c} \quad \text{(rank one: probe along } c \text{ itself)}, \qquad c^\top (M^*)^+\, c \;=\; \frac{c^\top\Gamma_{\mathrm{PO}}\,c}{B}.$$

Derivation: substitute $N = \Gamma_{\mathrm{PO}}^{1/2} M\, \Gamma_{\mathrm{PO}}^{1/2}$, so that $\operatorname{tr} N = \operatorname{tr}(\Gamma_{\mathrm{PO}} M) \le B$, and $w = \Gamma_{\mathrm{PO}}^{1/2} c$; the objective becomes $w^\top N^{-1} w$. By Cauchy–Schwarz,

$$\big(w^\top w\big)^2 \;=\; \big(w^\top N^{-1/2}\, N^{1/2}\, w\big)^2 \;\le\; \big(w^\top N^{-1} w\big)\big(w^\top N w\big),$$

and $w^\top N w \le \|w\|^2 \operatorname{tr} N \le \|w\|^2 B$, so $w^\top N^{-1} w \ge \|w\|^2/B$, attained at $N^* = B\, w w^\top/\|w\|^2$. Back-substituting,

$$M^* \;=\; \Gamma_{\mathrm{PO}}^{-1/2} N^* \,\Gamma_{\mathrm{PO}}^{-1/2} \;=\; B\, \frac{\big(\Gamma_{\mathrm{PO}}^{-1/2} w\big)\big(\Gamma_{\mathrm{PO}}^{-1/2} w\big)^\top}{\|w\|^2}, \qquad \Gamma_{\mathrm{PO}}^{-1/2} w = c, \quad \|w\|^2 = c^\top \Gamma_{\mathrm{PO}}\, c,$$

giving $M^* = B\, cc^\top/(c^\top \Gamma_{\mathrm{PO}}\, c)$ as stated.

The corrected trio is worth displaying side by side:

$$\text{A-opt: } M^* \propto \Gamma_{\mathrm{PO}}^{-1/2}, \qquad \text{D-opt: } M^* \propto \Gamma_{\mathrm{PO}}^{-1}, \qquad \text{c-opt: } M^* \propto c\, c^\top.$$

A-optimality shapes exploration as $\Gamma_{\mathrm{PO}}^{-1/2}$, D-optimality as $\Gamma_{\mathrm{PO}}^{-1}$, and c-optimality probes along the correction direction itself: the curvature does not tilt the direction at all, it only prices it. The value $c^\top \Gamma_{\mathrm{PO}}\, c/B$ says that the cost of knowing your correction is the Γ_{PO}-norm of what the correction asks for. The savings over the A-optimal value is the operational argument for correction-targeted probing: a desk does not need the whole Jacobian, and the geometry says exactly what it may skip.

Two caveats, recorded as they arose. First, an earlier draft of this derivation mis-stated the c-optimal direction as $\Gamma_{\mathrm{PO}}^{-1} c$; the substitution above corrects it. The error is recorded here per the project's falsification convention, and the corrected direction is locked by the V4 numerical check. Second, rank-one designs are singular (hence the pseudo-inverse in the value); in practice they are regularized by the trust region, and the paper states both the limiting and the regularized forms.

Temporal shaping is irrelevant (L3): design space collapse. Lemma L3 (temporal shaping is irrelevant). With response noise i.i.d. and independent of the deployment path, the information about the response parameters depends on the exploration process only through its empirical second moment $\sum_t d_t d_t^\top$. The computation is one line: the conditional log-likelihood is

$$\sum_t -\frac{\big(\tau_t - a - \theta^\top d_t\big)^2}{2\sigma^2},$$

and its Hessian in θ is $-\sum_t d_t d_t^\top/\sigma^2$, regardless of the temporal order or dependence structure of $\{d_t\}$. Hence temporally correlated exploration (AR-shaped, cyclic, burst) buys nothing beyond its stationary design measure, and the design problem is exactly the static problems treated above. Scope boundary: the lemma fails when the noise is serially correlated or feedback-coupled (the regime of section 4 of D2), where shaping does matter; this is stated as the boundary of L3 and left to the journal version.

Chebyshev counting (T6): why curvature is invisible to trajectories. This is the D5 portion of the record. The structural family of REFLEX theory 1.1 (`structural_response.py`) is

$$\tau(h) \;=\; C_0 + C_1\, e^{-ch}, \qquad \text{parameters } (C_0, C_1, c), \quad p = 3,$$

with sensitivity vector

$$s(h) \;=\; \big(1,\; e^{-ch},\; -C_1\, h\, e^{-ch}\big)^\top.$$

(In this paragraph c is the decay rate of the structural family, not the correction direction of the c-optimal problem.) The counting argument has four parts.

(i) $\{1,\, e^{-ch},\, he^{-ch}\}$ is an extended Chebyshev system on any interval: the relevant Wronskian-type determinants are nonzero (standard).

(ii) Consequently the Fisher matrix $M(\xi) = \sum_i w_i\, s(h_i)\, s(h_i)^\top$ of any design ξ with fewer than three distinct support points is singular, and the decay rate c is unidentified. The computation: $\operatorname{rank} M(\xi) \le \#\mathrm{support}(\xi)$, so a two-point design gives rank at most $2 < 3 = p$, hence $\det M(\xi) = 0$ and the c-direction lies in (or meets) the null space. The exact-arithmetic check in V5 finds

$\det M = 0$ to machine precision for every random two-point design and $\det M > 0$ for generic three-point designs.

(iii) Any noiseless RRM trajectory has effective support at most two (C1.2 of D1: geometric collapse for modulus below one, a period-two orbit at the boundary). Hence no retraining trajectory, at any modulus, identifies the curvature of its own response family.

(iv) D-optimal designs for this family on a trust-region interval $[h^* - r,\, h^* + r]$ are supported on exactly three points including both endpoints (a de la Garza-type argument; the interior point is computed numerically in the experiments).

This is the sharp form of the repository's empirical rule that exponential response fits need a wide spread range, and the formal reason the anti-echo freeze (the `identified` flag in `structural_response.py`) exists.

Verified numerically (V4, V5). Six checks anchor the results above. (1) A-optimality: random search over positive semidefinite M at fixed budget never beats $(\mathrm{tr}\ \Gamma_{\mathrm{PO}}^{1/2})^2/B$, and the closed-form M^* achieves it (run at $d = 5$ with random symmetric positive definite Γ_{PO}). (2) D-optimality: $M^* = (B/d)\,\Gamma_{\mathrm{PO}}^{-1}$ beats random feasible designs in $\log\det$. (3) c-optimality: random search never beats $c^\top \Gamma_{\mathrm{PO}}\, c/B$, and the rank-one construction approaches it. (4) The isotropic factor $F = d\ \mathrm{tr}(\Gamma_{\mathrm{PO}})/(\mathrm{tr}\ \Gamma_{\mathrm{PO}}^{1/2})^2$ matches the measured ratio. (5) Temporal lemma: AR-shaped and i.i.d. exploration with matched empirical second moment give identical OLS covariance. (6) Chebyshev: $\det M = 0$ for two-point designs versus $\det M > 0$ for three-point designs, and the c-direction variance is infinite or huge under two-point designs.

D6: Safe certainty equivalence (L4, R2, P7.1)

This derivation records the safety analysis of the certainty-equivalent corrected loop: the structural observation R2 (open-loop exploration cannot destabilize the linearized retraining dynamics), the perturbed-modulus lemma L4 (how C^1 estimation error in the response moves the closed-loop modulus), and the safety proposition P7.1 (the pessimism gate inherits uniform-in-time safety from an anytime-valid confidence sequence). The formal statements are recorded in the register under these IDs, and the proofs in the proofs appendix. Status of the record: R2 and L4 are derived and verified (V6); P7.1 is proved conditional on the validity of the confidence sequence; the $O(\sqrt{T})$ design-regret bound in the closing paragraph is stated with a proof strategy only and is labeled a journal-version item.

Where the stability constraint actually bites (R2). A fact discovered in the derivation and promoted to the paper: open-loop exploration cannot destabilize the linearized loop. The linearized deviation dynamics under a planned (history-independent) probe sequence (u_t) read

$$d_{t+1} \;=\; -m\, d_t \;+\; u_t,$$

and the homogeneous part keeps its modulus $m < 1$ regardless of the input u: bounded inputs give bounded outputs. This is R2. The stability risk of probing therefore enters through exactly two channels.

First, estimate feedback, the real one. The corrected update deploys

$$h_{t+1} \;=\; h_t + \eta\, \hat{\Phi}'(h_t), \qquad \hat{\Phi}' \;=\; G + \hat{\Delta},$$

where $\hat{\Delta}$ is built from the estimated response $\hat{\varepsilon}(\cdot)$. Estimation error changes the closed-loop map itself; this is where safety must be engineered, and it is the subject of L4.

Second, basin exit, the nonlinear channel: large probes can leave the contraction neighbourhood. This is handled by the trust region $|d_t| \le r$, a constraint the design problems of D4 already carry.

The perturbed-modulus lemma (L4). Let the corrected map be

$$\Psi(h) \;=\; h + \eta \left[\, G(h) - \beta\,(h - \psi)\,\hat{\varepsilon}(h) \,\right],$$

let $\hat{\rho}$ be its slope at the fixed point, and let ρ be the slope under the true response $\varepsilon(\cdot)$, namely $\rho = 1 - \eta\, \gamma_{\mathrm{PO}}$, the Newton-scaled contraction of the corrected update [Nagarajan and Ashok, 2026].

Lemma L4 (perturbed modulus). With $e(h) := \hat{\varepsilon}(h) - \varepsilon(h)$ the C^1 estimation error,

$$\big|\hat{\rho} - \rho\big| \;\le\; \eta\,\beta\,\Big(\|e\|_\infty + |h^* - \psi|\,\|e'\|_\infty\Big) \;=:\; \eta\,\beta\,L_{\mathrm{corr}}\,\|e\|_{C^1}.$$

The computation is short. The slope of the corrected map is $\Psi'(h) = 1 + \eta\,\hat{\Phi}''(h)$, so the modulus perturbation is η times the curvature error, and

$$\begin{aligned}\hat{\Phi}''(h) - \Phi''(h) &= -\beta\,\frac{d}{dh}\Big[\,(h-\psi)\,e(h)\,\Big]\\ &= -\beta\,\Big[\,e(h) + (h-\psi)\,e'(h)\,\Big].\end{aligned}$$

Taking absolute values and the supremum over the operating interval gives the bound.

For the structural family the C^1 error is controlled by the parameter error with explicit constants: with $e = \varepsilon(\cdot\,;\hat{\theta}) - \varepsilon(\cdot\,;\theta)$,

$$\|e\|_{C^1} \;\le\; L_\theta\,\|\hat{\theta} - \theta\|,$$

where L_θ is computable from (C_1, c, r). Consequently a confidence region for θ translates directly into an interval for the closed-loop modulus.

The pessimism rule and the safety proposition (P7.1). Maintain any anytime-valid confidence sequence Θ_t for the response parameters (from common-random-number machinery as in Nagarajan and Ashok [2026], or a standard self-normalized bound; its width shrinks as information accumulates, per D2).

Rule. At each step choose the pair (step size η_t, design) so that the worst-case modulus over Θ_t respects the margin,

$$\sup_{\theta\in\Theta_t}\,\big|\hat{\rho}(\theta)\big| \;\le\; 1 - c_{\mathrm{margin}},$$

that is, pessimistic on stability, while the design objective of D4 is evaluated optimistically. When Θ_t is too wide to certify the margin, the rule freezes the correction and explores under the blind loop, which is provably stable with modulus $m < 1$ by R2. This is exactly the anti-echo freeze of `structural_response.py`, now derived rather than engineered: this freeze is the pessimism gate.

Proposition P7.1 (safety). If the confidence sequence is valid at level $1-\delta$ uniformly over time, then with probability at least $1-\delta$ the realized modulus satisfies $|\rho_t| \le 1 - c_{\mathrm{margin}}$ for all t simultaneously. The argument is one line: on the event that $\theta \in \Theta_t$ for every t, which has probability at least $1-\delta$ by anytime-validity, the rule's supremum dominates the realized modulus at every step. The engineering reading: safety is inherited from the confidence sequence's uniform validity; there is no union bound over steps and no per-step failure accumulation.

Design regret (stated; journal deliverable). **Claim (to be proved in full).** The certainty-equivalent re-solving scheme (the D-optimal objective from D4, the pessimism rule above, and the trust region r) attains identification regret $O(\sqrt{T})$ against the budget-matched oracle design. The elliptical-potential argument of linear-bandit design theory supplies the rate, and the freeze episodes contribute an additive $O(\log T)$ term: each freeze ends when the confidence width halves, and the widths shrink geometrically in the accumulated design energy. As a remark on scope, this item is labeled partial for the workshop version: the algorithm and the safety guarantee are proved, while the regret constant and the freeze-episode accounting are the journal version's work.

Numerical verification (V6). Three checks, all in V6.

- Open-loop neutrality: injected bounded probe schedules never change the measured homogeneous contraction rate, read off as the slope of the deviation envelope, confirming R2.
- The perturbed-modulus lemma: corrected loops were run with deliberately corrupted $\hat{\varepsilon}$, the C^1 error swept over a grid; in every cell the measured fixed-point slope deviates from $1 - \eta\,\gamma_{\mathrm{PO}}$ by less than the bound of L4, and the bound is tight within a factor of about two at small errors.
- The pessimism rule: under a simulated valid confidence sequence, no trajectory in 400 seeded runs violates the margin; under a deliberately invalid (too-narrow) sequence, violations appear, so the validity assumption of P7.1 is necessary, not an artifact of the proof.

D7: The anchoring crossover (T7)

This subsection records the derivation behind T7: when does anchoring the response estimate to the structural family ("structure") beat a free-form nonparametric estimator ("capacity")? The headline of the derivation record is that the crossover is horizon-dependent, not budget-only. The formal statement is recorded as T7 in the theorem register. A status remark carried over from the derivation record: the derivation is complete, including the subtlety that changes the statement of the theorem (the nonparametric bias floor is horizon-dependent), and both the exact secant-bias constant and the two MSE formulas below are verified numerically in V7.

The question. REFLEX's central empirical finding is that anchoring beats a free-form estimator: the v3/v4 negative-then-positive result. When is that the right choice *in general*? The naive answer prices misspecification against the exchange rate of T2 alone:

$$\text{anchor} \quad \text{iff} \quad \delta_{\text{mis}}^2 \;<\; \frac{\gamma_{\text{PO}}\,\sigma^2}{2B}.$$

Working the derivation shows the naive answer is incomplete: the nonparametric alternative's bias is horizon-dependent, and the honest crossover involves $(\delta_{\text{mis}}, B, T)$ jointly.

The nonparametric side: local (secant) slope estimation. Estimate $\varepsilon = -\tau'(h^*)$ from symmetric probes at $h^* \pm w$, using n probe pairs, under the value budget and the horizon. Writing $S = 2nw^2$ for the design energy, the two constraints read

$$\begin{aligned} &\text{budget:} \quad \tfrac{1}{2}\,\gamma_{\text{PO}}\,S \;\le\; B, \\ &\text{horizon:} \quad 2n \;\le\; T \;\Longrightarrow\; w^2 \;=\; \frac{S}{2n} \;\ge\; \frac{S}{T}, \end{aligned}$$

and with the budget binding, $S = 2B/\gamma_{\text{PO}}$, the horizon forces fat probes:

$$w^2 \;\ge\; \frac{2B}{\gamma_{\text{PO}}\,T}.$$

Secant bias, with its exact leading constant. For the symmetric difference quotient,

$$\frac{\tau(h^*+w) - \tau(h^*-w)}{2w} \;=\; \tau'(h^*) \;+\; \frac{\tau'''(h^*)}{6}\,w^2 \;+\; O(w^4):$$

the even-order terms cancel by symmetry, so the second derivative does not enter at all. Symmetric probing is therefore already bias-optimal at its order, a point worth one sentence in the paper. For the structural family $\tau(h) = C_0 + C_1 e^{-ch}$ one has $\tau'''(h) = -c^3 C_1 e^{-ch}$, so

$$\text{bias}(w) \;=\; -\frac{c^3 C_1 e^{-ch^*}}{6}\,w^2 \;=\; -\frac{c^2}{6}\,\varepsilon\,w^2,$$

using $\varepsilon = \varepsilon(h^*) = -\tau'(h^*) = c\,C_1 e^{-ch^*}$: a curvature multiple of the slope itself, with the sign fixed by the family (only the magnitude enters the MSE below).

MSE at the budget- and horizon-constrained optimum. Each probe pair yields a difference quotient with variance $\sigma^2/(2w^2)$; averaging the n pairs gives

$$\text{Var} \;=\; \frac{\sigma^2}{2nw^2} \;=\; \frac{\sigma^2}{S} \;=\; \frac{\gamma_{\text{PO}}\,\sigma^2}{2B}$$

at the binding budget, independent of how S splits into n and w. The bias, in contrast, requires small w, and the horizon caps how small: $w^2 \ge 2B/(\gamma_{\text{PO}}T)$. Hence, at the constrained optimum,

$$\text{MSE}_{\text{np}} \;=\; \frac{\gamma_{\text{PO}}\,\sigma^2}{2B} \;+\; \Big(\frac{\tau'''(h^*)}{6}\Big)^2 \Big(\frac{2B}{\gamma_{\text{PO}}\,T}\Big)^2.$$

The discovered subtlety, and the reason T7 carries the horizon: the bias term *falls* as T grows at fixed budget. With more (cheaper, smaller) probes the nonparametric estimator approaches the parametric rate. Nonparametric identification is not uniformly worse; it is worse at *short horizons and fat probes*.

The parametric (anchored) side. For a correctly specified family, $\mathrm{MSE_p} = \kappa_p\, \gamma_{\mathrm{PO}}\, \sigma^2/(2B)$, where $\kappa_p \geq 1$ is the design efficiency for extracting $\tau'(h^*)$ through the family; c-optimal probing in the sense of T5(c), worked in D4, makes $\kappa_p \approx 1$, and the constant is computable. Misspecified by δ_{mis}, the C^1 distance from the true response to the family as in D6, the anchored estimator picks up an irreducible squared bias δ^2_{mis}:

$$\mathrm{MSE_{anchor}} \;=\; \kappa_p\, \frac{\gamma_{\mathrm{PO}}\, \sigma^2}{2B} \;+\; \delta^2_{\mathrm{mis}}.$$

The crossover. Anchor iff $\mathrm{MSE_{anchor}} < \mathrm{MSE_{np}}$, that is,

$$\delta^2_{\mathrm{mis}} \;<\; \Big(\frac{\tau'''(h^*)}{6}\Big)^2 \Big(\frac{2B}{\gamma_{\mathrm{PO}}\, T}\Big)^2 \;+\; (1-\kappa_p)\, \frac{\gamma_{\mathrm{PO}}\, \sigma^2}{2B};$$

since $\kappa_p \geq 1$, the second term is a (small) penalty on anchoring that vanishes under c-optimal probing through the family. The dominant first term gives the memorable form stated in T7:

$$\text{anchor} \quad \text{iff} \quad \delta_{\mathrm{mis}} \;<\; \frac{|\tau'''(h^*)|\, B}{3\, \gamma_{\mathrm{PO}}\, T} \quad \text{(to leading order).}$$

Readings. (i) Anchoring wins at short horizons, tight budgets, and strongly curved responses; the free-form route wins asymptotically in T at fixed budget. (ii) The formula retrodicts the v3 negative result *quantitatively*: the free-form loop failed at modest T with the trust region forcing fat effective probes, exactly the regime the formula assigns to anchoring. (iii) It also predicts when the finding would reverse (long horizons, generous probe counts), a regime no REFLEX experiment tests [Nagarajan and Ashok, 2026]: a falsifiable novel prediction. (iv) Delta over Lin–Zrnic (literature review item G5): their plug-in analysis has neither the budget nor the horizon; this formula prices both in the paper's single currency.

Numerical verification (V7). Three checks. (1) The secant-bias constant: the measured ratio $(\mathrm{secant} - \tau')/w^2$ converges to $\tau'''(h^*)/6$ for the exponential family (a deterministic check with exact function evaluations, three values of w extrapolated). (2) The $\mathrm{MSE_{np}}$ formula: Monte-Carlo secant estimation at the constrained optimum matches the two-term formula across a (B, T) grid. (3) The crossover: simulated anchored fits (correct family plus injected misspecification) against the secant estimator show an empirical crossover δ^*_{mis} tracking the formula within Monte-Carlo tolerance, including its B/T scaling (halving T doubles the crossover threshold).

D8/D9: The ROI of self-knowledge and the LQ and multi-bond instantiations (T8, P9.1, P9.2, P9.3)

This subsection records the derivations behind Theorem T8 (the return on investment of self-knowledge), the two consistency propositions P9.1 and P9.2, and the multi-bond curvature instantiation P9.3. Status of the underlying derivation document: the ROI closed forms are derived and verified (V8); both consistency propositions are proved (one-line proofs, verified); the multi-bond Γ_{PO} is derived exactly in the separable case, with the coupled extension stated to first order. The formal statements live in the theorem register and are quoted here only as needed for the algebra.

Ingredients. Three constants, all computable from earlier derivations and the primitives of the REFLEX market [Nagarajan and Ashok, 2026], enter the decision problem:

$$A = \tfrac{1}{2}\, \gamma_{\mathrm{PO}}\, (h_{\mathrm{SP}} - h_{\mathrm{PO}})^2, \qquad b = \tfrac{1}{2}\, \gamma_{\mathrm{PO}}\, \sigma^2, \qquad a = \tfrac{1}{2}\, \gamma_{\mathrm{PO}}\, \kappa^2.$$

Here A is the echo-chamber gap, the per-step value of running the corrected deployment h_{PO} instead of the naive h_{SP}; b is the exchange rate of T2, under which estimator accuracy v costs b/v in performative value (derivation D2); and a is the residual gap per unit of estimator variance: an error in $\hat{\varepsilon}$ of variance v displaces the corrected deployment by $\kappa^2 v$ in squared spread units, where $\kappa = |\, dh_{\mathrm{PO}}/d\varepsilon\,|$ is the sensitivity of the corrected optimum to the response slope, available in closed form from the market model's fixed-point equation [Nagarajan and Ashok, 2026].

The decision problem. Explore once to accuracy v (paying the exchange-rate price b/v up front), then run the corrected deployment forever, discounting the perpetual stream at rate ρ_{disc}. The corrected stream is worth $A - av$ per step, so the net present value is

$$\mathrm{NPV}(v) \;=\; \frac{A - av}{\rho_{\mathrm{disc}}} \;-\; \frac{b}{v}, \qquad v \in (0, \infty).$$

Optimal accuracy and break-even patience (T8). Differentiating,

$$\mathrm{NPV}'(v) \;=\; -\frac{a}{\rho_{\mathrm{disc}}} + \frac{b}{v^2} \;=\; 0 \qquad\Longrightarrow\qquad v^* \;=\; \sqrt{\frac{b\,\rho_{\mathrm{disc}}}{a}} \;=\; \frac{\sigma}{\kappa}\,\sqrt{\rho_{\mathrm{disc}}},$$

where the second form follows by substituting a and b: the common factor $\frac{1}{2}\gamma_{\mathrm{PO}}$ cancels, so the optimal accuracy does not involve the curvature at all. Since $\mathrm{NPV}''(v) = -2b/v^3 < 0$ on $(0,\infty)$, the objective is concave there and v^* is the global maximizer. At the optimum the two v-dependent terms are equal,

$$\frac{a\,v^*}{\rho_{\mathrm{disc}}} = \sqrt{\frac{a\,b}{\rho_{\mathrm{disc}}}} = \frac{b}{v^*},$$

so the optimal value is

$$\mathrm{NPV}(v^*) \;=\; \frac{A}{\rho_{\mathrm{disc}}} \;-\; 2\sqrt{\frac{a\,b}{\rho_{\mathrm{disc}}}}.$$

Exploring at all is worthwhile iff $\mathrm{NPV}(v^*) > 0$, which rearranges to

$$A \;>\; 2\sqrt{a\,b\,\rho_{\mathrm{disc}}} \qquad\Longleftrightarrow\qquad \rho_{\mathrm{disc}} \;<\; \rho^* \;:=\; \frac{A^2}{4\,a\,b} \;=\; \frac{(h_{\mathrm{SP}}-h_{\mathrm{PO}})^4}{4\,\kappa^2\,\sigma^2}.$$

The last equality is the substitution $A^2/(4ab) = \big(\frac{1}{2}\gamma_{\mathrm{PO}}\big)^2 (h_{\mathrm{SP}}-h_{\mathrm{PO}})^4 \,/\, \big(4\cdot\frac{1}{2}\gamma_{\mathrm{PO}}\kappa^2\cdot\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2\big)$, in which γ_{PO} cancels completely.

Three readings of the closed forms. (i) The break-even patience $\rho^* = (h_{\mathrm{SP}} - h_{\mathrm{PO}})^4/(4\kappa^2\sigma^2)$ involves only the deployment gap, the correction's sensitivity, and the response noise: the curvature drops out, a clean and slightly surprising closed form (it cancels between the gap's value and the exploration price). (ii) Everything on the right-hand side is computable from the primitives of REFLEX theory before exploring, which is the point of the result as a decision rule. (iii) v^* is the right amount of self-knowledge: more accuracy than v^* is vanity, less is blindness.

Frontier consistency (P9.1). For any deterministic design d_t with energy $S_T = \sum_{t\le T} d_t^2$, the pathwise exchange-rate product is

$$\frac{\sigma^2}{S_T} \times \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,S_T \;=\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2 :$$

the S_T cancels, so nothing about the schedule matters. In particular the $d_t \sim c/\sqrt{t}$ schedules with $S_T = c^2\log T$, the Lai–Robbins cost-$O(\log T)$ regime of adaptive design, satisfy the exchange rate of T2 with equality: their celebrated cheapness buys proportionally little information, and they sit exactly on the frontier.

Isotropy contrast (P9.2). In the d-dimensional problem, the ratio of the isotropic design's risk to the A-optimal design's risk at matched budget is the curvature-dispersion factor of D4 and C5.1,

$$F \;=\; \frac{d\ \mathrm{tr}(\Gamma_{\mathrm{PO}})}{\big(\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2}\big)^2} \;\ge\; 1,$$

with $F = 1$ iff Γ_{PO} is a multiple of the identity. That is exactly the LQR-like isotropic case in which naive isotropic exploration is optimal (Simchowitz–Foster 2020); generically $F > 1$ and grows with the spread of the curvature spectrum. The performative setting is generically anisotropic (the bond-level curvatures γ_a span decades across the calibrated universe), so naive exploration is generically suboptimal: this is the contrast between the two regimes stated in the main text.

Separable multi-bond curvature (P9.3, exact). In the multi-bond market with per-bond decisions h_a and separable flows, the response Jacobian is diagonal, $E = \mathrm{diag}(\varepsilon_a)$, and the performative objective separates across bonds. Applying the scalar curvature computation [Nagarajan and Ashok, 2026] bond by bond gives, exactly,

$$(\Gamma_{\mathrm{PO}})_{aa} \;=\; \gamma_a \;+\; \beta\,\varepsilon_a\,\big(2 + c_t\,\psi_a - c_t\,h_a\big), \qquad (\Gamma_{\mathrm{PO}})_{ab} = 0 \;\;(a \ne b).$$

All quantities on the right are the calibrated per-bond constants of the REFLEX market [Nagarajan and Ashok, 2026]. Consequently the dispersion factor F of P9.2 is directly computable on the calibrated universe (170 CUSIPs with at least twelve months of returns; the REALDATA record); that computed value is the number quoted in the paper's experiments.

Factor-coupled case (construction, first order). With factor-coupled responses, E diagonal plus low rank as in REFLEX theory 1.5, the same differentiation at the operating point gives

$$\begin{aligned}\Gamma_{\mathrm{PO}} \;&=\; \Gamma \;+\; \beta\Big[\,2\,E_{\mathrm{sym}} + c_t\big(\,\mathrm{diag}(\psi)\,E_{\mathrm{sym}} - H\,E_{\mathrm{sym}}\big)\Big] \;+\; O\big(\|E_{\mathrm{offdiag}}\|^2\big),\\ E_{\mathrm{sym}} \;&=\; \tfrac{1}{2}\,(E + E^\top), \qquad H = \mathrm{diag}(h_a),\end{aligned}$$

which is exact when E is symmetric and is evaluated at the operating point. The low-rank structure of E is inherited by the correction, so $\Gamma_{\mathrm{PO}}^{-1/2}$ and $\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2}$ remain computable in $O(dk^2)$ via the same Woodbury and eigen-split machinery already used in REFLEX theory 1.5: the instantiation adds no new computational burden.

Caveat (honesty convention). The paper states the separable case as an exact result (P9.3) and the coupled case only as a first-order construction carrying its explicit $O(\|E_{\mathrm{offdiag}}\|^2)$ error term, matching the reporting conventions of Nagarajan and Ashok [2026].

Numerical verification (V8). Five checks. (1) $v^* = \sqrt{b\,\rho_{\mathrm{disc}}/a}$ and the $\mathrm{NPV}(v^*)$ formula match a direct numerical scan over v on three parameter sets. (2) Root-finding $\mathrm{NPV}(v^*)(\rho_{\mathrm{disc}}) = 0$ in ρ_{disc} recovers $\rho^* = (h_{\mathrm{SP}} - h_{\mathrm{PO}})^4/(4\kappa^2\sigma^2)$ to 10^{-10}, confirming the γ_{PO} cancellation. (3) The frontier proposition: $c/\sqrt{t}$ schedules measure exactly on-frontier. (4) The dispersion factor on a synthetic curvature spectrum spanning two decades: F matches the measured isotropic-to-optimal risk ratio (cross-checked against V4). (5) The separable Γ_{PO}: a finite-difference Hessian of the separable objective matches the per-bond formula entrywise.

D Experimental details

This appendix records the configuration behind every number in Section 5 and Table 1: the register-market constants, the per-experiment settings of the measured-vs-predicted harness, the reporting metric and its stability audit, the baseline protocol, the real-data pipeline, and the compute footprint. The harness (`experiments/run_all.py`) stamps each row PASS, DRIFT, or FAIL: PASS means the measurement lands inside a preregistered relative tolerance of the register's closed form, DRIFT marks a deliberately out-of-scope cell whose deviation is measured and reported rather than excused, and the exit code fails on FAIL only. The shipped run has 36 rows with 35 PASS, one DRIFT, and zero FAIL. The roster runs E1–E8 and E10; there is no E9 family in the harness: the adversarial design search behind Theorem 3 runs separately in `experiments/run_open1.py`, and the real-data leg in `experiments/run_realdata.py`.

Register-market constants

The simulation environment is the structural family of Section 2.3 exactly: benign flow $U(h) = Ae^{-kh}$, toxic flow $\tau(h) = C_0 + C_1 e^{-ch}$, and frozen-level profit $J(h;\tau) = h\,U(h) + (h-\psi)\,\tau - w(h - h_{\mathrm{ref}})^2$. This is the REFLEX 1.1 market [Nagarajan and Ashok, 2026] with its gate and participation constants (P, ρ, $\bar{g}$) set to one and folded into the flow scales. Table 2 lists the defaults. Experiments that sweep the modulus hold every other constant fixed and rescale C_1 by bisection until the stable-point modulus $m = \varepsilon(h_{\mathrm{SP}})/\gamma(h_{\mathrm{SP}})$ hits the target: targets 0.3/0.6/0.85 in E1, 0.3/0.6 in E2, 0.5 in E5, and 0.6 in E4, E6, and E7.

The experiments

E1 (saturation cap and feedback floor; T1). Measures the identifying energy of the blind retraining loop against the transient cap $d_0^2/(1-m^2)$ and the feedback excitation floor $(\sigma/\gamma)^2/(1-m^2)$, at moduli 0.30, 0.60, 0.85. The cap cell runs the noiseless cobweb ($\sigma = 0$) from an excursion $d_0 = 0.10$ above h_{SP} for 200 steps; measured energies land within 1.2%, 4.0%, and 6.3% of the cap (tolerance 10%). The floor cell runs the noisy loop at $\sigma = 0.05$, measures cumulative energy at horizons 300 and 1200, and compares the affine growth rate to the floor; gaps are 0.7%, 0.2%, and 6.4% across the three moduli (tolerance 30%).

E2 (the exchange rate; T2, Theorem 1). Measures $\mathrm{Var}(\hat{\varepsilon}) \times C_T$ against $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ on the stationary retraining-plus-exploration loop, at moduli 0.3 and 0.6 crossed with exploration amplitudes $s_e \in$

Table 2: Register-market constants (defaults of the pipeline's `Market` dataclass). C_1 is rescaled per cell to hit each experiment's modulus target.

Symbol	Role	Value
A	benign arrival scale	1.0
k	benign demand elasticity	1.5
C_0	toxic base level	0.6
C_1	toxic slope scale (the feedback lever)	0.9
c	toxic spread decay (the curvature parameter of T6)	1.5
ψ	adverse severity per unit toxic flow	0.4
w	quoting-cost convexity	0.25
h_{ref}	quoting-cost anchor	1.0
σ	response-noise standard deviation	0.25
$[h_{\mathrm{lo}}, h_{\mathrm{hi}}]$	admissible quote interval	$[0.05,\ 4.0]$

$\{0.05, 0.15\}$, horizon $T = 5000$ with the first 300 steps discarded. Retraining consumes flow aggregated over 25 draws (noise $\sigma/5$, the REFLEX `n_episodes` convention) while the estimator's tape is single-draw; $\mathrm{Var}(\hat{\varepsilon})$ is the plug-in σ^2/S_{xx} on mean-centered deviations, and cost is $\sum_t \Phi(\bar{h}) - \Phi(h_t)$ at the mean deployed spread. Measured products: 0.010621 and 0.010696 against the predicted 0.010558 at $m = 0.3$; 0.0054946 and 0.0056856 against 0.0053539 at $m = 0.6$ (tolerance 15%), flat in both amplitude and modulus as the theorem requires. The drift cell replaces the environment with a saturating (tanh-capped, cap 0.9) response at $m = 0.6$: the product rises to 0.0062244, a measured $+16.3\%$ over the rate, recorded DRIFT, outside A1's scope by construction.

E3 (shaped versus isotropic exploration; T5a, C5.1). Measures the isotropic/A-optimal risk ratio against the dispersion factor $F = d \operatorname{tr}\Gamma_{\mathrm{PO}}/(\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2$ in $d = 8$, with curvature eigenvalues log-spaced from 0.1 to 10 in a random orthogonal basis, budget $B = 1.0$, noise $\sigma = 0.5$, $T = 4000$ probes per run, and 200 seeds per arm. The ratio comes out 1.529 against $F = 1.5063$, and the A-optimal risk itself matches its closed form $\sigma^2(\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2/(TB)$: 0.0064248 measured against 0.0068504 (both at tolerance 12%).

E4 (identification and safety; T6, L4). Two offline arms of $N = 600$ deployments over 60 seeds at $m = 0.6$ compare narrow jitter (amplitude 0.05 at h_{SP}) with the D-optimal three-point support of radius 0.8: the design arm's $\mathrm{sd}(\hat{c}) = 0.23364$ matches the Fisher prediction 0.24784, while the jitter/design sd ratio is 7.8 against the preregistered threshold of 3. The online cell runs SafeD-PerfGD ($r = 0.6$, margin 0.3, refit every 10 rounds) for $T = 4000$: the 300-step tail mean is 1.2575 against $h_{\mathrm{PO}} = 1.2964$, the trust region is never violated (maximum step-to-cap ratio 0.88, where the cap allows consecutive deployments to span the design support $2r$ plus one center move), and the correction stays frozen for a freeze fraction of 0.33 of the run: the gate clears after roughly 1,300 deployments, with 2,672 corrected steps.

E5 (the anchoring crossover; T7). At $m = 0.5$, budget $B = 0.6$, horizon $T = 120$, and 700 seeds, the nonparametric arm runs a secant slope estimator for $T/2$ probe pairs at width $w = \sqrt{S/T}$ with $S = 2B/\gamma_{\mathrm{PO}}(h_{\mathrm{SP}})$, measuring MSE 0.013205 against the predicted two-term optimum 0.012318 (tolerance 15%). The anchored arm fits the structural family on a three-point design of radius $w\sqrt{3/2}$ under a linear leak $-ah$ that the exponential family cannot represent, for $a \in \{0, 0.05, 0.1, 0.2, 0.35\}$; the anchored MSE rises from 0.0107 at $a = 0$ to 0.0126 at $a = 0.35$, crossing the nonparametric level 0.0123 between $a = 0.2$ and $a = 0.35$: anchoring wins when well-specified and loses under gross misspecification, in the direction and order T7 predicts. (An earlier misspecification, ae^{-2ch}, had decayed to numerical zero at the operating spread and was rightly invisible to the fit; the recorded pivot moved the leak to the operating point.)

E6 (published baselines). The baseline protocol and its results are described below.

E7 (ablation grid). The 2^3 grid design $\times$ anchor $\times$ gate runs at $T = 4500$ and $m = 0.6$ on a common environment seed, with the noisy c-identification comparison re-measured over six seeds. The full architecture reaches h_{PO} with tail error 0.044 (threshold 0.1); the design-off cell also converges (0.018) at matched transit energy, and the matched-energy RMS ratio of c errors is 0.7, inside the preregistered parity band $[0.3, 3.5]$: T6's identification cliff is a support and amplitude phenomenon (it bites E4's narrow priced jitter and E8's collinear probes), not a shape phenomenon.

Anchor-off degrades the best attainable final error, and only gated cells ever freeze. Two earlier versions of this experiment were pivots: a single-draw c comparison that inverted between profiles, and a matched-energy arm that had been funded at half the design's realized energy.

E8 (second economy: LQ performative pricing). The exchange-rate identity is re-measured in the LQ pricing model at $g \in \{0.4, 0.7\}$ crossed with $s_e \in \{0.2, 0.5\}$, horizon $T = 5000$ with 300 burn-in steps, under exact-intercept myopic retraining plus exploration noise. All four cells come out exact at the reported precision: 0.384 at $g = 0.4$ and 0.192 at $g = 0.7$, both amplitudes (tolerance 12%); A1 is global in an LQ model, so the drift of E2 vanishes. The transplanted agent runs at $g = 0.6$ for $T = 2500$ with randomized probe signs (deterministic alternating probes make the deployed price and its lag collinear, the T6 degeneracy recorded as pivot six): it reaches a tail average of 12.653 against $p_{\mathrm{PO}} = 12.5$, starting from $p_{\mathrm{SP}} = 7.14$, with a frozen fraction of 0.37.

E10 (multi-bond vector agent). The $d = 4$ market runs the vector agent with $\Gamma_{\mathrm{PO}}^{-1/2}$-shaped versus isotropic exploration at matched budget inside the running corrected loop, $T = 4000$, with 24 seeds on the dispersed universe and 12 on the flat control; steady-state risk is measured by holding the operating point and averaging ten independent windows, and anchored bonds (diagonal curvature above 0.3 at h_{PO}) are scored for convergence. The isotropic/shaped steady-state risk ratio is 1.20 on the dispersed universe (static $F = 1.45$; preregistered threshold 1.15) and 1.00 on the flat control (band $|\cdot - 1| < 0.25$), and the shaped agent reaches the anchored-bond optima with error 0.046 (threshold 0.2): the allocation effect appears exactly where the dispersion factor says it should, and nowhere else.

Reporting metric and stability audit

Every "final" value, in every arm of every experiment, is the same statistic: the mean over the last 300 steps of the deployed path, never a last iterate. The reason is recorded as a pivot: unanchored cells limit-cycle, a last-iterate reading samples the cycle at an arbitrary phase, and two E7 cells moved by a factor of 2–4 under a one-step horizon change before the metric was unified. The repository's stability report re-measures each audited cell at perturbed horizons around its base: the E7 ablation grid at five horizons ($T = 4498$–4502), the E4 SafeD cell around $T = 4000$, and the E8 pricing cell around $T = 2500$. The worst observed spread across all cells is 0.0062 (the E8 pricing agent), inside the preregistered tolerance of 0.01.

Baseline protocol (E6)

Four comparison arms run on the identical environment interface, at $m = 0.6$ and $T = 4000$, over a common bank of twelve seeds (99 through 110): *FD-PerfGD*, a finite-difference performative gradient method in the style of Izzo et al. [2021]; *ZO-PerfOpt*, two-point zeroth-order optimization on noisy P&L; *UCB-Grid*, a performative-confidence-bounds explorer in the spirit of Jagadeesan et al. [2022] on the quote interval $[0.4, 3.6]$; and *BlindRRM*, the retraining cobweb of Perdomo et al. [2020]. Each arm is scored with the same two numbers: the 300-step tail distance to h_{PO} and the oracle-measured cumulative performative regret $\sum_t \Phi(h_{\mathrm{PO}}) - \Phi(h_t)$. Table 3 reports medians with interquartile ranges on both axes. The Pareto criterion is preregistered at the medians; the seed-by-seed count is recorded, not asserted. On medians, no arm dominates SafeD-PerfGD on (final error, regret) jointly; seed by seed, SafeD is dominated in one of twelve seeds (seed 106, by UCB-Grid). The unsafe gradient baseline pays a median regret of 9586.79 against SafeD's 1007.86, more than five times. We deliberately report that UCB-Grid attains lower raw regret (315.92) than SafeD: certification is a purchased cost, and its visibility is the thesis. An earlier version of this experiment scored one arm with a tail average and the rest with last iterates, from a single seed each; the recorded fix is one metric, many seeds.

Real-data pipeline

The real-data leg (`experiments/run_realdata.py`) ports the REFLEX unit mapping and the 1.1/1.2 closed forms to the pipeline (numpy only), validates the port cell by cell against the published paper-grade REFLEX run [Nagarajan and Ashok, 2026], and then computes this paper's objects on the calibration. The universe is 10 rating $\times$ regime cells, $\{$IG, HY$\}$ crossed with $\{$calm, normal, elevated, stress, crisis$\}$. Validation requires h^*, $\varepsilon^* = \gamma$, and m to reproduce the published run's `calibrated_boundaries.csv`: all $10/10$ cells match, with worst per-quantity relative discrepan-

Table 3: E6 baselines, twelve matched seeds: median [Q1, Q3] on both axes.

Arm	Final error $\lvert h - h_{\mathrm{PO}}\rvert$	Cumulative regret
SafeD-PerfGD	0.0599 [0.0398, 0.0687]	1007.86 [910.00, 1130.79]
FD-PerfGD	0.0263 [0.0251, 0.0283]	9586.79 [9580.86, 9593.83]
ZO-PerfOpt	0.3868 [0.3089, 0.4693]	952.33 [915.92, 963.12]
UCB-Grid	0.1547 [0.1036, 0.1990]	315.92 [309.40, 341.71]
BlindRRM	1.2293 [1.2192, 1.2429]	2115.54 [2101.94, 2125.89]

cies of 0.23% on h^*, 0.62% on ε^*, and 0.76% on m, inside the 0.8% aggregate quoted in Section 5. The port uses the published run's YAML constants for the toxic channel ($\alpha = 0.5$, feedback 5.0); the package's dataclass defaults (0.15, 0.22) put the moduli off by a factor of 75.8 and matched $0/10$ cells, a trap the validation harness caught (pivot eight). The extension then computes, per cell, γ_{PO} (spanning 510.6 at IG-calm down to 2.3 at HY-crisis), the exchange rate $\frac{1}{2}\gamma_{\mathrm{PO}}$ per unit response variance (from 255.3 to 1.2), the echo-chamber gap $h_{\mathrm{SP}} - h_{\mathrm{PO}}$ (1.9–4.3% of the anchor spread), and the modulus (0.0547–0.0663 across cells). The dispersion factor is $F = 1.6252$ across the ten-cell portfolio (isotropic exploration overpays the A-optimal shape by about 63%) against $F = 1.0020$ within the IG-normal cell alone (0.2%): the shaping gain lives across ratings and regimes. The per-bond leg uses 170 CUSIPs with at least 12 months of returns, with per-bond annualized volatility spanning 0.001–0.148 (5th–95th percentile). Provenance is inherited and binding: only (A, k, σ, h) are data-identified, the toxic channel is structurally scaled with documented ratios, the panel is not trade-level TRACE, and the validation target is REFLEX's own published run, not market ground truth; the real-data leg validates machinery, not effect size.

Compute

The entire pipeline is numpy-only, CPU-only, and deterministic from named seeds; no GPU is used anywhere, and all output is ASCII. As stated in the repository README, the nine unit tests run in about one minute, the fast harness profile in about ten minutes, the full profile (the numbers reported here) in about twenty-five minutes, and the OPEN-1 premise check in about one minute, on a single CPU process. The real-data leg requires the REFLEX tree and skips gracefully without it. Continuous integration re-runs the verification suites on every push, and the companion derivation folder carries its own suite of 38 numerical checks (all passing) alongside the pipeline's 36 preregistered rows.

E Additional figures

F Reproducibility

All results are deterministic from seeds, numpy-only, CPU-only; reported final errors are 300-step tail averages over the deployed path, stable within 0.01 under horizon perturbation (worst measured spread 0.0062; the repository's stability report audits every cell at perturbed horizons). The repository contains: the derivation documents and theorem register (26 results; 38 numerical derivation checks), the pipeline (environments, estimators, designs, agents, baselines; 36 measured-vs-predicted rows; 9 unit tests), the real-data leg with its 10/10 port validation, and continuous integration running the verification suites on every push. The register of eleven measurement-forced pivots, each recorded in the code where it happened, with the failed prediction and the fix, is in the repository README.

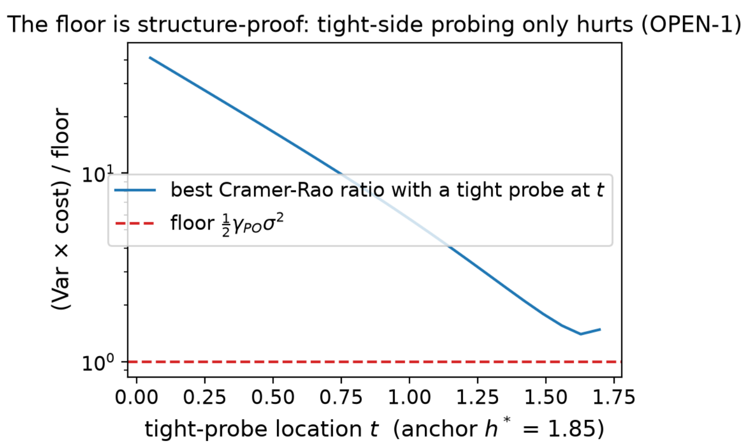

Figure 6: The coarse adversarial scan behind the old structure-proofness verdict: best Cramér–Rao ratio with one tight probe at t over the coarse weight grid ($w \geq 0.05$). At appreciable weight, tight probes only hurt. The below-reach violation of Theorem 3(ii) lives outside this family (a vanishing-weight probe, 3×10^{-4}, inside an asymmetric cluster), which is exactly why the coarse scan missed it and why the $1.005\times$ verdict stood until the proof said where to look.